Dell PowerFlex 5.1.x and PowerFlex Manager 5.1.x

CLI Reference Guide

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Rev. 1.1

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

This document provides a reference for using the PowerFlex command-line interface (CLI) to work with your system.

The PowerFlex CLI is intended for users who want to interactively manage a PowerFlex system, or to use commands in scripts for automating routine tasks.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

2

Revision history

Table 2. Revisions

Date	Document revision	Description of changes
August 2026	1.1	Removed query_oscillating_failure_counter_parameters command.
June 2026	1.0	Initial release

I

PowerFlex Manager CLI

This section explains the basic concepts of using the PowerFlex CLI, including how to access the SCLI and log in.

3

Accessing CLI

The `powerflex_cli` file is available in the Installer node in the folder `/opt/dell/pfmp`. You must download the bundle from the Dell Technologies support site and open it to locate the file. The powerFlex management platform bundle includes the meta data manager and other components. PowerFlex CLI is a tool that connects to the MDM, which in turn, sends commands or requests information about the system. You can use the PowerFlex CLI to run commands that are related to the management and operations cluster—the management layer, that is, events and alerts, and services in the management and operations cluster.

Logging in to CLI

To log in to CLI, it is required to have a server address and enter the username and password. The command options are:

- `server_address`—(Required) For other commands, the `server_address` is taken from the session file.
- `server_port`—(Optional) Default value: PowerFlex Default port: 443
- `session_name`—(Optional) Default value is "default". It is a user-provided name for the session.
- `cert_file`—(Required) when you are logging in for the first time. However, it is possible to log in using a certificate that exists from a prior PowerFlex CLI session.

The CLI commands are derived from the RESTful API. For example:

- Accessing help

```
powerflex_cli -help
```

- Running a query

```
powerflex_cli alerts query --session_token de98deb6b9927fc5 --id "3af3ab8277783029"
```

Login command examples

```
./powerflex_cli login --username scaleio --password scaleio --cert_file /tmp/
my_ca_cert.crt --server_address dellpowerflex.com
```

Try to log in to dellpowerflex.com using the certificate, `/tmp/my_ca_cert.crt`. The certificate is copied automatically to `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`. For details about how to get the certificate from the PowerFlex Manager server, see *Certificate*. If the login succeeds, an encrypted session file `~/.powerflex_cli/cli/default.token` is generated.

A session token is returned, which you can use for subsequent CLI sessions.

NOTE: You must have a session token to use the CLI.

Login command with session name

An example of a login command with a session name is given in the following.:

```
./powerflex_cli login --username scaleio --password scaleio --cert_file /tmp/
my_ca_cert.crt --server_address dellpowerflex.com -session_name some_script_name
```

1. Copy `/tmp/my_ca_cert.crt` to `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`
2. Try to log in to dellpowerflex.com using the certificate. If the login succeeds:
   - An encrypted session file `~/.powerflex_cli/cli/some_script_name.token` is generated.

A session token is returned.

Login command without certificate

An example of a login command with a session name is given in the following:

```
./powerflex_cli login --username scaleio --password scaleio --server_address
dellpowerflex.com
```

1. Try to log in to dellpowerflex.com using the certificate `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`.
2. If the certificate does not exist, then login without the certificate.

Certificate

The CLI must provide a certificate at least one time per server. The certificates can be provided more than once too. When the certificate is provided, the CLI saves it in `~/.powerflex_cli/certificates/<server>/ca.crt`. The certificate is used again if a user logs in to the same server without a certificate.

You can set the certificate using a flag in the login command.

To get the certificate from the PowerFlex Manager server, run the command:

```
kubectl get secret platform-ca -o json | jq -r '.data["ca-bundle"]' | base64 --decode
```

Sessions

If the login succeeds, the CLI generates an encrypted session file and saves it in `~/.powerflex/cli/`. The name of the session token file is in the format, `<session_name>.token` if the session name is given or `default.token` if the session name is not given. The session name allows you to run more than one script that calls the login command simultaneously and to be logged in to more than one server at the same time. The session file contains the name of the server, server port, and keycloak access token in an encrypted format.

Session token

The CLI returns a user session token, which is the key to open the session file. The session token is mandatory in any call to the CLI, except login. It expires if the session is idle for 30 minutes (default value). While in use, the session token is valid for 10 hours (default value). The CLI refreshes the session automatically. If a session name other than `default`, is passed with the log in operation, the same session name must be passed in every command with the session token.

Filters and operators

You can use filters for querying alerts, alert templates, and events.

You can use filters for a command many times. This is applicable to filters that have a logical AND between them.

Following are some examples of filter formats:

- query to get all alerts filtered by name=HEALTH_CHECK_FAILED -

  ```
  ./powerflex_cli alerts query --session_token 0541629989f63e34 --filter 'name eq
  HEALTH_CHECK_FAILED'
  ```

- query alerts on multiple filters - name = HEALTH_CHECK_FAILED and severity=CRITICAL

  ```
  ./powerflex_cli alerts query --session_token 0541629989f63e34 --filter 'name eq
  HEALTH_CHECK_FAILED' --filter 'severity eq CRITICAL'
  ```

- query events -

```
./powerflex_cli events query --session_token a909dcc30ddca7a9
```

- query events with filters -

```
./run_cli.sh events query --session_token a909dcc30ddca7a9 --filter 'code eq 20010001'
```

You can use the following with the filters:

- eq—
- ne—means 'not equal' - can be used in the same way of "eq"
- like—when you want to query all the items that their description contains a substring. and you can use "%" as a wildcard.

  For example:

  filter=description like "%this is %"

  It returns all the items with the substring "this is " anywhere in the string. the usage and behavior of like operator and "%" is exactly as in SQL.

- in—to retreive elements that the field value is one of that list. filter=name in (storage nodes, PROTECTION_DOMAIN)
- gt—greater than (>)
- lt—less then (<)
- ge—greater then equal (>=)
- le—less then or equal (<=)

The greater than, less than, greater than or equal to, and less than or equal to operators behave the same as the corresponding SQL operator in brackets. They are used in the syntax similarly to equal to or not equal to operators.

4

Alerts commands

Command line options for managing system alerts.

Table 3. Command options

Command option	Description
`--acknowledge`	Acknowledges an alert.
`--clear`	Clears an alert.
`--disacknowledge`	Disacknowledges an alert.
`--query`	Fetches a list of alerts.

Acknowledge alerts

Acknowledges an alert.

Syntax

```
./powerflex_cli alerts --[parameters] --acknowledge
```

Table 4. Parameter table

Parameter	Use	Descriptions
`-h, --h`	Optional	Help for acknowledging alerts.
`--session_token [string]`	Required	Secret session token. For more information, click Session token.
`--session_name [string]`	Optional	Session name. For more information, click Sessions.
`--id [string]`	Required	ID of an alert that needs to be acknowledged.

Example

This sample command uses a session token to locate and acknowledge a specific alert by its ID.

```
./powerflex_cli alerts --session token a909dcc30ddca7a9 --id c857efs4194wer --acknowledge
```

Clear alerts

The clear alerts command is used to clear a specific alert.

Syntax

```
./powerflex_cli alerts --[parameters] --clear
```

Table 5. Parameter table

Parameter	Use	Descriptions
-h, --h	Optional	Help for clearing alerts.
--session_token [string]	Required	Secret session token. For more information, click Session token.
--session_name [string]	Optional	Session name. For more information, click Sessions.
--id [string]	Required	ID of an alert that needs to be cleared.

Example

This sample command uses a session token to locate and clear a specific alert by its ID.

```
./powerflex_cli alerts --session token a909dcc30ddca7a9 --id c857efs4194wer --clear
```

Disacknowledge alerts

The disacknowledge alerts command is used to disacknowledge an alert, and it requires a secret session token and the ID of the alert to be disacknowledged.

Syntax

```
./powerflex_cli alerts --[parameters] --disacknowledge
```

Table 6. Parameter table

Parameter	Use	Descriptions
-h, --h	Optional	Help for disacknowledging alerts.
--session_token [string]	Required	Secret session token. For more information, click Session token.
--session_name [string]	Optional	Session name. For more information, click Sessions.
--id [string]	Required	ID of an alert that needs to be disacknowledged.

Example

This sample command uses a session token to locate and disacknowledge a specific alert by its ID.

```
./powerflex_cli alerts --session token a909dcc30ddca7a9 --id c857efs4194wer --
disacknowledge
```

Query alerts

The query alerts command is used to fetch a list of alerts.

Syntax

```
./powerflex_cli alerts --[parameters] --query
```

Table 7. Parameter table

Parameter	Use	Descriptions
`-h, --h`	Optional	Help for querying of alerts.
`--session_token [string]`	Required	Secret session token. For more information, click Session token.
`--session_name [string]`	Optional	Session name. For more information, click Sessions.
`--id [string]`	Required	ID of an alert that needs to be fetched.
`--filter [string]`	Optional	Conditions for filtering alerts.
`--limit [int32]`	Optional	Limit the number of returned objects.
`--offset [int]`	Optional	Starting row of the result set, which is used with limit for paging. The value is zero based (offset=0 starts the response from the first row of results).
`--order [string]`	Optional	Sorts the result set by ascending or descending order. Ascending order is default if not specified. The syntax to sort the result is: • **Ascending**: `<fieldname>.asc` • **Descending**: `<fieldname>.dsc` **NOTE:** To fetch records that are ordered by timestamp in ascending order: `--order timestamp.asc`

Example

This sample command uses the list of alerts and also sorts the ordered records by timestamp in ascending order.

```
./powerflex_cli alerts --session token a909dcc30ddca7a9 --order timestamp.asc --query
```

5

Alert template commands

This section contains commands for managing alert templates.

Syntax

```
powerflex_cli alert-templates [command]
```

Parameters

-h, --help
Help for querying dispatch sources.

--session name [string]
Session name.

--session token [string]
Secret session token.

Command options

modify
Updates an alert template.

query
Returns a list of alert templates

Modify alert template

The modify alert template command is used to update an alert template.

Syntax

```
powerflex_cli alert-templates modify [parameters]
```

Parameters

-h, --help
Help for modifying alert templates.

--id [string]
ID of alert template that needs to be updated.

--severity [int32]
Indicates the severity of the alert. The options are:

- INFORMATION
- MINOR

- MAJOR
- CRITICAL

--session_name [string]

Session name. The default value is "default".

--session_token [string]

Token of the secret session.

Query alert template

The powerflex_cli alert-templates query command fetches a list of alert templates, allowing for filtering, sorting, and pagination of results.

Syntax

```
powerflex_cli alert-templates query [parameters]
```

Parameters

-h, --help

Help for query.

--filter [string]

Conditions for filtering alerts.

--id [string]

ID of alert template to return.

--limit [int32]

Limit number of returned objects.

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order [string]

Sorts the result set by ascending or descending order. Ascending order is default if not specified. The syntax to sort the result is:

```
<fieldname>.asc
```

Or

```
<fieldname>.dsc
```

Example: To fetch records ordered by timestamp in ascending order:

```
--order timestamp.asc
```

--session_name string

Session name. The default value is "default".

--session_token [string]

Token of the secret session.

6

PowerFlex CLI version

This command displays the version of PowerFlex CLI.

Syntax

```
powerflex_cli version [parameters]
```

Parameters

-h, --help

Help for PowerFlex CLI.

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

7

Directory service commands

This section contains commands for managing directory services.

Syntax

```
powerflex_cli directory-services [command option]
```

Command options

create
: Create a directory service.

delete
: Delete a directory service by ID.

modify
: Modify a directory service by ID.

query
: Queries a directory service.

sync
: Synchronizes a directory service.

Create directory services

Creates a directory service.

Syntax

```
powerflex_cli directory-services create [parameters]
```

Parameters

-h, --help
: Help for creating directory services.

--configuration [string]

--directory_service_type [string]
: Options are:
- ad
- other

--group_configuration [string]

--name [string]
: Name of the directory service.

--state [string]

--user_search_setting [string]

--session_name [string]
Session name. The default name is "default".

--session_token [string]
Secret token of the session.

Modify directory services

Modifies a directory service by ID.

Syntax

```
powerflex_cli directory-services modify [parameters]
```

Parameters

-h, --help
Help for modifying directory services.

--configuration [string]

--group_configuration [string]

--id [string]
ID of an existing directory service.

--name [string]
Name of the directory service.

--state [string]

--user_search_setting [string]

--session_name [string]
Session name. The default name is "default".

--session_token [string]
Secret token of the session.

Query directory services

Query directory services.

Syntax

```
powerflex_cli directory-services query [parameters]
```

Parameters

-h, --help
Help for querying directory services.

--id [string]

ID of an existing directory service.

--session_name [string]

Session name. The default name is "default".

--session_token [string]

Secret token of the session.

Sync directory services

Synchronizes directory services.

Syntax

```
powerflex_cli directory-services sync [parameters]
```

Parameters

-h, --help

Help for synchronizing directory services.

--id [string]

--session_name [string]

Session name. The default name is "default".

--session_token [string]

Secret token of the session.

Delete directory services

Deletes a directory service by ID.

Syntax

```
powerflex_cli directory-services delete [parameters]
```

Parameters

-h, --help

Help for deleting directory services.

--id [string]

Id of an existing directory service.

--session_name [string]

Session name. The default name is "default".

--session_token [string]

Secret token of the session.

8

Dispatch destination commands

This section contains commands for managing dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations [command]
```

Parameters

-h, --help
Help for dispatch destinations.

--session_name [string]
Session name.

--session_token [string]
Secret session token.

Command options

modify
Modifies a dispatch destination.

query
Queries dispatch destination.

delete
Deletes a dispatch destination.

Query dispatch destinations

Returns a list of dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations query [parameters]
```

Parameters

-h, --help
Help for querying dispatch destinations.

--id [string]
ID of dispatch destination.

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Modify dispatch destinations

Modifies dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations modify [parameters]
```

Parameters

-h, --help

Help for modifying dispatch destinations.

--id [string]

--data [string]

--description [string]

--name [string]

--type [string]

Options:

- SNMPV2C
- SNMPV3
- SYSLOG
- WEBHOOKS
- Dell Technologies connectivity services
- EMAIL

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Delete dispatch destinations

Deletes dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations delete [parameters]
```

Parameters

-h, --help

Help for deleting dispatch destinations

--id [string]

Unique identifier of the dispatch destination.

--session_name [string]

Session name.

--session_token [string]

Secret session token.

9

Dispatch policies commands

This section contains commands for managing dispatch policies.

Syntax

```
powerflex_cli dispatch-policies [command options]
```

Parameters

-h, --help
Help for querying dispatch policies.

--session_name [string]
Session name.

--session_token [string]
Secret session token.

Command options

modify
Modifies a dispatch policies.

query
Queries a list of dispatch policies.

delete
Deletes dispatch policies

Query dispatch policies

Syntax

```
powerflex_cli dispatch-policies query [parameters]
```

Parameters

-h, --help
Help for querying dispatch policies.

--id [string]
Unique identifier of dispatch policy.

--session_name [string]
Session name.

--session_token [string]

Secret session token.

Modify dispatch policies

Syntax

```
powerflex_cli dispatch-policies modify [parameters]
```

Parameters

-h, --help

Help for modifying dispatch policies.

--id [string]

Unique identifier of dispatch policy.

--description [string]

--domain [string]

Values are:

- MANAGEMENT
- BLOCK
- NETWORK
- PLATFORM
- COMPUTE
- SECURITY

--name [string]

--severity [string]

--source [string]

Values are:

- SNMPV2C
- SNMPV3
- SYSLOG

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Delete dispatch policies

Syntax

```
powerflex_cli dispatch-policies delete [parameters]
```

Parameters

-h, --help

Help for deleting dispatch policies.

--id [string]

Unique identifier of dispatch policy.

--session_name [string]

Session name.

--session_token [string]

Secret session token.

10

Dispatch sources commands

This section contains commands for managing dispatch sources.

Syntax

```
powerflex_cli dispatch-sources [command options]
```

Parameters

-h, --help
Help for querying dispatch sources.

--session_name [string]
Session name.

--session_token [string]
Secret session token.

Command options

modify
Modifies a dispatch sources.

query
Queries a list of dispatch sources.

delete
Deletes dispatch sources.

Query dispatch sources

Syntax

```
powerflex_cli dispatch-sources query [parameters]
```

Parameters

-h, --help
Help for querying dispatch sources.

--id [string]
Unique identifier of dispatch sources.

--session_name [string]
Session name.

--session_token [string]

Secret session token.

Modify dispatch sources

Syntax

```
powerflex_cli dispatch-sources modify [parameters]
```

Parameters

-h, --help

Help for modifying dispatch sources.

--id [string]

Unique identifier of dispatch sources.

--data [string]

--description [string]

--name [string]

--type [string]

Values are:

- SNMPV2C
- SNMPV3
- SYSLOG

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Delete dispatch sources

Syntax

```
powerflex_cli dispatch-sources delete [parameters]
```

Parameters

-h, --help

Help for deleting dispatch sources.

--id [string]

Unique identifier of dispatch sources.

--session name [string]

Session name.

--session token [string]

Secret session token.

11

Event commands

This section contains commands for managing events.

Syntax

```
powerflex_cli events [command options]
```

Parameters

-h, --help
Help for events.

--session name [string]
Session name.

--session token [string]
Secret session token.

Command options

query
Returns a list of events

Query events

Returns a list of events.

Syntax

```
powerflex_cli events query [flags]
```

Parameters

-h, --help
Help for querying events

--id [string]

--filter [string]
Conditions for filtering alerts

--limit [int32]
Limit number of returned objects

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order [string]

Sorts the result set by ascending or descending order. Ascending order is default if not specified. The syntax to sort the result is:

```
<fieldname>.asc
```

Or

```
<fieldname>.dsc
```

Example: To fetch records ordered by timestamp in ascending order:

```
--order timestamp.asc
```

--session name [string]

Session name.

--session token [string]

Secret session token.

12

Login banner commands

This section contains commands to manage login banners.

Syntax

```
powerflex_cli login-banners [command options]
```

Parameters

-h, --help
Help for login-banners.

--session name [string]
Session name.

--session token [string]
Secret session token.

Command options

create
Creates a group.

delete
Delete a login banner by ID.

modify
Modify a login banner by ID.

query
Queries a login banner.

text
Text for a login banner.

Create login banners

Creates login banners.

Syntax

```
powerflex_cli login-banners create [parameters]
```

Parameters

-h, --help

Help for creating login banners.

--login_banner [string]

--session name [string]

Session name.

--session token [string]

Secret session token.

Modify login banner

Modify a login banner by ID.

Syntax

```
powerflex_cli login-banners modify [parameters]
```

Parameters

-h, --help

Help for modifying login banners.

--id [string]

ID of an existing login banner.

--login_banner [string]

--session name [string]

Session name.

--session token [string]

Secret session token.

Query login banner

Queries login banner.

Syntax

```
powerflex_cli login-banners query [parameters]
```

Parameters

-h, --help

Help for querying login banners.

--id [string]

ID of an existing login banner.

--session name [string]

Session name.

--session token [string]

Secret session token.

Delete login banner

Delete a login banner by ID.

Syntax

```
powerflex_cli login-banners delete [parameters]
```

Parameters

-h, --help
Help for deleting login banners.

--id [string]
ID of an existing login banner.

--session name [string]
Session name.

--session token [string]
Secret session token.

13

SMTP service commands

This section contains commands to manage SMTP services.

Syntax

```
powerflex_cli smtp-services [command options]
```

Parameters

-h, --help
Help for file interfaces.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

send-test-email
Sends test email to the destination.

test-connection
Tests SMTP connection.

SMTP services to send test email

Syntax

```
powerflex_cli smtp-services send-test-email [parameters]
```

Parameters

-h, --help
Help for sending test emails.

--password [string]

--recipient_1 [string]

--recipient_2 [string]

--recipient_3 [string]

--recipient_4 [string]

--recipient_5 [string]

--sender_address [string]

--server_type [string]

--username [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

SMTP services to test connection

Syntax

```
powerflex_cli smtp-services test-connection [parameters]
```

Parameters

-h, --help

Help for testing connections.

--password [string]

--recipient_1 [string]

--recipient_2 [string]

--recipient_3 [string]

--recipient_4 [string]

--recipient_5 [string]

--sender_address [string]

--server_type [string]

--username [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

14

PowerFlex storage node commands

This section contains commands to manage PowerFlex storage nodes.

Syntax

```
powerflex_cli storage-nodes [command options]
```

Parameters

-h, --help
Help for storage nodes.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

abort
Abort a running storage-nodes job.

progress
The progress of the running job of an operation.

reboot
Reboot block nodes on co-res installation.

run-script-on-host
Run script on host.

PowerFlex storage nodes abort

Stops a running storage nodes job.

Syntax

```
powerflex_cli storage-nodes abort [parameters]
```

Parameters

-h, --help
Help for abort.

--operation_name [string]

Storage node operations names enum:

- REBOOT
- RUN_SCRIPT_ON_HOST

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Storage nodes progress

Check the progress of the running job of an operation.

Syntax

```
powerflex_cli storage-nodes progress [parameters]
```

Parameters

-h, --help

Help for progress.

--operation_name [string]

storage node operations Names enum:

- REBOOT
- RUN_SCRIPT_ON_HOST

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Storage nodes reboot

Reboot storage nodes on integrated PowerFlex management (non-KVM) installation.

Syntax

```
powerflex_cli storage-nodes reboot [parameters]
```

Parameters

-h, --help

Help for reboot.

--all_pfmps

Reboot all PowerFlex Manager platform nodes.

--block_configuration [string]

--force_run_with_non_ready_pods

When set, the node is rebooted even if some pods are not ready. In such case, after the reboot the workflow does not wait for pods that were not ready before the reboot, to be ready.

--is_verification_script_required

A verification script should be run after the reboot. The verification script is expected to already be available on the nodes. The script should be located on each node at `/opt/emc/scaleio/lia/bin/verification_script`.

--node_ips strings

PowerFlex Manager platform nodes list to reboot.

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Storage nodes run script on host

Run the script on host.

Syntax

```
powerflex_cli storage-nodes run-script-on-host [parameters]
```

Parameters

-h, --help

Help for running script on host.

--block_configuration [string]

--execute_on_all_mdms

--execute_on_all_storage_nodes

--is_reboot_required

--is_running_in_parallel_on_pds

--is_stop_processing_on_script_failure

--is_upload_file_needed

--is_verification_script_required

--maintenance_mode_type [string]

--patch_script_file_path [string]

--verification_script_file_path [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

15

Dell Technologies connectivity services configuration commands

This section contains commands to manage Dell Technologies connectivity services configuration.

Syntax

```
powerflex_cli support-assist-config [command options]
```

Parameters

-h, --help
Help for file interfaces.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

connection_state

connectivity

modify
Modifies Dell Technologies connectivity services commands.

query
Queries all Dell Technologies connectivity services commands.

send-test-alert
Sends test alerts.

status

system_mode

Dell Technologies connectivity services configuration connection state

Syntax

```
powerflex_cli support-assist-config connection_state [parameters]
```

Parameters

-h, --help
Help for connection state.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Dell Technologies connectivity services configuration connectivity

Syntax

```
powerflex_cli support-assist-config connectivity [parameters]
```

Parameters

-h, --help
Help for connectivity.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Modify Dell Technologies connectivity services configuration

Syntax

```
powerflex_cli support-assist-config modify [parameters]
```

Parameters

-h, --help
Help for modifying Dell Technologies connectivity services configuration.

--contacts [string]

--enabled [string]

--eventMuted [string]

--systemMode [string]

--session name [string]
Session name. The default name is "default".

--session token [string]

Secret session token.

Query Dell Technologies connectivity services configuration

Syntax

```
powerflex_cli support-assist-config query [parameters]
```

Parameters

-h, --help

Help for querying Dell Technologies connectivity services configuration.

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Send Dell Technologies connectivity services configuration test alert

Syntax

```
powerflex_cli support-assist-config send-test-alert [parameters]
```

Parameters

-h, --help

Help for sending Dell Technologies connectivity services configuration test alert.

--message [string]

--symptomid [string]

--type [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

SupportAssist Dell Technologies connectivity services configuration status

Syntax

```
powerflex_cli support-assist-config status [parameters]
```

Parameters

-h, --help
Help for Dell Technologies connectivity services configuration status.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Dell Technologies connectivity services configuration system mode

Syntax

```
powerflex_cli support-assist-config system_mode [parameters]
```

Parameters

-h, --help
Help for Dell Technologies connectivity services configuration system mode.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

16

Syslogs commands

This section contains commands to manage syslogs.

Syntax

```
powerflex_cli syslogs [command options]
```

Parameters

-h, --help
Help for syslogs.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

modify
Modify the syslogs commands.

query
Query syslogs.

delete
Deletes syslogs.

Modify syslogs

Modifies the syslogs.

Syntax

```
powerflex_cli syslogs modify [parameters]
```

Parameters

-h, --help
Help for modifying the syslogs.

--id [string]
Unique Identifier of the syslogs.

--description [string]

--destination_address [string]

--name [string]

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Query syslogs

Queries syslogs.

Syntax

```
powerflex_cli syslogs query [parameters]
```

Parameters

-h, --help
Help for modifying the syslog.

--id [string]
Unique Identifier of the syslog.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Delete syslogs

Deletes syslogs.

Syntax

```
powerflex_cli syslogs delete [parameters]
```

Parameters

-h, --help
Help for deleting the syslog.

--id [string]
Unique Identifier of the syslog.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

17

User management commands

This section contains commands to manage users.

Syntax

```
powerflex_cli users [command options]
```

Parameters

-h, --help
Help for users commands.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

create
Creates a user.

modify
Modify user settings.

query
Query all users.

delete
Delete a user.

reset-password
Resets the password.

Create users

Syntax

```
powerflex_cli users create [parameters]
```

Parameters

-h, --help
Help for creating users.

--email_address [string]

--first_name [string]

--last_name [string]

--is_enabled [string]

--is_temporary [string]

--username [string]

--password [string]

--role [string]

Values are:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician,Monitor
- Support
- NA

--type [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Modify users

Modify a user by ID.

Syntax

```
powerflex_cli users modify [parameters]
```

Parameters

-h, --help

Help for modifying users.

--id [string]

Unique Identifier of an existing user.

--is_enabled [string]

--first_name [string]

--last_name [string]

--role [string]

Values are:

- SuperUser
- SystemAdmin
- StorageAdmi
- LifecycleAdmin

- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support
- NA

--username [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Query users

Syntax

```
powerflex_cli users query [parameters]
```

Parameters

-h, --help

Help for querying users.

--id [string]

Unique Identifier of an existing user.

--filter [string]

Conditions for filtering users (username, first_name, last_name, type, is_enabled, email_address).

--limit [int32]

Limit number of returned objects.

--offset [string]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Delete users

Delete a user by ID.

Syntax

```
powerflex_cli users delete [parameters]
```

Parameters

-h, --help

Help for deleting users.

--id [string]

Unique identifier of an existing user.

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

II

PowerFlex SCLI

This section explains the basic concepts of using the PowerFlex SCLI, including how to access the SCLI and log in and lists the SCLI commands.

18

SCLI basics

This section explains the basic concepts of using the PowerFlex SCLI, including how to access the SCLI and log in.

Accessing SCLI

PowerFlex CLI (SCLI) is installed as part of the MDM component. It can be found in the following path:

Table 8. Operating systems and CLI paths

Operating system	Path
Linux	`scli`
ESXi	`scli`
Citrix Hypervisor (XenServer)	`siocli`

Before using most SCLI commands, you must log in using the following command:

```
scli --login --management_system_ip <MNO IP> --username <username> --password <password>
```

`--management_system_ip <MNO_IP>`—IP address of the management system.

`--username <NAME>`—User name to log in.

`--password <PASSWORD>`—User password. If the password is missing, the user is prompted to enter the password.

Command format

All SCLI commands use the following format:

Table 9. Operating systems and command format

Operating system	Command format
Linux	`scli [--mdm_ip <IP>] --<cmd_name>`
ESXi	`scli [--mdm_ip <IP>] --<cmd_name>`
Windows	`scli [--mdm_ip <IP>] --<cmd_name>`
Xen	`siocli [--mdm_ip <IP>] --<cmd_name>`

Where:

- `--mdm_ip <IP>`

  is one or more IP addresses of the servers running the primary MDM and secondary MDM. In a non-clustered environment, use the MDM IP address.

  If the CLI does not reside on the MDM, the `--mdm_ip` parameter must be added to every CLI command.

- `--<cmd_name>`

  Is the command to be run.

Example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --query_all
```

Additional rules:

- All parameters are preceded by --
- The order of the parameters in the command is insignificant.
- SCLI commands are lowercase and case-sensitive.

--mdm_ip

The `mdm_ip` parameter indicates the MDM designated to receive and run the command. If the command is run from the primary MDM, this switch may be omitted.

Example:

```
scli --query_all
```

To avoid using the `--mdm_ip` parameter in every command, or avoid having to install the CLI on other servers, use SSH, or RDM to log in to the shell running on the management server.

You cannot run SCLI commands on the secondary MDM. However, you can send a command from the secondary MDM by adding the `--mdm_ip` parameter with the primary MDM IP address to the command.

Syntax

SCLI syntax follows these rules:

Format

The following command syntax is required:

```
scli --cmd_name
```

Parentheses ( ) indicate a group of required parameters, from which you must choose one.

Angle brackets <> indicate an argument.

Braces { } indicate a series of options for parameter values, from which you must choose one.

A pipe | is used to separate items in a series of options (parameters or parameter values). Select one of them.

Square brackets [ ] indicate an optional parameter.

Usage

Each command entry uses the above syntax and looks similar to the following example:

```
scli --cmd_example --r1 (--r2 | --r3 <V1>) [o1 <V2> | o2 {--so1 <V3> | --so2 }] [Options]
```

Parameter description

`--r1`

The `r1` parameter description.

`--r2`

The `r2` parameter description.

--r3<*V1*>

The `r3` parameter description with possible <*V1*> input values.

--o1<*V2*>

The `o1` description with possible <*V2*> input values.

--o2 {--so1 <V3> | --so2 }

The `o2` description

[Options]

Additional optional parameters:

so1 <V3>

The `so1` description with possible *V3* input values.

so2

The `so2` description

Interpretation

- The text `scli --cmd_example r1` is mandatory. In commands that have no parameters, only `scli --cmd_example` is necessary.
- `(--r2 | --r3 <V1>)` indicates that you must choose one of the options that are separated by |. Selecting an option is required, as indicated by ( ).
- `[o1 <V2>| o2 {--so1 <V3> | --so2}]` indicates that you may choose one of the options that are separated by |. Selecting one of these options is optional, as indicated by [ ].
- `[Options]` indicates that you may choose from the following additional parameters. It is optional, as indicated by [ ].

Object names

Define object names according to the following rules:

1. Contains fewer than 32 characters.
2. Contains only alphanumeric and punctuation characters.
3. Is unique within the object type.

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID using a query or the property sheet of the object, available in the PowerFlex Manager user interface.

It is recommended that you give each object a meaningful name that is associated with its operational role.

NOTE: The system has a unique name as it is a single object.

Using SCLI in non-secure mode

If PowerFlex is running in non-secure mode, you must disable secure communications on every MDM server to enable execution of commands.

- To disable secure communications in Windows, on each MDM open the SCLI `conf.txt` file, and add the following line:

```
cli_use_secure_communication=0
```

- To disable secure communications in Linux, run the following on each MDM:

```
echo cli_use_secure_communication=0 >> ~/.scli/conf.txt
```

NOTE: Changing from secure mode to non-secure mode after renewing certificates is not supported.

Logging in

To access the CLI, you must first log in to the management system using a terminal application.

If the CLI and the MDM do not reside on the same server, add the `--mdm_ip` parameter to all CLI commands.

In a non-clustered environment, use the MDM IP address. In a clustered environment, use the IP addresses of the primary and secondary MDMs, separated by a comma. For example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --login --username supervisor1 --password password1
--management_system_ip 10.11.12.13
```

If you do not enter the password, you will be prompted to enter the password.

login

Log in to the system by either specifying management credentials or using a pre-generated p12 certificate. It is recommended to avoid the password options. If not used, the user is prompted to enter the passwords in a secure manner.

Once you are authenticated by the system, all commands can be run with the respective role until a logout is performed, or until the session expires by reaching one of the following timeouts:

- Maximum session length (default: 8 hours)
- Session idle time (default: 10 minutes)

Before you log in for the first time, run the following commands:

`cp/opt/emc/scaleio/mdm/cfg/mgmt_ca.pem/`—Copy the management certificate to the root location.

`scli --add_certificate --certificate_file /opt/emc/scaleio/mdm/cfg/mgmt_ca.pem`—Add the management certificate to the trusted certificate.

Syntax

```
scli --login ( (--management_system_ip <MNO_IP> --username <NAME> [--password
<PASSWORD>] ) | ( [--p12_path <P12_PATH>] [--p12_password <P12_PASSWORD>] ) ) [--
accept_banner_by_scripts_only]
```

Parameters

`--management_system_ip <MNO_IP>`

IP address of the management system.

`--username <NAME>`

User name to log in.

`--password <PASSWORD>`

User password. If the password is missing, you are prompted to enter the password.

NOTE: In Linux, to prevent the password from being recorded in the history log, type the command without password parameter and then enter the password in the new line when it is prompted.

`--p12_path <P12_PATH>`

Path to p12 certificate file. If not specified, uses same default path as in generate_login_certificate.

`p12_password <P12_PASSWORD>`

Password for p12 certificate. If not specified, you are prompted to enter the password.

`--accept_banner_by_scripts_only`

Preemptive approval of login banner.

Examples

```
scli --login --management_system_ip dellpowerflex.com --username myuser --password 2@3#4A
```

If a login banner is configured and enabled on your system, press any key to display the banner. Enter **q** to quit the login banner, and then enter **y** to approve the banner.

generate_login_certificate

Generate a p12 certificate for CLI.

Syntax

```
scli --generate_login_certificate --management_system_ip <MNO_IP> --username <NAME> [--
password <PASSWORD>] [--p12_password <P12_PASSWORD>] [--p12_path <P12_CERT_PATH>]
```

Parameters

--management_system_ip <MNO_IP>

IP address of the management system.

--username <NAME>

User name to log in.

--password <PASSWORD>

User password. If the password is missing, the you are prompted to enter the password.

NOTE: In Linux, to prevent the password from being recorded in the history log, leave out the `password` flag and enter the password interactively.

--p12_password <P12_PASSWORD> (Optional)

The P12 password is used to encrypt the certificate. If missing, you are prompted to enter the password.

--p12_path <P12_CERT_PATH> (Optional)

The path where the p12 certificate is stored. If not specified, use the same default path as in login.

Examples

```
scli --login --management_system_ip dellpowerflex.com --username myuser --password
2@3#4A --p12_path /home/myuser/my_certificate.p12 --p12_password "password"
```

logout

Log out the current user of the system.

Syntax

```
scli --logout
```

Example

```
scli --logout
```

Getting help

The PowerFlex CLI supports auto-completion. To complete a command or parameters, press the **Tab** key while typing CLI commands.

NOTE:

In Windows, PowerFlex does not support auto-completion.

Alternately, you can run the `help` command.

Help

Display help for the CLI commands.

Syntax

NOTE: To view all help options, run the command without any additional options:

```
scli --help
```

Options

`--mdm`
Displays help for MDM commands.

`--sdt`
Displays help for SDT commands.

`--vol`
Displays help for volume commands.

`--sdc`
Displays help for SDC commands.

`--host`
Displays help for host.

`--general`
Displays help for general commands for example, license and obfuscation.

`--user`
Displays help for commands related to user management.

`--pds`
Displays help for PDS commands.

`--dgwt`
Displays help for DGWT commands.

`--storage_node`
Displays help for storage node commands.

`--all`
Displays help for all commands.

Example

```
scli --help --mdm
```

CLI command help syntax

- `Text`—Required
- <>—Argument
- `()`—Required element
- `[]`—Optional element
- `|`—Select from options A | B | C

Example command:

```
scli --cmd_example --req1 (--req2 | --req3 <VAL1>) [opt1 <VAL2> | opt2] Options
```

Some description of what `cmd_example` does.

Parameters:

`--req1`
req1 description

`--req2`
req2 description

`--req3 <V1>`
req3 description with possible VAL1 input values

`--opt1 <VAL2>`
opt1 description with possible VAL2 input values

`--opt2`
opt2 description

Options:
CHOOSE ONE/SEVERAL

`--spec_opt1 <VAL3>`
spec_opt1 description with possible VAL3 input values

`--spec_opt2`
spec_opt2 description

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Device gateway target commands

This section contains commands for managing device gateway target (DGWT).

add_dgwt

Adds a device gateway target (DGWT) to a specified storage node.

Syntax

```
scli --add_dgwt (--storage_node_id <ID> | --storage_node_name <NAME> | --storage_node_ip
<IP> [--storage_node_port <PORT>]) [--dgwt_name <NAME>] [--force_clean]
```

Parameters

--storage_node_name <NAME>

Storage node name.

--storage_node_id <ID>

Storage node ID.

--storage_node_ip <IP>

Storage node IP address.

--storage_node_port <PORT>

Port assigned to the device gateway target. on the storage node.

--dgwt_name <NAME>

A name to be assigned to the device gateway target.

--force_clean

Clean a previous device gateway target. configuration.

Examples

- `scli --add_dgwt --storage_node_name node_1 --dgwt_name dgwt_1`
- `scli --add_dgwt --storage_node_id b511d77f00000000 --dgwt_name dgwt_1 --force_clean`
- `scli --add_dgwt --storage_node_ip 10.20.30.40 --storage_node_port 9030`

remove_dgwt

Removes a device gateway target (DGWT).

Syntax

```
scli --remove_dgwt (--dgwt_id <ID> | --dgwt_name <NAME> | --storage_node_id <ID> |
--storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) [--
remove_all_devices] [--force]
```

Parameters

--dgwt_name <NAME>

Device gateway target name.

--dgwt_id <ID>

Device gateway target ID.

--storage_node_name <NAME>

Storage node name.

--storage_node_id <ID>

Storage node ID.

--storage_node_ip <IP>

Storage node IP address.

--storage_node_port <PORT>

Port assigned to the device gateway target on the storage node.

--remove_all_devices

Remove all devices first, then remove device gateway target. If not specified, device gateway target is expected to be empty of devices.

--force

Force devices removal despite miscellaneous unfavorable system conditions.

Examples

- `scli --remove_dgwt --dgwt_id 15b2fde000000000 --remove_all_devices`
- `scli --remove_dgwt --dgwt_name dgwt_1 --remove_all_devices --force`
- `scli --remove_dgwt --storage_node_name node_1`

abort_remove_dgwt

Aborts removal of a device gateway target (DGWT).

Syntax

```
scli --abort_remove_dgwt (--dgwt_id <ID> | --dgwt_name <NAME> | --storage_node_id <ID> |
--storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])
```

Parameters

--dgwt_name <NAME>
: Device gateway target name.

--dgwt_id <ID>
: Device gateway target ID.

--storage_node_name <NAME>
: Storage node name.

--storage_node_id <ID>
: Storage node ID.

--storage_node_ip <IP>
: Storage node IP address.

--storage_node_port <PORT>
: Port assigned to the device gateway target on the storage node.

Examples

- `scli --abort_remove_dgwt --dgwt_id 15b2fde000000000`
- `scli --abort_remove_dgwt --dgwt_name dgwt_1`
- `scli --abort_remove_dgwt --storage_node_name node_1`

query_dgwt

Retrieves information about a device gateway target.

Syntax

```
scli --query_dgwt (--dgwt_id <ID> | --dgwt_name <NAME> | --storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])
```

Parameters

--dgwt_name <NAME>
: Device gateway target name.

--dgwt_id <ID>
: Device gateway target ID.

--storage_node_name <NAME>
: Storage node name.

--storage_node_id <ID>
: Storage node ID.

--storage_node_ip <IP>
: Storage node IP address.

--storage_node_port <PORT>
: Port assigned to the device gateway target on the storage node.

Examples

- `scli --query_dgwt --dgwt_id 2bafbb0a00000000`
- `scli --query_dgwt --dgwt_name dgwt_0`
- `scli --query_dgwt --storage_node_id 0c3f5b6400000000`
- `scli --query_dgwt --storage_node_name node_0`
- `scli --query_dgwt --storage_node_ip 127.0.0.1,127.0.0.2`
- `scli --query_dgwt --storage_node_ip 127.0.0.1,127.0.0.3 --storage_node_port 9022`

query_all_dgwts

Retrieve information about all device gateway targets.

Syntax

```
scli --query_all_dgwts
```

query_all_devices

Retrieves information about all devices.

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Device commands

This section contains commands for managing devices.

query_all_device_latency_meters

Retrieve device latency meters for all the devices, or for devices in the specified protection domain.

NOTE: To query the devices in a specific storage node, use the `query_device_latency_meters` command.

Syntax

```
scli --query_all_device_latency_meters
  [(--protection_domain_id <ID> | --protection_domain_name <NAME>)]
```

NOTE: If the background device scanner is enabled, several device read statistics are dramatically affected.

Parameters

`--protection_domain_id <ID>`
Protection domain ID

`--protection_domain_name <NAME>`
Protection domain name

Example

```
scli --query_all_device_latency_meters --protection_domain_name pd10
```

Sample output

```
Protection Domain: fcc9af5500000000, Name: pd1 contains 4 SDS
Storage Pool: 3df1cd9600000000, Name: sp1 contains 8 devices
  SDS sds_idd_1:
    Device: /dev/sdb Average latency read: 836 write: 921 (microseconds)
    Device: /dev/sdc Average latency read: 755 write: 843 (microseconds)

  SDS sds_idd_2:
    Device: /dev/sdb Average latency read: 658 write: 815 (microseconds)
    Device: /dev/sdc Average latency read: 689 write: 732 (microseconds)

  SDS sds_idd_4:
    Device: /dev/sdb Average latency read: 842 write: 811 (microseconds)
    Device: /dev/sdc Average latency read: 852 write: 918 (microseconds)

  SDS sds_idd_5:
    Device: /dev/sdb Average latency read: 844 write: 698 (microseconds)
    Device: /dev/sdc Average latency read: 795 write: 801 (microseconds)
```

```
   Devices with maximal read latency:
     SDS: sds_idd_4 Device: /dev/sdc  Latency: 852 (microseconds)
     SDS: sds_idd_5 Device: /dev/sdb  Latency: 844 (microseconds)
     SDS: sds_idd_4 Device: /dev/sdb  Latency: 842 (microseconds)
   Devices with maximal write latency:
     SDS: sds_idd_1 Device: /dev/sdb  Latency: 921 (microseconds)
     SDS: sds_idd_4 Device: /dev/sdc  Latency: 918 (microseconds)
     SDS: sds_idd_1 Device: /dev/sdc  Latency: 843 (microseconds)

Protection Domain: fcc9af5600000001, Name: pd2 contains 1 SDS
Storage Pool: 3df1cd9700000001, Name: sp2 contains 1 devices
  SDS sds_idd_3:
    Device: /dev/sdb Average latency read: 0 write: 0 (microseconds)


Protection Domain: fcc9d66500000002, Name: rrr contains 0 SDS
```

query_device_test

Retrieve the storage node device test results. Every storage node saves results for the last two tests per device.

Syntax

```
scli --query_device_test ((--device_id <ID> | ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])
(--device_name <NAME> | --device_path <PATH>))) | (--storage_node_id <ID> | --
storage_node_name <NAME> | --stroage_node_ip <IP> [--storage_node_port <PORT>]) | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
| --storage_pool_id <ID>) | (((--protection_domain_id <ID> | --protection_domain_name
<NAME>))))
```

Parameters

--device_id *<ID>*
: Device ID.

--device_id *<ID>*
: Device ID.

--device_name *<NAME>*
: Storage node storage device name or file name.

--device_path *<PATH>*
: File path to storage device.

--storage_node_name <NAME>
: Storage node name.

--storage_node_id <ID>
: Storage node ID.

--storage_node_ip <IP>
: Storage node IP address.

--storage_node_port <PORT>
: Port assigned to the storage node.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --query_device_test
--sds_ip 192.168.1.6 --device_name /dev/sdb
```

add_device

Adds a mirroring or EC device.

Syntax

```
scli --add_device --device_name --device_path (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --device_group_name <NAME>) | --device_group_id <ID>)
(--storage_node_id <ID> | --storage_node_name <NAME> | --storage_node_ip <IP> [--
storage_node_port <PORT>]) --media_type <TYPE> [--test_time] [Test Options]
```

Parameters

--storage_node_name <NAME>

Storage node name.

--storage_node_id <ID>

Storage node ID.

--storage_node_ip <IP>

Storage node IP address.

--storage_node_port <PORT>

Port assigned to the storage node.

--protection_domain_id <ID>

Protection domain ID.

--protection_domain_name <NAME>

Protection domain name.

--device_group_name <NAME>

Device group name.

--device_group_id <ID>

Device group ID.

--device_path <PATH>

Storage device path or file path. A device can be a disk, unmounted partition or a file which represents free space on a mounted device.

--device_name <NAME>

Device name

--force_device_takeover

Force adding the device, ignoring other storage nodes holding the device. Use with caution, as others holding the device may write to it as well.

--media_type <TYPE>

Media type of the device (in mirroring, default is inheriting the type from the storage pool). TYPE can be one of the following:

- SSD - The device is an SSD device
- PMEM - The device is an PMEM device (EC only)

--test_time <TIME>

The maximum test run-time in seconds. If not specified, the test will be limited by IO size only. When --no_test is selected, this switch is ignored

Test Options

Choose one:

--test_only

Devices are tested but not used. Later, use the command activate_sds_device to start using their capacity.

--no_test

Devices capacity is used without any device testing.

<BLANK>

Read and write test is run on devices before their capacity is used.

activate_device

Activate one or all storage node or node devices which completed their tests.

Syntax

```
scli --activate_device (--device_id <ID> | ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--
device_name <NAME> | --device_path <PATH> | --activate_all_devices)))
```

Parameters

--device_id <ID>

Device ID

--device_name <NAME>

Device name

--device_path <PATH>

Storage node storage device path or file path

--storage_node_name <NAME>

Storage node name (EC only)

--storage_node_id <ID>

Storage node ID (EC only)

--storage_node_ip <IP>

Storage node IP address (EC only)

--storage_node_port <PORT>

Port assigned to the storage node (EC only)

--activate_all_devices

Activate all the devices of the specified storage node. If no devices can be activated, this command fails.

Examples

- `scli --activate_device --device_id 4e3b519200000000`
- `scli --activate_device --storage_node_id 568452366548654a --activate_all_devices`
- `scli --activate_device --storage_node_name sn1 --device_name dev1`

- `scli --activate_device --storage_node_ip 192.168.40.40 --device_path /dev/ssd3`
- `scli --activate_device --storage_node_ip 192.168.40.41 --storage_node_port 1855 --device_name dev2`

rename_device

Assign a name to, or rename, a storage node device.

Syntax

```
scli --rename_device (--device_id <ID> | ((--storage_node_id <ID> | --storage_node_name
<NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--device_name <NAME> |
--device_path <PATH>))) --new_name <NAME>
```

Parameters

--device_id *<ID>*
: ID of the storage node device.

--storage_node_name <NAME>
: Storage node name (EC only).

--storage_node_id <ID>
: Storage node ID (EC only).

--storage_node_ip <IP>
: Storage node IP address (EC only).

-storage_node_port <PORT> -
: Port assigned to the storage node (EC only).

--device_name *<NAME>*
: Name of the storage node device.

--new_name *<NAME>*
: New name to be assigned to the storage node device.

query_device

Queries device information.

Syntax

```
scli --query_device (--device_id <ID> | ((--storage_node_id <ID> | --storage_node_name
<NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--device_name <NAME> |
--all_devices)))
```

Parameters

--device_id <ID>
: Device ID. Can also be an RFcache device ID.

--device_name <NAME>
: Device Name. Can also be an RFcache device name.

--all_devices
Query all devices.

--storage_node_name <NAME>
Storage node name (EC only).

--storage_node_id <ID>
Storage node ID (EC only).

--storage_node_ip <IP>
Storage node IP address (EC only).

--storage_node_port <PORT>
Port assigned to the storage node (EC only).

query_all_devices

Retrieves information about all devices.

modify_device_capacity

Modifies the capacity that is used on a storage node.

Syntax

```
scli --modify_device_capacity (--device_id <ID> | ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--
device_name <NAME> | --device_path <PATH>))) --size_gb <SIZE>
```

Parameters

--storage_node_name <NAME>
Storage node name (EC only).

--storage_node_id <ID>
Storage node ID (EC only).

--storage_node_ip <IP>
Storage node IP address (EC only).

--storage_node_port <PORT>
Port assigned to the storage node (EC only).

--device_id <ID>
Device ID.

--device_name <NAME>
Device name.

--size_gb <SIZE>
Size of device to use in GB.

clear_device_error

Clear a storage node or storage node device error.

Syntax

```
scli --clear_device_error (--device_id <ID>) | ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--
device_name <NAME> | --clear_all))
```

Parameters

--storage_node_name <NAME>
Storage node name (EC only)

--storage_node_id <ID>
Storage node ID (EC only)

--storage_node_ip <IP>
Storage node IP address (EC only)

--storage_node_port <PORT>
Port assigned to the storage node (EC only)

--device_id <ID>
Device ID

--device_name <NAME>
Device name

--clear_all
Clear errors on all the devices

remove_device

Initiates removal of a device from a storage node or a storage node.

Syntax

```
scli --remove_device (--device_id <ID> | ((--storage_node_id <ID> | --storage_node_name
<NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--device_name <NAME> |
--device_path <PATH>)))
```

Parameters

--storage_node_name <NAME>
Storage node name (EC only)

--storage_node_id <ID>
Storage node ID (EC only)

--storage_node_ip <IP>
Storage node IP address (EC only)

--storage_node_port <PORT>
Port assigned to the storage node (EC only)

--device_id <ID>

Device ID

--device_name <NAME>

Device name

abort_remove_device

Cancels the removal of a storage device.

Syntax

```
scli --abort_remove_device (--device_id <ID> | ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--
device_name <NAME> | --device_path <PATH>)))
```

Parameters

--storage_node_name <NAME>

Storage node name (EC only)

--storage_node_id <ID>

Storage node ID (EC only)

--storage_node_ip <IP>

Storage node IP address (EC only)

--storage_node_port <PORT>

Port assigned to the storage node (EC only)

--device_id <ID>

Device ID

--device_name <NAME>

Device name

set_device_led

Turn the LED on a storage or cache device on or off.

Syntax

```
scli --set_device_led (--device_id <ID> | ((--storage_node_id <ID> | --storage_node_name
<NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]) (--device_name <NAME> |
--device_path <PATH>))) (--led_on | --led_off)
```

NOTE: This feature works only if LIA is enabled in the conf file of DGWT.

```
conf param: dgwt_hw_aware_enabled
```

Parameters

--device_id <ID>

ID assigned to the storage device.

--device_id *<ID>*

ID of a device or RF cache.

--device_name *<NAME>*

Name assigned to the storage device or RF cache.

--device_path *<PATH>*

Device path or file path of a storage device or RF cache.

--storage_node_name *<NAME>*

Storage node name (EC only).

--storage_node_id *<ID>*

Storage node ID (EC only).

--storage_node_ip *<IP>*

Storage node IP address (EC only).

--storage_node_port *<PORT>*

Port assigned to the Storage node (EC only)

--led_on

Turn the LED on.

--led_off

Turn the LED off.

Examples

- `scli --set_device_led --device_id fd7fa17a00030000 --led_on`
- `scli --set_device_led --storage_node_ip 127.0.0.1 --device_path /dev/sim_dev_0_0 --led_off`

start_device_test

Initiate a device test for one or all storage node devices.

Syntax

```
scli --start_device_test ((--device_id <ID> | ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])
(--device_name <NAME> | --all_devices))) | (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --device_group_name <NAME>) | --device_group_id <ID>))
[Options] [Control] [--i_am_sure]
```

Parameters

--device_id *<ID>*

Device ID.

--device_name *<NAME>*

Storage node device name or file.

--device_path *<PATH>*

Storage node device path or file path.

--protection_domain_id *<ID>*

Protection domain ID.

--protection_domain_name *<NAME>*

Protection domain name.

--device_group_name <NAME>

Device group name.

--device_group_id <ID>

Device group ID.

--storage_node_name <NAME>

Storage node name.

--storage_node_id <ID>

Storage node ID.

--storage_node_ip <IP>

Storage node IP address.

--storage_node_port <PORT>

Port assigned to the storage node.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Options

Choose several:

--io_type_write

Use write I/O type (default is read). If the storage device is in use, only read testing can be performed.

--io_pattern_squential

Use sequential I/O pattern (default is random).

--io_size_kb *<SIZE>*

The read or write I/O size in KB. Range is 4-512 KB. Default: 8 KB.

Control

Choose several:

--test_time *<TIME>*

Maximum test run time in seconds. If not specified, the test will be limited by I/O size only.

--io_total_size_mb *<SIZE>*

The total size of the test reads or writes in MB. Range is 1-32768 MB, which specifies the total I/O size in MB. Default: 128 MB

<blank>

Use defaults

Example

```
scli --mdm_ip 192.168.1.200 --start_device_test
--sds_ip 192.168.1.6 --device_path /dev/sdb --io_type_write --io_size_kb 8
```

set_secondary_io_policy

Set the secondary I/O policy for a protection domain.

Syntax

```
scli --set_secondary_io_policy (--protection_domain_id <ID>
| --protection_domain_name <NAME>) --policy <POLICY> [--
overall_concurrent_io_limit <LIMIT>] [--bandwidth_limit_overall_ios <BANDWIDTH>]
[--bandwidth_limit_bg_dev_scanner <BANDWIDTH>] [--bandwidth_limit_garbage_collector
```

```
<BANDWIDTH>] [--bandwidth_limit_singly_impacted_rebuild <BANDWIDTH>] [--
bandwidth_limit_doubly_impacted_rebuild <BANDWIDTH>] [--bandwidth_limit_rebalance
<BANDWIDTH>] [--bandwidth_limit_node_network <BANDWIDTH>] [--i_am_sure]
```

Parameters

--protection_domain_id <ID>

Protection domain ID.

--protection_domain_name <NAME>

Protection domain name.

--policy <POLICY>

Policy to use for secondary I/O priority. Can be one of the following:

- no_limit- Secondary I/Os are not limited.
- limit_concurrent_io - Limit the number of allowed secondary I/Os to the value defined for concurrent_io_limit.
- limit_throughput - Limit the secondary IO throughput.
- favor_application_io - Limit the number of allowed secondary I/Os to the values defined for concurrent_io_limit and bandwidth_limit, regardless of user I/Os.

--overall_concurrent_io_limit <LIMIT>

The maximum number of concurrent secondary I/Os per device. Valid range: 1-16. Default: 4

This limit is effective if the policy is limit_concurrent_io or favor_application_io.

--bandwidth_limit_overall_ios <BANDWIDTH>

The maximum bandwidth of secondary I/Os, in MB/s, per device. Valid range: 1-1024. Default: 400

Define the overall limit to all the secondary IOs together.

--bandwidth_limit_bg_dev_scanner <BANDWIDTH>

The maximum bandwidth of background device scanner I/Os, in MB/s, per device. Valid range: 1-1024. Default: 10

This limit is effective if the policy is Limit_throughput or favor_application_io.

--bandwidth_limit_garbage_collector <BANDWIDTH>

The maximum bandwidth of background garbage collector I/Os, in MB/s, per device. Valid range: 1-1024. Default: 10

This limit is effective if the policy is Limit_throughput or favor_application_io.

--bandwidth_limit_singly_impacted_rebuild <BANDWIDTH>

The maximum bandwidth of Singly Impacted Rebuild I/Os, in MB/s, per device. Valid range: 1-1024. Default: 400

--bandwidth_limit_doubly_impacted_rebuild <BANDWIDTH>

The maximum bandwidth of Doubly Impacted Rebuild I/Os, in MB/s, per device. Valid range: 1-1024. Default: 400

--bandwidth_limit_rebalance <BANDWIDTH>

The maximum bandwidth of Rebalance I/Os, in MB/s, per device. Valid range: 1-1024. Default: 20

--bandwidth_limit_node_network <BANDWIDTH>

The maximum Node Network Bandwidth (per direction RX/TX) in Gbits/s. Valid range: 25-400. Default: 25

--i_am_sure

Preemptive approval

21

Device group commands

This section contains commands for managing device groups.

add_device_group

Adds a device_group to specified protection domain.

Syntax

```
scli --add_device_group (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --media_type <TYPE> [--device_group_name <NAME>] [--spare_storage_node_count
<COUNT>] [--spare_device_count <COUNT>] [--enable_device_hardware_checks | --
disable_device_hardware_checks]
```

Parameters

--protection_domain_id <ID>

Protection domain ID.

--protection_domain_name <NAME>

Protection domain name.

--device_group_name <NAME>

A name to be assigned to the device group.

--media_type <TYPE>

Media type of the devices in the device group where TYPE can be one of the following:

- PMEM - All devices in this device group are PMEMs
- SSD - All devices in this device group are SSDs

--spare_storage_node_count <COUNT>

The amount of storage_nodes to allocate for spare (0-2).

--spare_device_count <COUNT>

The amount of devices to allocate for spare (0-24) .

If no spare count is specified, then spare_storage_node_count gets 0 and spare_device_count gets 2. If only one is specified, the other gets 0.

Spare capacity is calculated according to the sum of the sizes of the largest storage nodes and largest devices regardless of whether one of the largest devices included in one of the largest storage nodes. If spare_storage_node_count or spare_device_count is smaller than the number of storage nodes or devices in accordance, the spare capacity is calculated according to the storage nodes and devices that exist.

--enable_device_hardware_checks

Enable device hardware checks for the device group

--disable_device_hardware_checks

Disable device hardware checks for the device group. The options are:

- Enable
- Disable

If neither is specified, the default is Enable.

modify_device_group

Modify the name, spare capacity, or device hardware checks policy of the device group.

Syntax

```
scli --modify_device_group (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --device_group_name <NAME>) | --device_group_id <ID>) [--
spare_storage_node_count <COUNT> | --spare_device_count <COUNT>] [--new_name <NAME>] [--
enable_device_hardware_checks | --disable_device_hardware_checks]
```

Parameters

--protection_domain_id <ID>

Protection domain ID

--protection_domain_name <NAME>

Protection domain name

--device_group_name <NAME>

Device group name

--device_group_id <ID>

Device group ID

--spare_storage_node_count <COUNT>

The number of storage_nodes to allocate for spare (0-2)

--spare_device_count <COUNT>

The number of devices to allocate for spare (0-24). If only one spare parameter is specified, the other gets 0

--new_name <NAME>

A name to be assigned to the device group

--enable_device_hardware_checks

Enable device hardware checks for the device group

--disable_device_hardware_checks

Disable device hardware checks for the device group

remove_device_group

Removes a device group.

Syntax

```
scli --remove_device_group (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --device_group_name <NAME>) | --device_group_id <ID>)
```

Parameters

--protection_domain_id <ID>

Protection domain ID.

--protection_domain_name <NAME>

Protection domain name.

--device_group_name <NAME>

Device group name.

--device_group_id <ID>

Device group ID.

query_device_group

Retrieves information about a device group.

Syntax

```
scli --query_device_group (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --device_group_name <NAME>) | --device_group_id <ID>)
```

Parameters

--protection_domain_id <ID>

Protection domain ID.

--protection_domain_name <NAME>

Protection domain name.

--device_group_name <NAME>

Device group name.

--device_group_id <ID>

Device group ID.

query_all_device_group

Retrieves information about all device groups.

query_available_capacity

Retrieves information about the available capacity of a device group according to a specific protection scheme, if specified. Or according to all protection schemes if not specified.

Syntax

```
scli --query_available_capacity (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --device_group_name <NAME>) | --device_group_id
<ID>) [--data_protection_scheme <PROTECTION_SCHEME>] [--wrc_device_group_name | --
wrc_device_group_id]
```

Parameters

--protection_domain_id <ID>

Protection domain ID.

--protection_domain_name <NAME>

Protection domain name.

--device_group_name <NAME>

Device group name.

--device_group_id <ID>

Device group ID.

--data_protection_scheme <DATA_PROTECTION_SCHEME>

The data protection scheme.

--wrc_device_group_name <NAME>

Write cache device group name (EC only).

--wrc_device_group_id <ID>

Write cache device group ID (EC only).

22

General CLI commands

This section contains general CLI commands, including commands relating to licenses, certificates, login banner, and syslog.

add_trusted_ca

Upload the certificate from the specified file as a trusted certificate.

Syntax

```
scli --add_trusted_ca --certificate_file <FILE> --comment <COMMENT>
```

Parameters

--certificate_file *<FILE>*

The certificate file to upload as trusted

--comment *<COMMENT>*

Optional free text to describe the certificate

Example

```
scli --mdm_ip 192.168.1.200 --add_trusted_ca --certificate_file system2_ca.cer --comment
"Certificate for system 2"
```

list_trusted_ca

Display a list of the currently trusted certificates in the system.

Syntax

```
scli --list_trusted_ca
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --list_trusted_ca
```

query_license

Retrieve information about the current license and restrictions.

Syntax

```
scli --query_license
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_license
```

query_disconnections

Query SDC-storage node, SDC-PDS, SDT-storage node, SDT-PDS disconnections.

Syntax

```
scli --query_disconnections [--disconnection_type <sdc_sds | sdc_pds| sdt_sds | sdt_pds
| >] [--print_all] [--print_proxy]
```

Parameters

--disconnection_type <TYPE>

Print disconnections of the given type only. The following options are available:

sdc_sds

SDC-storage node connections

--print_all

Use this flag to print all disconnections

--print_proxy

When proxy access mode is active, print storage nodes that can not serve as a proxy.

Example

```
scli --mdm_ip 192.168.1.200 --query_disconnections --disconnection_type sdc_sds
```

query_remote_syslog

Retrieve the remote syslog module state and configuration.

Syntax

```
scli --query_remote_syslog
```

Parameters

None.

Example

```
scli --query_remote_syslog
```

remove_trusted_ca

Remove the certificate with the given fingerprint.

Syntax

```
scli --remove_trusted_ca --fingerprint <FINGERPRINT>
```

Parameters

--fingerprint <FINGERPRINT>

The fingerprint of the certificate to be removed

Example

```
scli --mdm_ip 192.168.1.200 --remove_trusted_ca --fingerprint
B5:F0:43:1E:9B:7B:55:7F:05:98:B7:D9:7E:50:D4:9F:6D:1B:A7:5F
```

set_cli_login_banner_acceptance_mode

Set the login banner acceptance mode.

Syntax

```
scli --set_cli_login_banner_acceptance_mode (--password_only | --accept_banner)
```

Parameters

--password_only

Allow CLI users to accept the login banner by entering a valid password.

--accept_banner

Require CLI users to accept the login banner by pressing 'y'.

Example

```
scli --set_cli_login_banner_acceptance_mode --password_only
```

set_cli_login_banner_preemptive_acceptance

Enable or prevent CLI users from preemptively accepting the login banner.

Syntax

```
scli --set_cli_login_banner_preemptive_acceptance (--enable | --disable)
```

Parameters

--enable

Allow CLI users to preemptively accept the login banner

--disable

Prevent CLI users from preemptively accepting the login banner

Example

```
scli --set_cli_login_banner_preemptive_acceptance --enable
```

set_login_banner

Set the login banner that is displayed at every login to the system, or remove an existing login banner.

The maximum file size for the banner is 16 KB. Only ASCII text files are supported.

Syntax

```
--set_login_banner (--filename <FILE> | --remove_banner)
```

Parameters

--filename *<FILE>*

File containing the login banner to set

--remove_banner

Remove the login banner.

Example

```
scli --set_login_banner --filename login_banner.txt
```

set_remote_read_only_limit_state

Configure the PowerFlex system to restrict access from remote clients to read-only operations on the MDM. When this restriction is enabled, remote client users cannot change any configurations on the MDM. However, users on a local host may still make configuration changes while the restriction is enforced.

NOTE: A local user is one that successfully communicates with the MDM using the IP address 127.0.0.1 (the default IP address used by SCLI). If you try to issue commands from the local machine using other local IP addresses, you will not be able configure the system.

Syntax

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state {enabled |
disabled}
```

Parameters

--remote_read_only_limit_state {enabled | disabled}

Enable or disable restricted state. Default: disabled.

Example

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state enabled
```

set_syslog_facility

Set the facility field of the syslog events. Legal values are 0 to 23.

Syntax

```
scli --set_syslog_facility --remote_syslog_server_ip <IP> --syslog_facility <FACILITY>
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

Example

```
scli --set_syslog_facility --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

start_remote_syslog

Start posting events to a remote syslog server.

Syntax

```
scli --start_remote_syslog --remote_syslog_server_ip <IP>
  [--remote_syslog_server_port <PORT>]
  [--syslog_facility <FACILITY>]
  [--attach_event_code]
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--remote_syslog_server_port *<PORT>*

Syslog server port. Default is 1468.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

--attach_event_code

Add the posted event code to the event message. This parameter is disabled by default.

Example

```
scli --start_remote_syslog --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

stop_remote_syslog

Stop posting events to a remote syslog server.

Syntax

```
scli --stop_remote_syslog (--remote_syslog_server_ip <IP> | --all_remote_syslog_servers)
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--all_remote_syslog_servers

Stop posting events to all remote syslog servers.

Example

```
scli --stop_remote_syslog --remote_syslog_server_ip 192.168.1.201
```

test_remote_system_connectivity

Perform a connectivity test between the specified SDTs of the protection domain and remote system.

Syntax

```
scli --test_remote_system_connectivity (--remote_system_id <ID> | --remote_system_name
<NAME>) (--protection_domain_id <ID> | --protection_domain_name <NAME>) [--
destination_volume_nsid <NSID>] [--full_output]
```

Parameters

--remote_system_id *<ID>*
: The ID of the remote system

--remote_system_name *<NAME>*
: The name of the remote system

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--destination_volume_nsid *<NSID>*
: The destination volume NSID. If not specified, only an indication of the success of discovery information retrieval is returned.

--full_output
: Print a list of IO connections to the remote system, If not specified, only the number of healthy connections is returned.

add_network_set

Add a network set to the PowerFlex system.

Syntax

```
scli --add_network_set [--network_set_name <NAME>] [--network_id <ID> | --network_name
<NAME>]
```

Parameters

--network_set_name *<NAME>*
: A name to be assigned to the network set

--network_id *<ID>*
: A comma separated list of system network IDs to be associated with the network set

--network_name *<NAME>*

A comma separated list of system network names to be associated with the network set

modify_network_set

Modify the list of system networks of an existing network set.

Syntax

```
scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [--
network_id <ID> | --network_name <NAME>]
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

--network_id *<ID>*
A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*
A comma separated list of system network names to be replaced with the old one

query_network_set

Retrieve information about a network set.

Syntax

```
scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

query_network_set

Retrieve information about a network set.

Syntax

```
scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

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PDS commands

This section contains commands for managing PDS.

add_pds

Add a PDS to the specified storage node.

Syntax

```
scli --add_pds (--storage_node_id <ID> | --storage_node_name <NAME> | --storage_node_ip
<IP> [--storage_node_port <PORT>]) [--pds_name <NAME>] [--force_clean] [--profile
<PROFILE>] [--i_am_sure]
```

Parameters

--storage_node_name <NAME>

Storage node name

--storage_node_id <ID>

Storage node ID

--storage_node_ip <IP>

Storage Node IP address

--storage_node_port <PORT>

Port assigned to the DGWT on the storage node

--pds_name <NAME>

A name to be assigned to the PDS

--force_clean

Clean a previous PDS configuration

--profile <PROFILE>

Set the performance profile from the following options:

- compact
- high_performance

The default is high_performance

--i_am_sure

Preemptive approval

query_pds

Retrieve information about a PDS.

Syntax

```
scli --query_pds (--pds_id <ID> | --pds_name <NAME> | --storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])
```

Parameters

--pds_id <ID>

PDS ID

--pds_name <NAME>

PDS name

--storage_node_name <NAME>

Storage vode name

--storage_node_id <ID>

Storage node ID

--storage_node_ip <IP>

Storage node IP address

--storage_node_port <PORT>

Port assigned to the DGWT on the storage node

query_all_pds

Retrieve information about all PDSes.

remove_pds

Initiate a removal of a PDS.

Syntax

```
scli --remove_pds (--pds_id <ID> | --pds_name <NAME> | --storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])
```

Parameters

--pds_id <ID>

PDS ID

--pds_name <NAME>

PDS name

--storage_node_name <NAME>

Storage node name

--storage_node_id <ID>

Storage node ID

--storage_node_ip <IP>

Storage Node IP address

--storage_node_port <PORT>

Port assigned to the DGWT on the storage node

--ShutDown

Indicates if PDS shall shutdown gracefully after the clean up is done. This is an optional parameter.

Example

```
scli --remove_pds --pds_id <pds_id> --shut_down
```

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Protection domain commands

This section contains commands for managing protection domains.

activate_protection_domain

Activate a protection domain.

Protection domains are activated by default. Use this command to activate a protection domain that was inactivated.

Syntax

```
scli --activate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_activate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--force_activate

Activates the protection domain, ignoring current system errors

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --activate_protection_domain --protection_domain_name pd1
```

add_protection_domain

Add a protection domain.

Syntax

```
scli --add_protection_domain [--protection_domain_name <NAME>] [--core_count_per_node
<COUNT>] [--socket_count_per_node <COUNT>]
```

Parameters

--protection_domain_name *<NAME>*

Unique name to assign the protection domain

--core_count_per_node <COUNT>

Number of cores per storage node. Valid range: 32-256. Default: 64

--socket_count_per_node <COUNT>

Number of sockets per storage node. Valid range: 1-8. Default: 2

Example

```
Example
scli --mdm_ip 192.168.1.200 --add_protection_domain
--protection_domain_name rack_1.1
```

Protection domain names

Assign each protection domain a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the ID of the volume.

Each protection domain name should conform to the following rules:

1. Contains fewer than 32 characters.
2. Contains only alphanumeric and punctuation characters.
3. Is unique within the object type.

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

inactivate_protection_domain

Inactivate the specified protection domain. Using this command is an effective way to shut down nodes and is preferable to shutting them down manually.

NOTE: When you inactivate a protection domain, the data remains on the storage nodes. It is preferable to remove a protection domain if you no longer need it.

When this feature is in effect, the system may perform the following activities in the background:

- Determine whether there are any current rebuild/rebalance activities taking place. If so, the shutdown is delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Pause application I/O and disable access to volumes
- Move the DRL mode of all storage nodes to harden in preparation for rebooting the server.
- Reload all storage nodes before re-enabling data access.

Syntax

```
scli --inactivate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_inactivate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--force_inactivate

Inactivate the protection domain, ignoring current system errors.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --inactivate_protection_domain --protection_domain_name pd1
```

query_port_flapping_status

Query the system port flapping status.

Syntax

```
scli --query_port_flapping_status [--weak] (--all | --protection_domain_id <ID> | --
protection_domain_name <NAME> | --sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID>
| --sdc_ip <IP> | --apply_to_mdm | --peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID> | --
sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>] | --storage_node_id
<ID> | --storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>]
| --pds_id <ID> | --pds_name <NAME>)
```

Parameters

--weak

Include reported paths with weak ERT violations.

--all

Query all protection domains in the system.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--apply_to_mdm
: Display MDM only

--sdt_id *<ID>*
: SDT ID

--sdt_name *<NAME>*
: SDT name

--sdt_ip *<IP>*
: SDT IP address

--sdt_port *<PORT>*
: Port assigned to the SDT

--pds_id *<ID>*
: PDS ID

--pds_name *<NAME>*
: PDS name

--storage_node_name *<NAME>*
: Storage node name

--storage_node_id *<ID>*
: Storage node ID

--storage_node_ip *<IP>*
: Storage node IP address

--storage_node_port *<PORT>*
: Port assigned to the storage node

--apply_to_mdm
: Display MDM only

query_port_flapping_unsupported_components

Query components that do not support the port flapping feature, and/or have it disabled.

Syntax

```
scli --query_port_flapping_unsupported_components [(--full | --protection_domain_id <ID>
| --protection_domain_name <NAME>) [--raw]] [(--only_disabled | --only_unsupported)]
```

Parameters

--full
: Display a full unsummarized report

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--raw
: Outputs results in raw format with one component displayed per line

--only_disabled
: Query only the the components that have the port flapping feature disabled

--only_unsupported

Query only the the components that do not support the port flapping feature. Omit components that have the feature disabled.

Example

```
scli --mdm_ip 192.168.1.200 --query_port_flapping_unsupported_components --full
```

query_protection_domain

Retrieve aggregated information about all objects in a single protection domain.

Syntax

```
scli --query_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --query_protection_domain --protection_domain_name samplePD
```

Sample output

```
Protection Domain samplePD , (Id: 90dbe59a00000001), Type: EC, has 2 Device Groups, 1
storage pools, 5 Storage Nodes, 5 PDSs, 5 DGWTs, and 1 volumes
        Remaining volume logical provisioning: 4.6 TB (4735 GB), total volume logical
provisioning limit: 5.6 TB (5760 GB)
        Operational state is Active

Maximum IO Limits:
        Overall concurrent IO limit                              : 4 per device.
        Bandwidth limit overall IOs                              : 400 MB per device.
        Bandwidth limit background device scanner                : 10 MB per device.
        Bandwidth limit degraded                                 : 400 MB per device.
        Bandwidth limit critically degraded                      : 400 MB per device.
        Bandwidth limit rebalance                                : 40 MB per device.
        Bandwidth limit of all other secondary IOs               : 10 MB per device.
        Bandwidth limit of Node Network                          : 25 Gb per Node.
Secondary I/O policy: favor_application
Rebuild Enabled: TRUE
Rebalance Enabled: TRUE

Storage Pool sampleSP (Id: 9520a65800000000) has 1 volumes, 0 clones, 0 snapshots.
        Logical Provisioning:
                1500% Overprovisioning Factor
                5.6 TB (5760 GB) total volume logical provisioning limit
                4.6 TB (4735 GB) remaining volume logical provisioning
                1.0 TB (1025 GB) volume logical provisioned (17.8% of total)
```

```
        Data layout: Erasure Coding
        Compression method: Normal
        Device Group id:805e6ea900000000
        WrC Device Group id:805e6eaa00000001
        Data protection scheme: 2+2
        SP health state: PROTECTED
        Capacity alert thresholds: High: 80, Critical: 90
        Physical capacity:
                384.0 GB (393216 MB) total capacity
                        0 Bytes used capacity
                        381.5 GB (390656 MB) free capacity
                        2.5 GB (2560 MB) system capacity


Device Group 805e6ea900000000 Name: DeviceGroupGen2
        Media Type: SSD
        Protection Domain id:90dbe59a00000001
        Raw Capacity:
                Total                                   : 15.0 TB (15360 GB)
                Used                                    : 768.0 GB (786432 MB)
                Free                                    : 12.2 TB (12523 GB)
                Rebuild used                            : 0 Bytes
                Spare                                   : 2.0 TB (2068 GB)
                Spare Node Count                        : 0
                Spare Device Count                      : 2
        Capacity Health State: Protected
        Node Quantity: 5
        Device Quantity: 15
        Device HW Checks Enabled: TRUE
        Physical Capacity:
                Total                                   : 384.0 GB (393216 MB)
                Used                                    : 0 Bytes
                Free                                    : 381.5 GB (390656 MB)
                System                                  : 2.5 GB (2560 MB)
        Rebuild and Rebalance:
                Rebuild Required:                       : FALSE
                Rebuild Is Running:                     : FALSE
                Rebalance Is Running:                   : FALSE
        Raw Capacity Health:
                Raw health failed                       : 0 Bytes
                Raw health degraded                     : 0 Bytes
                Raw health degraded critical            : 0 Bytes
        Bandwidth:
                Backward-rebuild-reads                  : 0 IOPS 0 Bytes per second
                Backward-rebuild-writes                 : 0 IOPS 0 Bytes per second
                Forward-rebuild-reads                   : 0 IOPS 0 Bytes per second
                Forward-rebuild-writes                  : 0 IOPS 0 Bytes per second
                Rebalance-reads                         : 0 IOPS 0 Bytes per second
                Rebalance-writes                        : 0 IOPS 0 Bytes per second

Device Group 805e6eaa00000001 Name: PmemDeviceGroupGen2
        Media Type: PMEM
        Protection Domain id:90dbe59a00000001
        Raw Capacity:
                Total                                   : 320.0 GB (327680 MB)
                Used                                    : 10.3 GB (10560 MB)
                Free                                    : 309.6 GB (317040 MB)
                Rebuild used                            : 0 Bytes
                Spare                                   : 0 Bytes
                Spare Node Count                        : 0
                Spare Device Count                      : 0
        Capacity Health State: Protected
        Node Quantity: 5
        Device Quantity: 5
        Device HW Checks Enabled: TRUE
        Physical Capacity:
                Total                                   : 3.4 GB (3520 MB)
                Used                                    : 3.4 GB (3520 MB)
                Free                                    : 0 Bytes
                System                                  : 0 Bytes
        Rebuild and Rebalance:
                Rebuild Required:                       : FALSE
                Rebuild Is Running:                     : FALSE
```

```
                Rebalance Is Running:                    : FALSE
        Raw Capacity Health:
                Raw health failed                        : 0 Bytes
                Raw health degraded                      : 0 Bytes
                Raw health degraded critical             : 0 Bytes
        Bandwidth:
                Backward-rebuild-reads                   : 0 IOPS 0 Bytes per second
                Backward-rebuild-writes                  : 0 IOPS 0 Bytes per second
                Forward-rebuild-reads                    : 0 IOPS 0 Bytes per second
                Forward-rebuild-writes                   : 0 IOPS 0 Bytes per second
                Rebalance-reads                          : 0 IOPS 0 Bytes per second
                Rebalance-writes                         : 0 IOPS 0 Bytes per second


Storage Node Summary:
        Total 5 Storage Node Nodes

PDS Summary:
        Total 5 PDS Nodes
        5 PDS nodes have membership state 'Joined'
        5 PDS nodes have connection state 'Connected'

DGWT Summary:
        Total 5 DGWT Nodes
        5 DGWT nodes have membership state 'Joined'
        5 DGWT nodes have connection state 'Connected'

Raw capacity:
        15.0 TB (15360 GB) total (raw)
        2.0 TB (2068 GB) spare (raw)
Physical capacity:
        384.0 GB (393216 MB) total
        0 Bytes used
        381.5 GB (390656 MB) free

Bandwidth:
        total device Pmem read bandwidth         0 IOPS 0 Bytes per second
        total device Pmem write bandwidth        0 IOPS 0 Bytes per second
        total WrC read bandwidth                 0 IOPS 0 Bytes per second
        total WrC write bandwidth                0 IOPS 0 Bytes per second
        total device read bandwidth              0 IOPS 0 Bytes per second
        total device write bandwidth             0 IOPS 0 Bytes per second
        host read bandwidth                      0 IOPS 0 Bytes per second
        host write bandwidth                     0 IOPS 0 Bytes per second
        host trim bandwidth                      0 IOPS 0 Bytes per second
        storage fe read bandwidth                0 IOPS 0 Bytes per second
        storage fe write bandwidth               0 IOPS 0 Bytes per second
        storage fe trim bandwidth                0 IOPS 0 Bytes per second

Volumes summary:
        1 volume. Total logical used size: 0 Bytes
        1.0 TB (1025 GB) all volumes size

SDT Summary:
        Total 0 SDT Nodes
```

remove_protection_domain

Remove a protection domain from the PowerFlex system. You can only remove a protection domain if no SDS and no storage pool are associated with it.

Syntax

```
scli --remove_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --remove_protection_domain
--protection_domain_name my_protection_domain
```

rename_protection_domain

Assign a name to, or rename, a protection domain.

Syntax

```
scli --rename_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--new_name *<NAME>*

Example

```
scli --mdm_ip 192.168.1.200 --rename_protection_domain --protection_domain_name pd1 --
new_name protection_domain_1.
```

Protection domain names

Assign the protection domain a meaningful name associated with its operational role. The protection domain name should conform to the following rules:

1. Contains fewer than 32 characters.
2. Contains only alphanumeric and punctuation characters.
3. Is unique within the object type.

set_rebalance_mode

Enable or disable rebalancing of the specified protection domain. Storage pool is required only for mirroring protection domains.

CAUTION: Rebalancing is an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. Disabling rebalance may cause the system to become unbalanced even if no capacity is added or removed—for example, during recovery from a DGWT or device failure.

Syntax

```
scli --set_rebalance_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebalance | --
disable_rebalance)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name. This parameter is required only for mirroring protection domains.

--storage_pool_id *<ID>*

Storage pool ID. This parameter is required only for mirroring protection domains.

--enable_rebalance

Enable rebalancing of the specified protection domain.

--disable_rebalance

Disable rebalancing of the specified protection domain.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_rebalance_mode --protection_domain_name pd10 --disable_rebalance
```

set_rebuild_mode

Enable or disable rebuilds in the specified protection domain. Storage pool is required only for mirroring protection domains.

CAUTION: Rebuilds are an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. If rebuilds are disabled, redundancy will not be restored after failures.

Syntax

```
scli --set_rebuild_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebuild | --
```

```
disable_rebuild)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name. This parameter is required only for mirroring protection domains.

--storage_pool_id *<ID>*

Storage pool ID. This parameter is required only for mirroring protection domains.

--enable_rebuild

Enable rebuilds in the specified protection domain.

--disable_rebuild

Disable rebuilds in the specified protection domain.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

- `scli --set_rebuild_mode --protection_domain_name pd10 --disable_rebuild`
- `scli --set_rebuild_mode --protection_domain_name pd10 --enable_rebuild`

25

Return messages

This section describes the contents and the types of return messages in the PowerFlex CLI.

Each return message contains the following:

- Name of the return message (return code)
- Message presented with the return code
- Commands that may return the relevant code
- Action or actions that may resolve the issue (if relevant)

Storage node device name changed

Parameters

Name

SDS_DEVICE_NAME_CHANGED

Message

The storage node detected that the device name has changed. Re-run the command using the device ID.

message

The storage node detected that the device name has changed. Re-run the command using the device ID.

Commands

`remove_sds_device`, `abort_remove_sds_device`

Description

In some situations (usually following a reboot), the operating system changes the device name. The storage node detects this change. `Remove_sds_device` is a time consuming operation. To ensure that you have chosen the correct device, PowerFlex alerts you to the name change.

action

Identify the device in the output of `query_sds` and use the device ID to run the command again. You can also resynchronize the device name.

Success

Parameters

Name

SUCCESS

Message

Success

commands

All

Description

Command was executed successfully

Action

None

No resources

Parameters

Name

NO_RESOURCES

Message

Insufficient resources

Commands

`add_volume`

Description

There is insufficient space to create a volume of the requested size.

Action

Add capacity to the system by adding another storage node/device, or create a smaller volume

Timeout

Parameters

Name

TIMEOUT

Message

A timeout occurred

Commands

All

Description

No reply received from the MDM within the specified timeout period

Action

Validate the requested operational state using queries

Already exists

Parameters

Name

ALREADY_EXISTS

Message

Already Exists

Commands

All object addition commands

Description

The added object already exists

Action

Check command parameters, and use queries to view existing objects' properties

Communication error

Parameters

Name
: COMMUNICATION_ERROR

Message
: Communication error

Commands
: Storage node related commands

Description
: Storage node has communication errors

Action
: Use `query_all` to determine which storage node is disconnected

Wrong reconfiguration mode

Parameters

Name
: WRONG_RECONF_MODE

Message
: Storage node being configured, retry command

Commands
: All storage node related commands

Description
: Storage node is being configured by the MDM

Action
: Retry the command

Storage node add device “open” error

Parameters

Name
: TGT_ADD_DEV_OPEN_ERROR

Message
: Could not open storage node device

Commands
: `add_sds`, `add_sds_device`

Description
: Storage node cannot open one of the specified devices.

Action
: Validate the command parameters. If correct, validate that the device can be accessed on the storage node.

Storage node add device I/O error

Parameters

Name

TGT_ADD_DEV_IO_ERROR

Message

Add storage node device I/O error

Commands

`add_sds`, `add_sds_device`

Description

The storage node was unable to write to one of the specified devices

Action

Validate the command parameters, and check the specified devices

Storage node add device size problem

Parameters

Name

TGT_ADD_DEV_SIZE_PROBLEM

Message

Storage node device size error

Commands

`add_sds`, `add_sds_device`

Description

The device size is smaller or larger than the allowed size.

Action

Validate the command parameters, and set the device size appropriately.

Storage node actively attach to this MDM

Parameters

Name

TGT_ACTIVE_ATTACH_TO_THIS_MDM

Message

Storage node is actively attached to this MDM

Commands

`add_sds`

Description

Storage node was already configured by this MDM. This usually implies that the provided IP address resolves to an already configured Storage node that is currently connected. “Actively” implies that the storage node is configured and communicating with the MDM.

Action

Check the provided IP address. Contact Dell Technologies Support.

Storage node already attached to this MDM

Parameters

Name

TGT_ALREADY_ATTACH_TO_THIS_MDM

Message

Storage node is already attached to this MDM

Commands

`add_sds`

Description

Storage node was previously configured by this MDM. This usually indicates that the storage node was previously removed using `remove_sds`, but was not cleaned up properly.

Action

Validate the storage node IP address and port, and check the MDM configuration to validate that the storage node is not already configured under a different IP address. If the storage node is not found in the MDM configuration, this might indicate that it was not cleaned up properly when it was removed. To add the storage node forcefully, thus overriding its configuration and data, use the `force_clean` flag.

Storage node active attach to other MDM

Parameters

Name

TGT_ACTIVE_ATTACH_TO_OTHER_MDM

Message

Storage node is actively attached to another MDM

Commands

`add_sds`

Description

Storage node is already managed by another MDM. "Actively" implies that the storage node is configured and communicating with the MDM.

Action

Check the command parameters to validate the storage node IP address and port. Check the configuration of any other MDMs installed that might be connected to this storage node. Contact Dell Technologies Support.

Storage node already attached to other MDM

Parameters

Name

TGT_ALREADY_ATTACH_TO_OTHER_MDM

Message

Storage node is already attached to another MDM

Commands

`add_sds`

Description

Storage node was previously configured by another MDM

Action

Check your command parameters to validate the storage node IP address and port. Check for other PowerFlex installations of MDMs. If the SDS is not found in any MDM configuration, this might indicate that it was not cleaned up properly when it was removed. To add the storage node, use the `force_clean` flag.

NOTE: This command erases the existing storage node configuration and data.

Storage node not found

Parameters

Name

TGT_NOT_FOUND

Message

Could not find storage node

Commands

All storage node related commands

Description

The storage node was not found

Action

Check the provided parameters. Use `query_all_sds` to view the configured storage nodes

Volume not found

Parameters

Name

VOL_NOT_FOUND

Message

Could not find the volume

Commands

All volume related commands

Description

The volume was not found

Action

Check the provided parameters. Use `query_all_volumes` to view the configured volumes.

SDC not found

Parameters

Name

INI_NOT_FOUND

Message

Could not find the SDC

Commands

All SDC related commands

Description

SDC was not found

Action

Use `query_all_sdc` to view the configured SDCs

Volume already mapped to SDC

Parameters

Name

VOL_ALREADY_MAPPED_TO_THIS_INI

Message

The volume is already mapped to this SDC

Commands

`map_volume_to_sdc` (when mapping to a single SDC)

Description

The specified volume is already mapped to the specified SDC

Action

Check the command parameters, and use `query_all_volumes` to validate the current mapping

Volume already mapped to all SDCs

Parameters

Name

VOL_ALREADY_MAPPED_TO_ALL_INIS

Message

The volume is already mapped to all SDCs

Commands

`map_volume_to_sdc` (when mapping to all SDCs)

Description

The specified volume is already mapped to all SDCs

Action

Check the command parameters, and use `query_all_volumes` to validate the current mapping

Volume not mapped to SDC

Parameters

Name

VOL_NOT_MAPPED_TO_INI

Message

The volume is not mapped to SDC

Commands

`unmap_volume_from_sdc,query_sdc_volume_limits,set_sdc_volume_limits`

Description

The specified volume is not mapped to the specified SDC

Action

Check the command parameters, and use `query_all_sdc` to validate the current mapping

Volume mapped to all SDCs

Parameters

Name

VOL_MAPPED_TO_ALL_INIS

Message

The volume is mapped to all SDCs

Commands

`map_volume_to_sdc, unmap_volume_from_sdc, set_sdc_volume_limits`

Description

The volume is mapped to all SDCs

Action

Check the command parameters, and use `query_all_volumes` to validate the current mapping

Too many mappings

Parameters

Name

TOO_MANY_MAPPINGS

Message

There are too many mappings

Commands

`map_volume_to_sdc`

Description

Mapping of an additional volume will exceed the system limit

Action

Validate if all existing volumes that are mapped are relevant. Contact Dell Technologies Support.

Invalid volume ID

Parameters

Name

INVALID_VOL_ID

Message

Invalid volume ID. Try again with a valid ID.

Commands

All volume-related commands.

Description

The specified volume ID is invalid.

Action

Check the specified volume ID. Use `query_all_volumes` to find the correct ID, or use the volume name.

Invalid storage node ID

Parameters

Name

INVALID_TGT_ID

Message

Invalid storage node ID. Try again with a valid ID.

Commands

All storage node commands.

Description

The specified storage node ID is invalid.

Action

Check the specified storage node ID. Use `query_all_sds` to find the correct ID, or use the storage node name or IP address.

Invalid SDC ID

Parameters

Name

INVALID_INI_ID

Message

Invalid SDC ID. Try again with a valid ID.

Commands

All SDC commands.

Description

The specified SDC ID is invalid.

Action

Check the specified SDC ID. Use `query_all_sdc` to find the correct ID, or use the SDC IP address.

Host not found

Parameters

Name

HOST_NOT_FOUND

Message

Could not find the host

Commands

Secret session token. All storage node commands

Description

The specified host name was not found

Action

Check the specified host name

Duplicate host name

Parameters

Name

DUP_HOST_NAME

Message

Duplicate host name

Commands

`add_sds`

Description

The list of host names includes two identical host names or IP addresses, or two host names are resolved to the same IP address

Action

Check the specified host name list

Address and port in use

Parameters

Name

TGT_ADDR_AND_PORT_IN_USE

Message

The storage node IP address and port already in use. Try with different ones.

Commands

`add_sds`

Description

The specified IP address and port are already in use by another defined storage node.

Action

Check the specified IP address and port.

Too many storage nodes in protection domain

Parameters

Name

TOO_MANY_TGTS_IN_FDOM

Message

There are too many storage nodes in a protection domain.

Commands

`add_sds`

Description

Adding the storage node exceeds the system limit of allowed storage nodes in a protection domain.

Action

Check the configuration.

Too many storage nodes

Parameters

Name

TOO_MANY_TGTS

Message

There are too many storage nodes

Commands

`add_sds`

Description

Adding the storage node exceeds the system limit of allowed storage nodes in the system

Action

Check configuration

Too many volumes in protection domain

Parameters

Name

TOO_MANY_VOLUMES_IN_FD

Message

Too many volumes in a protection domain.

Commands

`add_volume`

Description

Adding a volume exceeds the limit for the number of volumes allowed in a protection domain.

Action

Check the configuration.

Too many volumes

Parameters

Name

TOO_MANY_VOLS

Message

There are too many volumes

Commands

`add_volume`

Description

Adding a volume exceeds the limit for the number of volumes allowed in a system

Action

Check configuration

Storage node name in use

Parameters

Name

TGT_NAME_IN_USE

Message

Storage node name already in use. Use a different name.

Commands

`add_sds`, `rename_sds`

Description

The specified storage node name is already in use by another storage node.

Action

Use `query_all_sds` to validate existing names, and issue the command with another name.

Volume name in use

Parameters

Name

VOL_NAME_IN_USE

Message

Volume name already in use. Use a different name.

Commands

`add_volume`, `rename_volume`

Description

The specified volume name is already in use by another volume.

Action

Use `query_all_volumes` to validate existing names, and issue the command with another name.

Duplicate storage node device name

Parameters

Name

DUPLICATE_TGT_DEVICE_NAME

Message

Duplicate storage node device name

Commands

`add_sds`

Description

A device appears more than once in the `add_sds` command

Action

Validate command parameters

Duplicate SDC IP address

Parameters

Name

DUPLICATE_SDC_IP

Message

More than one SDC exists for the same IP address. Use name or ID instead, or remove the duplicate SDC.

Commands

`map_volume_to_sdc`, `unmap_volume_from_sdc`, `set_sdc_volume_limits` , `query_sdc_volume_limits`

Description

The specified SDC IP address is not unique. It is associated with more than one SDC.

Action

Use `query_all_sdc` to find the relevant ID. Use the SDC ID in the command instead of the SDC IP address.

Storage node not under removal

Parameters

Name

TGT_NOT_UNDER_REMOVAL

Message

Storage node is not being removed

Commands

`abort_remove_sds`

Description

The storage node is not being removed

Action

Check command parameters. If the parameters are correct, it is still possible that the storage node has already been removed.

Storage node being removed

Parameters

Name

TGT_BEING_REMOVED

Message

Storage node is being removed

Commands

Storage node related commands

Description

The storage node is currently being removed

Action

None

Volume mapped

Parameters

Name

VOL_MAPPED

Message

Volume is mapped

Commands

`remove_volume`, `remove_consistency_group`

Description

The volume (for `remove_volume`) or any volume in the group (`remove_consistency_group`) is mapped to an SDC

Action

Unmap before removing the volume or consistency group. Use `query_all_volumes` to check current mapping.

Use MDM IP address

Parameters

Name

USE_MDM_IP

Message

Use cluster MDM physical IP to access the MDM.

Commands

All cluster commands.

Description

The command reached the secondary MDM. Use the physical IP address to access the MDM.

Action

Add `--mdm_ip` to the commands.

Command only in cluster

Parameters

Name

CMD_ONLY_IN_CLUSTER

Message

Command can only be used in cluster mode.

Commands

`switch_mdm_ownership`

Description

This command can be issued only when in cluster mode.

Action

Use `query_cluster` to check the current configuration.

Command only in single mode

Parameters

Name

CMD_ONLY_IN_SINGLE

Message

Command can only be used in a single mode.

Commands

MDM cluster commands

Description

This command can be issued in a single mode only.

Action

Use `switch_mdm_ownership` to switch to single mode before issuing this command.

Command error when cluster degraded

Parameters

Name

CMD_ERROR_CLUSTER_DEGRADED

Message

Command cannot be used when a cluster is degraded

Commands

`switch_mdm_ownership`

Description

This command cannot be issued when the cluster is degraded

Action

Check the cluster state and hardware configuration to determine the reason for cluster degradation

Too many devices

Parameters

Name

TOO_MANY_DEVICES

Message

There are too many devices for storage node

Commands

`add_sds`, `add_sds_device`

Description

Too many devices specified (`add_sds`), or the storage node already has the maximum number of allowed devices

Action

None

Storage nodes device not found

Parameters

Name

TGT_DEVICE_NOT_FOUND

Message

Could not find the storage node device

Commands

Storage nodes device related commands

Description

Storage node device was not found

Action

Check the command parameters. Use `query_sds` to view the device details.

License too long

Parameters

Name

LICENSE_TOO_LONG

Message

The license key is too long. Check your key and enter it again.

Commands

`set_license`

Description

The specified license key is too long.

Action

Validate the specified license key with the one received from Dell.

License error

Parameters

Name

LICENSE_ERROR

Message

License key is corrupted.

Commands

`set_license`

Description

The specified license key is invalid or does not match this version. Contact Dell Technologies Support.

Action

Validate the specified license key with the one received from Dell.

Unknown license version

Parameters

Name

LICENSE_UNKNOWN_VERSION

Message

Unknown license version

Commands

`set_license`

Description

The specified license key is corrupted

Action

Validate the specified license key with the one received from Dell.

License not set

Parameters

Name

LICENSE_NOT_SET

LICENSE_NOT_SET

License has not been set

Commands

`query_license`, `add_sds`, `add_sds_device`, `set_data_copy_limit`

Description

License was not set. Product cannot be used.

Action

Set the license before continuing to work

Configuration exceeds new license

Parameters

Name

CONFIG_EXCEEDS_NEW_LICENSE

Message

The current system configuration exceeds the license entitlements.

Commands

`set_license`

Description

The specified license decreases the amount of licensed capacity. The current system capacity already exceeds the specified license.

Action

Contact Dell Technologies support to receive a license for a larger amount of capacity.

License capacity exceeded

Parameters

Name

CAPACITY_LICENSE_EXCEEDED

Message

The operation could not be completed. The license capacity has exceeded.

Commands

`add_sds`, `add_sds_device`

Description

Adding the storage node or device exceeds the current licensed capacity

Action

Contact Dell Technologies support to receive a license for a larger amount of capacity

License expired

Parameters

Name

LICENSE_TIME_EXCEEDED

Message

The license has expired

Commands

Most commands

Description

The system time-based license has expired.

Action

Contact Dell Technologies support to receive a new license.

Installation ID mismatch

Parameters

Name

INSTALL_ID_MISMATCH

Message

The license installation ID does not match the ID of this system

Commands

`set_license`

Description

The provided license key does not match the current installation.

Action

Contact Dell Technologies support

Customer ID mismatch

Parameters

Name

CUSTOMER_ID_MISMATCH

Message

License customer ID does not match this system.

Commands

`set_license`

Description

The provided license key does not match the current installation.

Action

Contact Dell Technologies support.

Unsupported license version

Parameters

Name

LICENSE_UNSUPPORTED_VERSION

Message

Unsupported license version.

Commands

`set_license`

Description

The provided license key does not match the current installation.

Action

Contact Dell Technologies support.

Volume decrease not supported

Parameters

Name

VOL_DECREASE_NOT_SUPPORTED

Message

Volume capacity can only be increased.

Commands

`modify_volume_capacity`

Description

The specified volume capacity is smaller than the existing volume capacity. Decreasing volume capacity is not supported.

Action

Use `query_all_volumes` to validate your current configuration.

Storage node device being removed

Parameters

Name

TGT_DEV_BEING_REMOVED

Message

The storage node device is being removed.

Commands

`remove_sds`, `remove_sds_device`, `activate_sds_device`, `start_sds_device_test`

Description

The specified storage node device is currently being removed. The command cannot be executed.

Action

Wait for the device to be removed

Cannot remove last storage node device

Parameters

Name

CANNOT_REMOVE_LAST_SDS_DEVICE

Message

Last remaining storage node device cannot be removed.

Commands

`remove_sds_device`

Description

The command attempts to remove the last remaining storage node device. This is an illegal operation.

Action

To remove the last remaining device, remove the storage node.

Device not under removal

Parameters

Name

DEV_NOT_UNDER_REMOVAL

Message

The storage node device is not being removed.

Commands

`abort_remove_sds`

Description

The specified device is not in the process of being removed.

Action

Check the command parameters.

Storage node already adding device

Parameters

Name

TGT_ALREADY_ADDING_DEV

Message

A device is being added to this storage node.

Commands

`add_sds_device`, `remove_sds`

Description

This operation cannot be performed while a device is being added to the storage node.

Action

Wait for the device to be added.

Invalid device ID

Parameters

Name

INVALID_DEV_ID

Message

The device ID is invalid. Use a valid device ID.

Commands

Storage node device-related commands.

Description

The specified storage node device ID is invalid.

Action

Use `query_sds` to determine the correct ID.

Too many protection domains

Parameters

Name

TOO_MANY_FDs.

Message

There are too many protection domains.

Commands

`add_protection_domain`

Description

Adding a protection domain exceeds the system limit of the allowed protection domains.

Action

None

Protection domain name in use

Parameters

Name

FD_NAME_IN_USE

Message

Protection domain name already in use. Try a different name.

Commands

`add_protection_domain`, `rename_protection_domain`

Description

The specified protection domain name is already in use by another protection domain.

Action

Use `query_all` to validate the existing names, and issue the command with another name.

Protection domain not found

Parameters

Name

FD_NOT_FOUND

Message

Could not find protection domain.

Commands

All protection domain-related commands.

Description

The specified protection domain name or ID do not match any existing protection domains.

Action

Check the command parameters. Use `query_all` to validate the existing protection domain names.

Protection domain has storage nodes

Parameters

Name

FD_HAS_TGTS

Message

Protection domain has storage nodes

Commands

`remove_protection_domain`

Description

The protection domain contains storage nodes and cannot be removed.

Action

Remove the storage nodes before attempting to remove the protection domain.

Not enough storage nodes in storage pool

Parameters

Name

NOT_ENOUGH_TGTS_IN_STORAGE_POOL

Message

Fewer than three devices from different storage nodes are defined in this storage pool. Add devices to the storage pool from additional storage nodes.

Commands

`add_volume`

Description

The storage pool must contain devices from at least three different storage nodes in order to create a volume.

Action

Add more devices to the storage pool before creating a volume.

Remove in progress

Parameters

Name

REMOVE_IN_PROGRESS

Message

Removal of the object is in progress. Please wait.

Commands

Multiple commands

Description

The operation cannot be performed, because the volume is being removed.

Action

Wait for the volume to be removed.

No network test results

Parameters

Name

NET_TEST_NOT_FOUND

Message

Could not find network test results

Commands

`query_sds_network_test_results`

Description

The specified storage node has no network test results.

Action

Check the command parameters to make sure that a `start_sds_network_test` was sent to the specified storage node. If this is the correct storage node, resend the `start_sds_network_test` command.

Network test in progress

Parameters

Name

NET_TEST_IN_PROGRESS

Message

Network test in progress.

Commands

`query_sds_network_test_results`

Description

The storage node network test has not been completed.

Action

Wait for the test to be completed. Use the same query periodically until results are displayed.

Storage node device already active

Parameters

Name

TGT_DEV_ALREADY_ACTIVE

Message

The storage node device is already active.

Commands

`activate_sds_device`

Description

At least one specified device is already active.

Action

Use `query_sds` to view the device status.

Storage node device wrong test mode

Parameters

Name

TGT_DEV_WRONG_TEST_MODE

Message

The storage node device is in incorrect test mode.

Commands

`activate_sds_device`

Description

At least one specified device is automatically activated when its test is completed.

Action

Use `query_sds` to view the device status.

Storage node device wrong test state

Parameters

Name

TGT_DEV_WRONG_TEST_STATE

Message

The storage node device test is not complete.

Commands

`activate_sds_device`

Description

At least one specified device test is not yet complete. This device cannot be activated.

Action

Use `query_sds` to view the devices' status.

Storage node device test in progress

Parameters

Name

TGT_DEV_TEST_IN_PROGRESS

Message

The storage node device test is in progress.

Commands

`start_sds_device_test`

Description

At least one specified device test is in-progress. A new test for this device cannot be started now.

Action

Use `query_sds` to view the devices' test status.

Volume allocation "busy" error

Parameters

Name

VOL_ALLOC_ERROR_BUSY

Message

The system is busy. Retry later.

Commands

`add_volume`

Description

The MDM is attempting to recover a connection with a storage node. You cannot create a volume now.

Action

Use `query_all` to view the system status. Retry the command when the situation has returned to normal.

Capacity highly unbalanced

Parameters

Name

CAPACITY_HIGHLY_UNBALANCED

Message

System capacity is unbalanced.

Commands

`add_volume`

Description

System capacity is unbalanced. This can be a temporary or permanent state.

- Temporary—Due to adding or removing capacity, or due to disconnections. In this case, the system automatically works in the background to correct the situation.
- Permanent—The capacity of the devices added to PowerFlex is highly unbalanced. This requires user intervention to correct the situation.

Action

If a rebalance is in progress, wait for it to finish and resubmit the command. If the situation persists, contact support.

Volume creation failed

Parameters

Name

VOL_CREATE_FAIL

Message

Could not create the volume.

Commands

`add_volume`

Description

The volume creation failed.

Action

Use `query_all` to validate the system state, and retry the command.

Invalid password

Parameters

Name

INVALID_PASSWORD

Message

Invalid password. Try again.

Commands

Most commands.

Description

The specified password is incorrect.

Action

Check that you are using the correct password in the command.

No permissions

Parameters

Name

NO_PERMISSIONS

Message

You do not have permission to perform this operation. Contact your system administrator for assistance.

Commands

Most commands.

Description

The specified user does not have permission to issue this command.

Action

Contact the system administrator.

Too many storage pools in protection domain

Parameters

Name

TOO_MANY_STORAGE_POOLS_IN_PROTECTION_DOMAIN

Message

There are too many storage pools in protection domain.

Commands

`add_storage_pool`

Description

Addition of another storage pool exceeds the system limit of allowed storage pools in a protection domain.

Action

Use `query_protection_domain` to view the storage pools in the protection domain.

Storage pool already exists

Parameters

Name

STORAGE_POOL_ALREADY_EXISTS

Message

Storage pool exists.

Commands

`add_storage_pool`

Description

A storage pool with the specified name exists in the specified protection domain.

Action

Use `query_protection_domain` to view the storage pools in the protection domain, and choose a different name.

Storage pool not found

Parameters

Name

STORAGE_POOL_NOT_FOUND

Message

Could not find a storage pool.

Commands

All storage pool–related commands.

Description

The specified storage pool was not found.

Action

Use `query_protection_domain` to view the storage pools in the protection domain, and check the issued parameters.

Storage pool has devices

Parameters

Name

STORAGE_POOL_HAS_DEVS

Message

The storage pool has storage node devices.

Commands

`remove_storage_pool`

Description

The specified storage pool has storage node devices that are associated with it.

Action

Remove the storage node devices before removing the storage pool.

Storage pool name already exists

Parameters

Name

STORAGE_POOL_NAME_ALREADY_EXISTS

Message

Storage pool name exists.

Commands

`rename_storage_pool`

Description

A storage pool with the specified new name exists in the protection domain.

Action

Use `query_protection_domain` to view the storage pools in the protection domain. Choose another name for the storage pool.

Invalid storage pool ID

Parameters

Name

INVALID_STORAGE_POOL_ID

Message

Invalid storage pool ID.

Commands

All storage pool-related commands.

Description

The specified storage pool ID is invalid.

Action

Check the specified storage pool ID. Use `query_protection_domain` to find the correct ID, or use the storage pool name.

Protection domain has storage pools

Parameters

Name

FD_HAS_STORAGE_POOLS

Message

Protection domain has storage pools.

Commands

`remove_protection_domain`

Description

The protection domain has more than one storage pool associated with it.

Action

Remove the storage pools before removing the protection domain.

SDC has mappings

Parameters

Name

INI_HAS_MAPPINGS

Message

SDC has mapped volume(s).

Commands

`remove_sdc`

Description

At least one volume is mapped to the specified SDC.

Action

Remove the mapping before removing the SDC.

Too many volumes in V-Tree

Parameters

Name

TOO_MANY_VOLS_IN_VTREE

Message

There are too many volumes in the V-Tree.

Commands

`snapshot_volume`

Description

Taking a snapshot of the specified volume exceeds the allowed system limit of snapshots per volume.

Action

Use `query_volume` to view the current snapshots. Consider deleting some snapshots.

Too many snapshots in V-Tree

Parameters

Name

TOO_MANY_SNAPS_IN_VTREE

Message

There are too many snapshots in the V-Tree.

Commands

`snapshot_volume`

Description

Taking a snapshot of the specified volume exceeds the allowed system limit of snapshots per volume.

Action

Contact Dell Technologies support.

No volume to delete

Parameters

Name

NOTHING_TO_DO

Message

No volume to delete.

Commands

`remove_volume`

Description

The combination of parameters resulted in an empty set of volumes to delete.

Action

Use `query_volume` to view the current state, and review the command parameters accordingly.

Too many devices in the storage pool

Parameters

Name

TOO_MANY_DEVICES_IN_STORAGE_POOL

Message

There are too many devices in storage pool.

Commands

`add_sds`, `add_sds_device`

Description

Adding a device to the specified storage pool exceeds the allowed system limit of storage node devices per storage pool.

Action

Use `query_sds` to find the association of devices to storage pools.

Too many IP addresses

Parameters

Name

TOO_MANY_IPS

Message

There are too many IP addresses for storage node.

Commands

`add_sds_ip`

Description

Addition of an IP address to the storage node exceeds the allowed system limit of IP addresses per storage node.

Action

Use `query_sds` to view the storage node IP addresses.

Cannot remove last storage node IP address

Parameters

Name

LAST_IP

Message

Storage node must have at least one IP address.

Commands

`remove_sds_ip`

Description

A storage node must have at least one IP address. Cannot remove the last remaining IP address.

Action

Use `query_sds` to view the storage node IP addresses.

Storage node IP address not found

Parameters

Name

SDS_IP_NOT_FOUND

Message

Storage node IP not found.

Commands

`remove_sds_ip`

Description

The specified storage node IP address was not found.

Action

Use `query_sds` to view the storage node IP addresses.

Volume size too large

Parameters

Name

VOL_SIZE_TOO_LARGE

Message

The volume size too large.

Commands

`add_volume`, `modify_volume_capacity`

Description

The specified volume size is too large.

Action

Use a smaller volume size.

Capacity too low for snapshot

Parameters

Name

CAPACITY_TOO_LOW_FOR_SNAP

Message

The capacity of the storage pool is too small to create new snapshots.

Commands

`snapshot_volume`

Description

Snapshots are disabled when the free capacity in the storage pool is very low. The threshold is defined using the `set_capacity_alerts_threshold` command. The default threshold is 10% below the predefined spare.

Action

Add storage capacity, or free up space by removing some snapshots.

IP address belongs to other storage nodes

Parameters

Name

IP_BELONGS_TO_OTHER_TGT

Message

The specified IP address belongs to another storage node. Use the correct IP address.

Commands

`add_sds`, `add_sds_ip`

Description

The specified IP address is already assigned to another storage node.

Action

Check command parameters.

Invalid port

Parameters

Name

INVALID_PORT

Message

The specified storage node port is not in the valid range. Try again with a valid port number.

Commands

`add_sds`, `add_sds_ip`

Description

The specified port must be greater than 1024.

Action

Check command parameters.

LIA package not installed

Parameters

Name

LIA_PACKAGE_NOT_INSTALLED

Message

The package is not installed.

Description

You are trying to uninstall the LIA package, but the package is not installed.

Action

Check why you are trying to uninstall a component that is not installed. Verify that you have the correct package name for the package that you want to uninstall.

LIA multiple packages installed

Parameters

Name

LIA_MULTIPLE_PACKAGES_INSTALLED

Message

More than one package is installed.

Description

More than one package of the specified component is installed.

Action

Uninstall the packages manually:

Linux:

1. Find the package name:

   ```
   rpm -qa | grep -i lia
   ```

   The LIA package name is displayed. For example:

   ```
   EMC-ScaleIO-lia-2.50-0.30.el6.x86_64
   ```

2. Uninstall the package:

   ```
   rpm -e <PACKAGE_NAME>
   ```

Windows:

- **Windows Control Panel > Add/Remove programs** > **EMC-ScaleIO-lia**

LIA command failed

Parameters

Name

LIA_COMMAND_FAILED

Message

The command failed. Try again.

Description

his is a general error message.

Action

Look for more information about the error in the LIA log:

Linux: `/opt/emc/scaleio/lia/logs`

Windows: `C:\Program Files\emc\scaleio\lia\logs`

Invalid password

Parameters

Name

INVALID_PASSWORD

Message

Invalid password. Try again.

Commands

Most commands.

Description

The specified password is incorrect.

Action

Check that you are using the correct password in the command.

LIA operation not permitted

Parameters

Name

LIA_OPERATION_NOT_PERMITTED

Message

The operation is not permitted by current configuration.

Description

LIA was configured to disallow the operation that you tried to perform.

Action

Contact your system administrator; change the LIA configuration in your system.

LIA file is too large

Parameters

Name

LIA_FILE_IS_TOO_LARGE

Message

The file is too large to send back to the user.

Description

The file returned by the `get_info` command is larger than 500 MB, and therefore cannot be sent back to you.

Action

Contact Dell Technologies support.

LIA wrong ID

Parameters

Name

LIA_WRONG_ID

Message

LIA installation ID verification failed.

Description

The LIA ID that was passed in the installation/uninstall package is different to the installation ID that LIA is currently using.

Action

Edit the LIA's configuration file with LIA token and restart the LIA. The configuration file is found at:

- Linux: `/opt/emc/scaleio/lia/cfg/cont.txt`
- Windows: `C:\Program Files\emc\scaleio\lia\cfg\conf.txt`

LIA disk does not exist

Parameters

Name

LIA_DISK_DOES_NOT_EXIST

Message

The disk does not exist in the system.

Description

The disk name that was passed with the `prepare_disk` command does not exist in the system.

Action

Check that you are using the correct the disk name.

Storage node IP address not found

Parameters

Name

TGT_IP_NOT_FOUND

Message

The storage node IP address cannot be found.

Commands

`modify_sds_ip`, `remove_sds_ip`

Description

You tried to remove or modify an storage node IP address that does not exist in the system.

Action

Verify that you are specifying the correct IP address in the command.

Bad threshold value

Parameters

Name

BAD_THRESHOLD_VALUE

Message

The threshold value is out of range. Enter a valid percentage value.

Commands

`set_capacity_alert_threshold`

Description

The legal range for the capacity alert threshold is a percentage value between 0 and 100. The value that is specified for critical alerts must be less than the value for high alerts.

Action

Specify a value between 0 and 100. Ensure that the value for critical alerts is less than the value for high alerts.

Storage pool has volumes

Parameters

Name

STORAGE_POOL_HAS_VOLS

Message

The storage pool has volumes. Remove all the volumes before removing the storage pool.

Commands

`remove_storage_pool remove_protection_domain`

Description

Only empty storage pools can be removed.

Action

Remove all volumes before removing the storage pool or protection domain.

Protection domain and storage pool mismatch

Parameters

Name

FD_AND_SP_MISMATCH

Message

The specified storage pool is not in the specified protection domain.

Commands

Any command which uses the combination of storage pool and protection domain.

Description

The operation could not be performed because the combination of storage pool and protection domain is incorrect and the storage pool could not be found.

Action

Use the correct combination of storage pool and protection domain in the command. Use the command `query_all` to list all the storage pools to find this information.

Storage node IP address already exists

Parameters

Name

TGT_IP_ALREADY_EXISTS

Message

This storage node IP address exists. Use a different IP address.

Commands

`add_sds_ip`

Description

The IP address that is used in the command is already in use in the system.

Action

Verify that you are using the correct IP address in the command, and that the address has not been assigned to any other storage nodes.

IP role already set

Parameters

Name

ROLE_ALREADY_SET

Message

The IP role for this storage node has already been configured.

Commands

`modify_sds_ip`

Description

The IP role for the specified storage node has already been configured.

Action

Verify that you are configuring the intended storage node, and that you are specifying the correct IP address.

Cluster ID mismatch

Parameters

Name

CLUSTER_ID_MISMATCH

Message

The system ID number that you provided does not match the actual system ID.

Commands

`query_properties`, `query_system`

Description

The system ID number has been modified, and does not match the ID specified in the command.

Action

No action required - this code is provided for information purposes for advanced users.

Device already exists

Parameters

Name

DEV_ALREADY_EXISTS

Message

A device with the given name already exists in the storage node.

Commands

`add_sds_device`

Description

You tried to add a device to the system, using a name that is already defined in the storage node.

Action

Verify that the specified device has not already been added to the system.

Volume size is illegal

Parameters

Name

VOL_SIZE_ILLEGAL

Message

The specified volume size is illegal. Volume size must be in 8 GB blocks.

Commands

`add_volume`, `modify_volume_size`

Description

You tried to define a volume size that is not a multiple of 8 GB. Volume size can only be defined in multiples of 8 GB.

Action

Specify volume size in blocks of 8 GB.

SDC must be disconnected

Parameters

Name

INI_MUST_BE_DISCONNECTED

Message

The SDC must be disconnected.

Commands

`remove_sdc`

Description

Only SDCs that are currently inactive (disconnected) can be removed.

Action

This command is intended to remove old, disconnected SDCs. Do not remove active SDCs.

Invalid percentage

Parameters

Name

INVALID_PERCENTAGE

Message

The percentage value that you provided is invalid. Use a value between 0—100.

Commands

`set_thick_volume_percentage`

Description

The value that you entered was less than 0 or greater than 100.

Action

Use a value between 1 and 100 in the command.

Wrong receive group

Parameters

Name

WRONG_RECEIVE_GROUP

Message

The command was sent to the wrong receive group due to component software mismatch.

Commands

All commands

Description

There are different software versions installed on some components in your system, and this is causing a software mismatch error.

Action

Verify that the same software version is installed on all components, and fix, if necessary.

Mismatched SW ID

Parameters

Name

MISMATCHING_SWID

Message

The license contains a mismatch of the SWID number. Contact Dell Technologies support.

Commands

`set_license`

Description

The SWID number that you entered does not match your license.

Action

Contact Dell Technologies support.

Invalid license issuer

Parameters

Name

INVALID_LICENSE_ISSUER

Message

The issuer of the license you are attempting to add does not match that of the product.

Commands

`set_license`

Description

There is a problem with your license.

Action

Contact Dell Technologies support.

Number of parallel messages is too low

Parameters

Name

NUM_PARALLEL_MSG_TOO_LOW

Message

The number of parallel messages is too low. Use a number in the range 1–6.

Commands

`start_sds_network_test`

Description

You used a number less than 1 in the command.

Action

Use a number in the range of 1 to 6 for parallel messages in the command.

Number of parallel messages is too high

Parameters

Name

NUM_PARALLEL_MSG_TOO_HIGH

Message

The number of parallel messages is too high. Use a number in the range 1- 6.

Commands

`start_sds_network_test`

Description

You used a number greater than six in the command.

Action

Use a number in the range of 1-6 for parallel messages in the command.

Network test size is too high

Parameters

Name

NETWORK_TEST_SIZE_TOO_HIGH

Message

The network test size is too high. Use a test size less than 10.

Commands

`start_sds_network_test`

Description

You used a number greater than 10 in the command.

Action

Use a number less than 10 for the test size in the command.

Network test length too long

Parameters

Name

NETWORK_TEST_LENGTH_TOO_HIGH

Message

The network test length is too long. Use a test length less than 43200 seconds.

Commands

`start_sds_network_test`

Description

You used a number greater than 43200 for the test length in the command.

Action

Use a number less than 43200 for test length in the command.

Network test must limit either time or size

Parameters

Name

NETWORK_TEST_MUST_LIMIT_EITHER_TIME_OR_SIZE

Message

The network test must limit either time or size.

Commands

`start_sds_network_test`

Description

Your command did not contain a value for time or size.

Action

Include a value for either time or size in the command.

Not enough storage nodes in protection domain for test

Parameters

Name

NOT_ENOUGH_TGTS_IN_FD_FOR_TEST

Message

To start the test, there must be at least two storage nodes in the protection domain.

Commands

`start_sds_network_test`

Description

There are fewer than two storage nodes in the protection domain, and therefore the test operation cannot be performed.

Action

Ensure that there are at least two storage nodes in the protection domain before attempting to test it.

Not allowed in storage pool with devices

Parameters

Name

NOT_ALLOWED_IN_SP_WITH_DEVS

Message

This operation is only allowed when there are no devices in the storage pool. Remove all devices from the storage pool.

Commands

`modify_zero_padding_policy`

Description

You can only modify the zero padding policy of empty storage pools.

Action

Ensure that all devices are removed from the storage pool before you attempt to modify its zero padding policy.

License capacity mismatch

Parameters

Name

LICENSE_CAPACITY_MISMATCH

Message

The license contains a mismatch of the capacity values for basic and advanced features. Contact Dell Technologies support.

Commands

`set_license`

Description

The capacity allowed for basic and advanced features in your license do not match your installation.

Action

Contact Dell Technologies support.

Exceeds system capacity limitations

Parameters

Name

EXCEED_SYSTEM_CAPACITY_LIMITATIONS

Message

The request exceeds system limitations.

Commands

`add_volume`, `modify_volume_capacity`

Description

The command specifies a volume larger in size than the maximum system capacity.

Action

Remove one or more volumes in the system, to make room for the new one.

SDC mapped to too many volumes

Parameters

Name

INI_MAPPED_TO_TOO_MANY_VOLS

Message

The SDC is mapped to too many volumes. Remove some mapping.

Commands

`map_volume_to_sdc`

Description

You attempted to map more than the maximum allowed number of volumes (8192) to one or more SDCs.

Action

Avoid mapping more than the maximum allowed volumes to a single SDC.

Not enough devices in storage pool for removal

Parameters

Name

NOT_ENOUGH_DEVICES_IN_STORAGE_POOL_FOR_REMOVAL

Message

Not enough devices in the storage pool.

Commands

`remove_sds`, `remove_sds_device`

Description

Each storage pool must contain at least two devices. If there are two or less devices in the storage pool and you try to remove one, the command fails.

Action

Add devices to the storage pool before trying to remove this one.

Invalid network limits

Parameters

Name

INVALID_TGT_NETWORK_LIMITS

Message

Bad limits given. The overall must be larger than both rebuild and rebalance and all limits must be over 5MB.

Commands

`sds_network_limits`

Description

The total (overall) limit must be greater than the sum of the rebuild limit and the rebalance limit. All limits must be greater than 5 MB.

Action

Ensure that limit values are greater than 5 MB, and that the overall limit value is greater than rebuild limit plus rebalance limit.

A device is being removed

Parameters

Name

ONE_TGT_DEV_BEING_REMOVED

Message

This command cannot be completed. A storage node device is being removed.

Commands

`activate_sds_device` with the parameter `activate_all_devices`.

Description

If a device is in the process of being removed, you cannot use the `activate_all_devices` parameter to activate the remaining ones.

Action

Wait until the device removal is complete, and try again.

A device is already active

Parameters

Name

ONE_TGT_DEV_ALREADY_ACTIVE

Message

This command cannot be completed; A storage node device is currently active.

Commands

`activate_sds_device` with the parameter `activate_all_devices`

Description

A device is already active.

Action

Do not use the `activate_all_devices` option in the command. Instead, specify each device individually in the command.

RAM cache wrong state

Parameters

Name

RMCACHE_WRONG_STATE

Message

Wrong RAM cache state.

Commands

`enable_sds_rmcache` , `disable_sds_rmcache`, `set_sds_rmcache_size`

Description

The cache is in transient state, so the requested operation cannot be executed.

Action

Wait for a few seconds and try again.

More than one mapping not allowed

Parameters

Name

NOT_ALLOWED_MORE_THAN_ONE_MAPPING

Message

Only a single SDC may be mapped to this volume at a time.

Commands

`map_volume_to_sdc`

Description

The volume is already mapped, and the `allow_multi_map` flag was not used in the command.

Action

The volume is already mapped. Use the `allow_multi_map` flag to map the volume to additional SDCs.

At least one storage node failed

Parameters

Name

AT_LEAST_ONE_TGT_FAILED

Message

At least one storage node failed.

Commands

`enable_sds_rmcache`, `disable_sds_rmcache`, `set_sds_rmcache_size`

Description

You tried to enable or disable, or change the size of cache for all storage nodes in the protection domain, and this failed for at least one of the storage nodes.

Action

Check the state of the cache in all the storage nodes, and try again for the specific storage node with the problem.

A device is in incorrect test state

Parameters

Name

ONE_TGT_DEV_WRONG_TEST_STATE

Message

This command cannot be completed; A storage node device is currently in incorrect test mode.

Commands

`activate_sds_device` with `activate_all_devices` parameter.

Description

A device was not added with the `test_only` option, so it cannot be activated with this command.

Action

It will activate itself on its own. No action required.

A device test is in progress

Parameters

Name

ONE_TGT_DEV_WRONG_TEST_MODE

Message

This command cannot be completed; A storage node device test is currently in progress.

Commands

`activate_sds_device` with `activate_all_devices` parameter.

Description

A device test is currently in progress, and the device cannot be activated until the test is finished.

Action

Wait for the device test to finish, and then try again.

Enterprise features not enabled

Parameters

Name

ENTERPRISE_FEATURES_NOT_ENABLED

Message

Enterprise features are not enabled for this system.

Commands

Any command that requires a special license.

Description

The command that you tried to use cannot be used with the type of license that you have activated in your system.

Action

Verify that you are using a license with enterprise features enabled. If problems persist, contact Dell Technologies support.

Volume already mapped to an SDC

Parameters

Name

VOL_ALREADY_MAPPED_TO_AN_INI

Message

The volume is already mapped to a SDC.

Commands

`map_volume_to_sdc` with the `all_sdcs` option.

Description

If the volume is already mapped to one or more SDCs, you cannot use the `all_sdcs` option.

Action

Remove the volume's mapping to the SDCs, and then try again.

Storage node already exists and is being removed

Parameters

Name

TGT_ALREADY_EXISTS_AND_BEING_REMOVED

Message

The storage node exists and is being removed. Wait for the operation to finish.

Commands

`add_sds`

Description

If you try to add a storage node that exists, and is in the process of being removed, this operation fails.

NOTE: If an `add_sds` command failed, the system tries to remove it automatically, without user intervention. Therefore, if you issue the same `add_sds` command twice in a row, and the first time failed, this return code may be generated (depending on why adding the storage node failed).

Action

Verify that you are adding the correct storage node. If you are adding a storage node which is in the process of being removed, wait for the removal process to finish before trying to add the storage node back to the system.

Invalid spare percentage

Parameters

Name

INVALID_SPARE_PERCENTAGE

Message

The spare percentage value that you provided is invalid. Use a value between 0 - 99.

Commands

`modify_spare_policy`

Description

The spare percentage that is required in the command is invalid.

Action

Verify that the correct input parameters (a value between 0-99) are used with the command.

Invalid background device scanner bandwidth limit

Parameters

Name

INVALID_SCANNER_BW_LIMIT

Message

The scanner bandwidth limit is invalid. Please use a value between 10KB - 1GB.

Commands

`enable_background_device_scanner`

Description

The bandwidth value that you provided for the bandwidth limit parameter is invalid.

Action

Verify that correct input parameters (a value between 10KB–1GB) are used with the command.

Zero padding with background device scanner data comparison mode

Parameters

Name

ZEROPAD_AND_SCANNER

Message

Zero padding must be enabled in order to set scanner to data comparison mode.

Commands

`enable_background_device_scanner`

Description

The background device scanner has two modes: device only, and data comparison mode. The storage pool must have zero padding enabled before you attempt to enable background device scanning in data comparison mode.

Action

Enable zero padding in the storage pool (you must do this before adding devices to the storage pool).

IP address already assigned to another storage node

Parameters

Name

ADDRESS_BELONGS_TO_DIFFERENT_TGT

Message

The given address belongs to a different storage node. Check the system configuration.

Commands

`add_sds`, `add_sds_ip`

Description

This message is returned if the specified IP address has already been assigned to another storage node in the system.

Action

Check storage node network configuration and IP address assignments.

Capacity too low for thin volumes

Parameters

Name

CAPACITY_TOO_LOW_FOR_THIN_VOL

Message

Storage pool reached critical capacity utilization. Unable to create new thin volumes.

Commands

`add_volume`, `modify_volume_capacity`

Description

Thin volumes cannot be allocated if the specified storage pool has reached the critical capacity level.

Action

Add more devices to the storage pool.

Too many storage pools in the system

Parameters

Name

TOO_MANY_STORAGE_POOLS

Message

There are too many storage pools in the system.

Commands

`add_storage_pool`

Description

Addition of another storage pool exceeds the system limit of allowed storage pools.

Action

Remove unused storage pools. If the problem is not solved, install a new system.

Remote access to the MDM is blocked

Parameters

Name

REMOTE_PERMISSION_DENIED

Message

Permission denied. Remote read-only limit state is enabled and is blocking this operation from a remote location. Run this command from the primary MDM host.

Commands

All

Description

Remote read-only limit state is enabled and is blocking this operation from a remote location.

Action

Run the command from the primary MDM host.

Cannot add a partitioned device

Parameters

Name

TGT_ADD_DEV_IS_PARTITIONED

Message

Unable to add a device that was already partitioned.

Commands

`add_sds`, `add_sds_device`

Description

The added device is partitioned and cannot be used by the system.

Action

Remove partitions from the device, or add the partitioned device to the system.

Cannot add a mounted device

Parameters

Name

TGT_ADD_DEV_IS_MOUNTED

Message

Unable to add a device that was already mounted.

Commands

`add_sds`, `add_sds_device`

Description

The added device is mounted and cannot be used by the system.

Action

Unmount the device.

Device header mismatch

Parameters

Name

INVALID_DEVICE_HEADER_SIGNATURE

Message

Found an invalid device header signature.

Commands

`add_sds`, `add_sds_device`

Description

The device was previously added to the system, or after storage node reboot, the signature of the header does not match the expected signature.

Action

Check the specific storage node devices configuration.

Changelog settings cannot be configured if checksum is stored on NVRAM

Parameters

Name

CHANGELOG_CANNOT_BE_CONFIGURED_IF_CHECKSUM_IS_STORED_ON_NVRAM_DEV

Message

Changelog can only be configured if checksum is not stored on NVRAM

Commands

`modify_checksum_per_device` , `enable_checksum`, `modify_checksum`

Action

If you want to modify changelog configurations, store checksums on a different device.

Checksum is already enabled on the storage pool

Parameters

Name

CHECKSUM_ALREADY_ENABLED_ON_SP

Message

The checksum feature is already enabled on the selected storage pool.

Commands

`enable_checksum`

Action

To change the feature settings, use the `modify_checksum` command.

Checksum cannot be relocated to NVRAM

Parameters

Name

CANNOT_RELOCATE_CHECKSUM_ON_NVRAM

Message

Checksum may not be relocated to NVRAM devices without first disabling the changelog (because there is no point in checksum and changelog both residing on NVRAM)

Commands

`modify_checksum_per_device`, `modify_checksum`

Action

Disable the changelog on NVRAM, and then try again.

Changelog bad pool type

Parameters

Name

CHANGELOG_BAD_POOL_TYPE

Message

The checksum changelog must reside on NVRAM devices.

Commands

`enable_checksum`, `modify_checksum`, `modify_checksum_per_device`

Changelog is not enabled in the Storage Pool

Parameters

Name

CHANGELOG_NOT_ENABLED_ON_SP

Message

The changelog feature is not enabled on the selected storage pool.

Commands

`modify_checksum`, `enable_checksum`

Action

Provide additional changelog fields to enable it on the storage pool.

Insufficient capacity for checksum on storage device

Parameters

Name

CHECKSUM_INSUFFICIENT_TOOTH_DEV_CAPACITY

Message

There is not enough free capacity on the selected device to store checksums.

Commands

`add_sds`, `add_sds_device`, `enable_checksum`, `modify_checksum`
`modify_checksum_per_device`

Invalid checksum calculation rate

Parameters

Name

INVALID_CHECKSUM_CALCULATION_RATE

Message

The calculation rate specified is not valid. Minimum: 10 KBs Maximum: 102400 KBs

Commands

`modify_checksum`, `enable_checksum`

Action

Choose a valid rate

Invalid changelog amortization

Parameters

Name

INVALID_CHANGELOG_AMORTIZATION

Message

The amortization specified is not valid. Minimum: 8 Maximum: 1000

Commands

`modify_checksum`, `enable_checksum`, `set_checksum_changelog_amortization`

Action

Choose a valid value.

Checksum mode is not persistent in the Storage Pool

Parameters

Name

CHECKSUM_MODE_IS_NOT_PERSISTENT_IN_SP

Message

The checksum mode of the storage pool is not set to one of the persistent options. To change the checksum mode, specify it explicitly in the command.

Commands

`modify_checksum`

Action

If you want the checksum to be persistent (recommended), modify the checksum mode accordingly.

Checksum is not configurable

Parameters

Name

CHECKSUM_NOT_CONFIGURABLE

Message

The checksum feature cannot be configured for the selected devices because they are still undergoing previous configuration changes.

Commands

`modify_checksum`, `modify_checksum_per_device`, `enable_checksum`, `disable_checksum`

Action

Wait until all configuration changes are complete, and then try again.

Checksum is not enabled on the device

Parameters

Name

CHECKSUM_NOT_ENABLED_ON_DEVICE

Message

The checksum feature is not enabled on the selected device.

Commands

`modify_checksum_per_device`

Action

Use the `enable_checksum` command instead, and include the required settings in the command.

Checksum is not enabled in the Storage Pool

Parameters

Name

CHECKSUM_NOT_ENABLED_ON_SP

Message

The checksum feature is not enabled on the selected storage pool.

Commands

`modify_checksum`

Action

Use the `enable_checksum` instead.

The checksum feature should be configured separately

Parameters

Name

CHECKSUM_SHOULD_BE_CONFIGURED_SEPARATELY

Message

The `enable_checksum` and `disable_checksum` flags are DEPRECATED. Please use the `enable_checksum` or `disable_checksum`command instead.

Commands

`add_storage_pool`

Action

Use the `enable_checksum` or `disable_checksum`command instead.

Device has user content

Parameters

Name

USER_CONTENT_DETECTED

Message

User content is present on the device.

Commands

`add_sds`, `add_sds_device`

Action

Clear the device from all content or partitions before adding the device.

Missing storage device media type

Parameters

Name

MISSING_MEDIA_TYPE

Message

The media type is not configured.

Commands

`add_storage_pool`, `add_sds`, `add_sds_device`, `set_storage_pool_media_type`

Action

Specify the media type of the storage devices when issuing the command. Storage pools must contain devices of the same media type.

Invalid media type

Parameters

Name

INVALID_MEDIA_TYPE

Message

The media type is invalid.

Commands

`add_storage_pool`, `add_sds`, `add_sds_device`, `set_storage_pool_media_type` , `set_sds_device_media_type`

Action

Issue the command with an appropriate media type.

Media type mismatch

Parameters

Name

MEDIA_TYPE_MISMATCH

Message

The device media type is not compatible with the storage pool media type.

Commands

`add_sds`, `add_sds_device`, `set_storage_pool_media_type`, `set_sds_device_media_type`

Action

Ensure that the storage device and the storage pool are the same media type. For example, storage pool defined for HDD can only contain HDD devices.

Metadata feature is already enabled on the storage pool

Parameters

Name

TGT_DEV_METADATA_ALREADY_ENABLED_ON_SP

Message

The storage nodes device metadata feature is already enabled on the selected storage pool. To change the feature settings, use the modify command.

Commands

`enable_metadata_acceration`

Action

None required

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NVMe host and SDC commands

This section contains commands for managing NVMe host and SDC.

Understanding the effect of network latency on NVMe over TCP latency counters

This section describes the possible effects of network latency between the host and storage data target (SDT) in an NVMe over TCP environment.

To access the performance counters, run the following commands:

- For storage data target

```
scli --query_properties --object_type SDT --all_objects --properties
SDT_DATA_READ_LATENCY,SDT_DATA_WRITE_LATENCY,SDT_HOST_READ_LATENCY,SDT_HOST_WRITE_LATE
NCY
```

  Output sample:

```
754135e800000000:
          SDT_DATA_READ_LATENCY             3595 IOPS 0.823 ms average latency
          SDT_DATA_WRITE_LATENCY          0 IOPS 0 us average latency
          SDT_HOST_READ_LATENCY             899 IOPS 82 microseconds average latency
          SDT_HOST_WRITE_LATENCY          899 IOPS 12 microseconds average latency
```

- For host:

```
scli --query_properties --object_type HOST --all_objects --properties
USER_DATA_SDC_READ_LATENCY,USER_DATA_SDC_WRITE_LATENCY
```

  Output sample:

```
f3e742f300010000:
          USER_DATA_SDC_READ_LATENCY        930 IOPS 69 microseconds average latency
          USER_DATA_SDC_WRITE_LATENCY       930 IOPS 12 microseconds average latency
```

Table 10. I/O latency measurement by counters

PowerFlex object	Performance counter	Measures I/O latency		
		Type	From	To
storage data target	SDT_DATA_**READ**_LATENCY	Read	Storage node	storage data target
	SDT_DATA_**WRITE**_LATENCY	Write	storage data target	Storage node
	SDT_HOST_**READ**_LATENCY	Read	storage data target	Host
	SDT_HOST_**WRITE**_LATENCY	Write	Host	storage data target
HOST	USER_DATA_SDC_**READ**_LATENCY	Read	Storage node	storage data target
	USER_DATA_SDC_**WRITE**_LATENCY	Write	storage data target	Storage node

NOTE: The term storage data client (SDC) used in the counter name is for backward compatibility purposes only.

Limitations of measuring latency between the host and SDT

In an NVMe over TCP setup, PowerFlex does not have a component that is installed on the host. As a result of the above fact there is no way of knowing when the host initiates the I/O request or when it receives requested data or a response about data it sent. Therefore, it is impossible to measure latency between the host and storage data target. The effect this has on the host-related counters are described below. Counters for traffic between storage node and storage data target (SDT) are reliable and reflect any network latency between those nodes.

Exception

Latency can be measured in one scenario. The following table describes how network latency affects the counters that measure latency between the host and the storage data target.

Table 11. Network latency effect on counters measuring latency between host and storage data target

I/O type	I/O size	I/O direction	Result
READ	All	Host to and from the storage data target	Values are estimates based on the I/O size. Network latency does not affect the values, and the values cannot provide any information about such issues.
WRITE	Less than or equal to 4 KB		
	More than 4 KB	Storage data target to the host	PowerFlex can accurately measure the network latency between the host and the storage data target. The network latency affects the counter value **twice**. That is, if network latency is increased by 100 microseconds then the counter value increases by 200 microseconds.
		Host to the storage data target	

Significance of large write requests

In all the host to storage data target flow cases, the host sends a read or write request to the storage data target.

The flow is complicated while writing large IOs in the host to the storage data target:

1. Host sends a write request to the storage data target.
2. Storage data target sends the host a request to send the write data.
3. Host sends the write data to the storage data target.

Since the system knows when it initiated step 2 and it knows when step 3 is completed it can measure the time of these steps. Any network latency affects both steps 2 and 3, and this is the reason that the latency affects twice as described in the table above.

Summary

To check for possible effects of network latency between the host and the storage data target in an NVMe over TCP setup, complete the following:

1. Run a write I/O load with I/O size larger than 4 KB and examine the **SDT_HOST_WRITE_LATENCY** counter.
2. If a baseline value is available to compare with, divide the delta of the two counter values by two for an estimate of the network latency.

add_sdc

Approve an SDC to map volumes and allow an SDC to access the MDM.

Syntax

```
scli --add_sdc --sdc_ip <ip> | --sdc_guid <guid>
  [--sdc_name <name>]
```

Parameters

--sdc_ip *<ip>*
SDC IP address

--sdc_guid *<guid>*
SDC GUID

--sdc_name *<name>*
Name to assign to the SDC

NOTE: You can find the GUID of the SDC using any of these methods:

- From the command line, run:

  ```
  scli --query_all_sdc
  ```

- View the properties of the SDC in PowerFlex Manager.
- Locally in `/opt/emc/scaleio/sdc/bin/drv_cfg`, with root access permission, run:

  ```
  drv_cfg --query_guid
  ```

Example

```
scli --add_sdc --sdc_ip 10.125.120.53 --sdc_name sdc53
```

remove_host

Remove a host from the system.

Syntax

```
scli --remove_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --remove_host --host_name host1
```

rename_host

Name or rename an existing host.

Syntax

```
scli --rename_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>) --new_name <NAME>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

---sdc_ip *<IP>*

SDC IP address

- --new_name *<NAME>*

New name assigned to the host.

Example

```
scli --rename_host --host_name host1 --new_name new_host_name
```

check_sdc_authentication_status

Print the SDC authentication password status for all SDCs.

Syntax

```
scli --check_sdc_authentication_status [--run_test] [--filename <FILENAME>]
```

Parameters

--run_test

Run a connectivity test to check whether the SDCs can successfully authenticate using CHAP.

--filename *<FILENAME>*

The filename and filepath where detailed SDC authentication information will be stored. This parameter is mandatory when using the `--run_test` option.

Example

```
scli --mdm_ip 192.168.1.200 --check_sdc_authentication_status --run_test --filename
report.txt
```

generate_sdc_password

Generate a new SDC authentication password for the specified SDC.

Syntax

```
scli --generate_sdc_password (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> |
--sdc_ip <IP>) [--reason <REASON>]
```

Parameters

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC GUID

--sdc_ip *<IP>*

SDC IP address

--reason *<REASON>*

Free text to describe the reason for the SDC password reset

Example

```
scli --mdm_ip 192.168.1.200 --generate_sdc_password --sdc_name SDC1 --reason
MonthlyUpdate
```

query_all_approved_sdc

Query the system when it is operating in restricted SDC mode for all SDCs that have been approved by the system.

NOTE: This query is useful when restricted SDC mode is enabled and you want to determine which SDCs can be used for volume mapping.

Syntax

```
scli  --query_all_approved_sdc
```

Parameters

None.

Example

```
scli --query_all_approved_sdc
```

query_host

Retrieve information about a host.

Syntax

```
scli --query_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe Host NQN

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

query_all_hosts

Retrieve information about all SDCs and NVMe hosts.

Syntax

```
scli --query_all_hosts
```

query_potential_nvme_hosts

Retrieve information about all potential NVMe hosts. The query result can be optionally filtered by protection domain.

Syntax

```
scli --query_potential_nvme_hosts (--all | --protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--all

Retrieve information about all potential hosts.

--protection_domain_id *<ID>*

Protection domain ID

-protection_domain_name *<NAME>*

Protection domain name

query_all_sdc

Retrieve information about all SDCs in the system.

Syntax

```
scli --query_all_sdc [--agent_status] [--mdm_ip_addrs_current] [--mdm_ip_addrs_hardened]
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sdc
```

Sample output

```
MDM restricted SDC mode: NONE
Query all SDC returned 5 SDC nodes.
SDC ID: 30b5c18600000000 Name: N/A IP: 10.55.137.119 State: Connected GUID:
96CCE42A-6ADB-45EA-8E3F-B71C1F67808C  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
```

```
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18700000001 Name: N/A IP: 10.55.137.114 State: Connected GUID:
C08C87B5-7E28-421A-A82C-C9C5B3070FF2  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18800000002 Name: N/A IP: 10.55.137.115 State: Connected GUID:
89947FF3-6E84-4DF9-B869-481BCB8E3B3D  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18900000003 Name: N/A IP: 10.55.137.116 State: Connected GUID: 7F5BC40E-
A1FB-44EC-A8B0-76C63A7B1123  OS Type: LINUX Loaded Version: 3.6.0 Installed Version:
3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18a00000004 Name: N/A IP: 10.55.137.117 State: Connected GUID:
34EE9476-65D3-44C1-83DC-3CF6912CCB86  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
```

query_restricted_sdc_mode

Query whether restricted SDC mode is enabled or disabled in the PowerFlex system.

If restricted SDC mode is enabled, you must register to the system each SDC to which you want to map volumes using the `--add_sdc` command.

Syntax

```
scli --query_restricted_sdc_mode
```

Example

```
scli --query_restricted_sdc_mode
```

query_sdc

Retrieve information about the specified SDC.

Syntax

```
scli --query_sdc --sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>
  [--show_oscillating_failures]
```

Parameters

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

--show_oscillating_failures

Show extended oscillating failure information if there are one or more failures.

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc --sdc_id c27ab52600000001
```

Sample output

```
SDC ID: c27ab52600000001 Name: N/A IP: 10.55.137.119 State: Connected GUID:
C6533636-224B-44E3-A9EB-3CD651116CF5  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  525 IOPS 262.4 MB (268697 KB) per-second
    Writes: 503 IOPS 251.2 MB (257222 KB) per-second

Volume information (total 2 volumes):
         1: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB) Access mode: READ_WRITE
         2: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB) Access mode: READ_WRITE
```

query_sdc_volume_limits

Retrieve the IOP and bandwidth limits that one SDC generates for the specified volume.

Syntax

```
scli --query_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc_volume_limits --volume_id f6351e3700000003 --
sdc_id c27ab52600000001 --mdm_port 6611
```

Sample output

```
Successfully queried limits from SDC with ID c27ab52600000001 to volume with ID
f6351e3700000003: IOPS limit is unlimited. Bandwidth limit is unlimited.
```

reset_sdc_connections

Resets all SDC connections.

Syntax

```
scli --reset_sdc_connections --i_am_sure
```

Parameters

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --reset_sdc_sds_connections --i_am_sure
```

set_restricted_sdc_mode

Enable or disable restricted SDC mode. When enabled, SDCs must be approved (by GUID or IP address) before volumes can be mapped to them.

NOTE: In a system that is upgraded and already has volumes mapped to SDCs, if you want to enable restricted SDC mode, you must first approve the SDCs and only then enable restricted SDC mode. Use the `add_sdc` command to approve SDCs. After approving SDCs, add all the SDCs to the approved list before enabling the restricted mode.

Syntax

```
scli --set_restricted_sdc_mode --restricted_sdc_mode {approved_ip | guid | none}
```

Parameters

--restricted_sdc_mode {approved_ip | guid | none

Set the restricted SDC mode to one of the following:

- `approved_ip` — Volumes can be mapped only to approved SDCs that are also approved by IP address.
- `guid` — Volumes can be mapped only to approved SDCs.
- `none` — Restricted SDC mode is disabled and volumes can be mapped to any SDC.

Example

```
scli --add_sdc --sdc_ip approved_ip | guid | none
```

```
scli --set_approved_sdc_ips --sdc_id --sdc_ips approved_ip | guid | none
```

```
scli --set_restricted_sdc_mode --restricted_sdc_mode approved_ip | guid | none
```

set_sdc_authentication

Enable or disable SDC authentication. This feature ensures security by enforcing password-based authentication of the SDC to the MDM for access to mapped volumes. This prevents the SDC from accessing unauthorized volumes.

Syntax

```
scli --set_sdc_authentication (--enable | --disable) [--allow_old_sdc] [--
allow_no_sdc_password]
```

Parameters

`--enable`

Enable SDC authentication and authorization

`--disable`

Disable SDC authentication and authorization

`--allow_old_sdc`

Allow enabling SDC authentication and authorization when there is an old SDC. All old SDCs will be disconnected

`--allow_no_sdc_password`

Allow enabling SDC authentication and authorization when there is an SDC without a password. All SDCs without a password will be disconnected.

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_authentication --enable --allow_old_sdc
```

set_sdc_volume_limits

Set limits to the IOPS and bandwidth that one SDC generates for the specified volume. This enables you to control the quality of service (QoS).

Syntax

```
scli --set_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID> |
--sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>) (Options)
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

Options

Choose at least one:

--limit_iops *<NUMBER>*

Limit the volume IOPS. The number of IOPS must be larger than 10. 0 is unlimited.

--limit_bandwidth *<NUMBER>*

Limits the volume network bandwidth. The bandwidth is in MB/s. 0 is unlimited

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_volume_limit
--volume_name vol1 --sdc_ip 192.168.1.3 --limit_iops 100
```

set_approved_sdc_ips

Set a list of up to four IP addresses that are approved for the given SDC.

Syntax

```
scli --set_approved_sdc_ips (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --
sdc_ip <IP>) --sdc_ips <IPs>
```

Parameters

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC GUID

--sdc_ip *<IP>*
IP address of an SDC

--sdc_ips *<IPS>*
A comma-separated list of SDC IP addresses. Omit the space after each comma.

Example

```
scli --set_approved_sdc_ips --sdc_ips 10.76.60.10,10.76.60.11,10.76.60.12,10.76.60.13
```

add_nvme_host

Add an NVMe host to the system.

Syntax

```
scli --add_nvme_host --nvme_host_nqn <NQN> [--host_name <NAME>] [--max_number_paths
<VALUE>] [--max_number_sys_ports <VALUE>] [--host_os_full_type <OS>] [--host_group_name
<ID> | --host_group_id <NAME>] | --force
```

Parameters

--nvme_host_nqn *<NQN>*
NVMe host NQN

--host_name *<NAME>*
Host name

--max_number_paths *<VALUE>*
Maximal number of paths allowed per mapped volume

Valid range: 2 to 8

Dafault: 4

-max_number_sys_ports *<VALUE>*-

Maximal number of system ports allowed per protection domain

Valid range: 2 to 128

Dafault: 128 (for PowerFlex Data Mobility). Else, the default is 10

--host_os_full_type *<OS>*

OS type. Used for OS with known NVMe host issues

Options:

- generic (default)
- powerflex

--host_group_id *<ID>*

Host group ID.

--host_group_name *<NAME>*

Host group name.

--force

Force NQN, ignore all NQN formating rules.

Syntax

```
scli --add_nvme_host --host_name host1 --nvme_host_nqn
nqn.2014-08.org.nvmexpress:uuid:11111111-1111-2222-3333-444455556666
```

modify_nvme_host_max_paths

Modify the maximum number of paths per mapped volume for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_max_paths (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn
<NQN>) --max_number_paths <VALUE>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--volume_id *<ID>*

Volume ID

--max_number_paths *<VALUE>*

Maximal number of paths allowed per mapped volume

Valid range: 2-8

Default: 4

Example

```
scli --modify_nvme_host_max_paths --host_name host1 --max_number_paths 6
```

modify_nvme_host_max_sys_ports

Modify the maximum number of system ports per protection domain for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_max_sys_ports (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) --max_number_sys_ports <VALUE>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--max_number_sys_ports *<VALUE>*

Maximal number of system ports allowed per protection domain

Valid range: 2-128

Default: 10

Example

```
scli --modify_nvme_host_max_sys_ports --host_name host1 --max_number_sys_ports 6
```

modify_nvme_host_os_full_type

Modify the operating system (OS) type and version for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_os_full_type (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) --host_os_full_type <OS>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe Host NQN

--host_os_full_type *<OS>*

OS type. Used for OS with known NVMe host issues

Options:

- generic (default)
- powerflex

set_manual_nvme_host_connectivity

Set a manual connectivity goal for NVMe host to a protection domain.

Syntax

```
scli --set_manual_nvme_host_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--protection_domain_id <ID> | --protection_domain_name <NAME>) (--
system_port_ip <IP> | --system_port_id <ID>) --force
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--system_port_ip *<IP>*

A comma separated list of system port IP addresses.

--system_port_id *<ID>*

A comma separated list of system port IDs (4 hexadecimal digits).

--force

Force non-standard connectivity settings

reset_manual_nvme_host_connectivity

Reset manual connectivity goal for NVMe host to a protection domain.

Syntax

```
scli --reset_manual_nvme_host_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--protection_domain_id <ID> | --protection_domain_name <NAME>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

set_manual_nvme_mapping_connectivity

Set manual connectivity for mapping of a volume to an NVMe host.

Syntax

```
scli --set_manual_nvme_mapping_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--volume_id <ID> | --volume_name <NAME>) [--protection_domain_id
<ID> | --protection_domain_name <NAME>] (--system_port_ip <IP> | --system_port_id <ID>)
--force
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--system_port_ip *<IP>*
A comma separated list of system port IP addresses.

--system_port_id *<ID>*
A comma separated list of system port IDs (4 hexadecimal digits)

--force
Force non standard connectivity settings

reset_manual_nvme_mapping_connectivity

Reset manual connectivity of volume mapping to an NVMe host.

Syntax

```
scli --reset_manual_nvme_mapping_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--volume_id <ID> | --volume_name <NAME>) [--protection_domain_id
<ID> | --protection_domain_name <NAME>]
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

reset_manual_nvme_mapping_connectivity

Reset manual connectivity of volume mapping to an NVMe host.

Syntax

```
scli --reset_manual_nvme_mapping_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--volume_id <ID> | --volume_name <NAME>) [--protection_domain_id
<ID> | --protection_domain_name <NAME>]
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

rebalance_nvme_host_connectivity

Set connectivity goal and system ports balancing on all system NVMe hosts.

Syntax

```
scli --rebalance_nvme_host_connectivity
```

query_nvme_discovery_info

Get the information required to set up a discovery service on hosts.

Syntax

```
scli --query_nvme_discovery_info (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --service <TYPE> --transport <TYPE>
```

Parameters

--protection_domain_id *<ID>*
The failure domain ID

--protection_domain_name *<NAME>*
The failure domain name

--service *<TYPE>*
The service type

Option: discovery

--transport *<TYPE>*
The transport type

Option: TCP

--remote_system_description *<DESCRIPTION>*
Description of the remote system

--transport *<TRANSPORT_TYPE>*
Currently only TCP is supported

clear_nvme_reservation

Clear NVMe reservation and registration from the given volume. If there is a host that is a registrant and has an active controller, the command fails unless the `--force` flag is used.

Syntax

```
scli --clear_nvme_reservation (--volume_id <ID> | --volume_name <NAME>) [--wait] [--
force] [--i_am_sure]
```

Parameters

--volume_id *<ID>*
: Volume ID

--volume_name *<NAME>*
: Volume name

--wait
: Wait for the command to complete. If specified, the return code is the actual return code of the command

--force
: Force clearing NVMe reservation

--i_am_sure
: Preemptive approval

add_host_group

Add a host group to the system.

Syntax

```
scli --add_host_group [--host_group_name <NAME>] [--network_id <ID> | --network_name
<NAME>] [--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN>] [--force]
```

Parameters

--host_group_name *<NAME>*
: A name to be assigned to the host group

--network_id *<ID>*
: A comma separated list of system network IDs to be associated with the group

--network_name *<NAME>*
: A comma separated list of system network names to be associated with the group

--host_id *<ID>*
: A comma separated list of NVMe host IDs to be added to the group

--host_name *<NAME>*
: A comma separated list of NVMe host names to be added to the group

--nvme_host_nqn *<NQN>*
: A comma separated list of NVMe host NQNs to be added to the group

--force

Force assigning NVMe hosts to group regardless of system networks accessibility

assign_host_to_host_group

Assign a host to a host group.

Syntax

```
scli --assign_host_to_host_group (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn
<NQN>) (--host_group_id <ID> | --host_group_name <NAME>) [--force]
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--host_group_id *<ID>*

Host group ID

--host_group_name *<NAME>*

Host group name

--force *<NAME>*

Force assigning NVMe host to host group regardless of system networks accessibility

query_all_host_groups

Retrieve information about all host groups.

query_host_group

Retrieve information about a host group.

Syntax

```
scli --query_host_group (--host_group_id <ID> | --host_group_name <NAME>)
```

Parameters

--host_group_id *<ID>*

Host group ID

--host_group_name *<NAME>*

Host group name

modify_host_group

Modify the list of system networks of an existing host group.

Syntax

```
scli --modify_host_group (--host_group_id <ID> | --host_group_name <NAME>) [--network_id
<ID> | --network_name <NAME>]
```

Parameters

--host_group_id *<ID>*
: Host group ID

--host_group_name *<NAME>*
: Host group name

--network_id *<ID>*
: A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*
: A comma separated list of system network names to be replaced with the old one

rename_host_group

Name or rename a host group.

Syntax

```
scli --rename_host_group (--host_group_id <ID> | --host_group_name <NAME>) --new_name
<NAME>
```

Parameters

--host_group_id *<ID>*
: Host group ID

--host_group_name *<NAME>*
: Host group name

--new_name *<NAME>*
: A new name to be assigned to the host group

remove_host_from_host_group

Remove a host from a host group.

Syntax

```
scli --remove_host_from_host_group (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

remove_host_group

Remove a host group from the system.

Syntax

```
scli --remove_host_group (--host_group_id <ID> | --host_group_name <NAME>)
```

Parameters

--host_group_id *<ID>*
Host group ID

--host_group_name *<NAME>*
Host group name

set_auto_update_sdc_with_mdm_ip_addresses

Enable or disable SDC automatic update with MDM IP addresses.

Syntax

```
scli --set_auto_update_sdc_with_mdm_ip_addresses ((--enable [--force] (--
mdm_control_ip_addrs | --mdm_management_ip_addrs) | disable)
```

Parameters

--enable
Enable SDC automatic update with MDM IP addresses.

--disable
Disable SDC automatic update with MDM IP addresses.

--force
Enable update with MDM IP addresses though some SDCs are disconnected

--mdm_control_ip_addrs
Set auto update of SDC with MDM IP addresses to control IPs

--mdm_management_ip_addrs
Set auto update of SDC with MDM IP addresses to management IPs

NOTE: To enable this command, you must deploy the SDC hardening agent. For more information, see the *Dell PowerFlex Manager MDM IP Address Dynamic Updates TechNote*.

restart_connections_to_realign_net_path

Set a restart connection to realign the network path interval.

Syntax

```
scli --restart_connections_to_realign_net_path [--disable | --enable | --force | --
interval_hour <hours> | --enable --interval_hour <hours>]
```

Parameters

--enable

Restart connections to realign network path. The default interval is 24 hours.

--disable

Disable periodic connections realignment.

--interval_hour *<INTERVAL_HOUR>*

Set the restart connection interval, 1-168 hours.

--force

Force on time refresh, even if periodic connections realignment has been disabled.

27

SDC operations

PowerFlex uses different methods to add external SDCs to an existing PowerFlex system, depending on the operation system of the server. For existing SDCs, many operations use the drv_cfg command line, which is a local utility that affects only the client on which the SDC is running.

Add an external SDC to an existing PowerFlex system

Add an external SDC to an existing PowerFlex system.

During manual installation of an SDC, you can install the SDC according to the operating system-specific instructions in the following section, and it will be connected to the existing PowerFlex system.

Install the SDC on an ESXi-based server and connect it to PowerFlex using esxcli

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Customer support site)
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID needs to conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

The following procedure explains how to manually install an external SDC on an ESXi-based using esxcli in command line.

NOTE: This procedure requires two server reboots.

Steps

1. On the ESXi-based server on which you are installing the SDC, set the acceptance level:

   ```
   esxcli software acceptance set --level PartnerSupported
   ```

2. Install the SDC:

Option	Description
ESXi 9	`esxcli software component apply -d <path_to_zip> -n scaleio-sdc-esx9`

Option	Description
	where *<path_to_zip>* is the path to the SDC zip file.
ESXi 8.x	`esxcli software vib install -d <full_path_to_offline_bundle>` where *<full_path_to_offline_bundle>* is the absolute path to the SDC VIB file.

3. Get permission to reboot the ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.
4. Set the IP address of the MDM:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the "+" symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one: `10.20.30.40,50.60.70.80+11.22.33.44`
   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.
5. Back up the parameters:

   ```
   /bin/auto-backup.sh
   ```
6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

   ```
   vmkload_mod -l | grep scini
   ```

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

   The SDC component is installed on the ESXi-based server.

Results

The SDC is installed on the ESXi-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Linux-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:
- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).
- Secure Boot is not supported; ensure that Secure Boot is disabled in the BIOS.

About this task

The following procedure explains how to manually install an external SDC on a Linux-based server. External SDCs can be deployed by the PowerFlex Installer.

The steps in the procedure below are relevant for Linux Red Hat Enterprise Linux/CentOS/Oracle Linux operating systems.

Table 12. Modifications required for deployment on other operating systems

Operating system	Modifications required
Ubuntu/OL	Before installing PowerFlex, ensure that you have followed the required preparation procedures relating to various types of Linux operating systems. ● PowerFlex component packages are delivered as TAR files. Before installing, perform the following: 1. Untar all the packages: `tar -xvf <tar_file>`. This yields SIOB files. 2. Extract the DEB from the SIOB files: `siob_extract <siob_file>` 3. The DEB file is in the following format: `EMC-ScaleIO-sdc-.<operating_system>.X.X.x86_64.deb` Use the extracted DEB files for the installation. ● Some commands are a bit different, noted where applicable.

Steps

1. Install the GPG key on every server on which SDC must be installed. From the PowerFlex installation folder, run the following command on every server:

```
rpm --import RPM-GPG-KEY-ScaleIO
```

2. In the command line, install the SDC:
   - Red Hat Enterprise Linux/CentOS /Oracle Linux/SLES

```
MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
```

   - Ubuntu (These files must be extracted before use, as described above.)

```
MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.
3. You can use the following command to verify if the driver is loaded:
   - SLES 15:

```
service sdc status
```

   - All other Linux flavors:

```
service scini status
```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks that are described in this guide.

In existing systems, map volumes to the new SDCs that were added to the system.

Install SDC on an AIX-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. These instructions are relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).

About this task

The following procedure explains how to manually install an external SDC on an AIX-based server. The PowerFlex Installer cannot be used.

Steps

1. In the command line, install the SDC:

   ```
   MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
   ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.

     The SDC package is in a format similar to this:
     `EMC-ScaleIO-sdc-.aix7.aix7.2.ppc.rpm`
2. Optional: For SDC authentication to the MDM for access, in the command line, enter:

   ```
   chdev -l scini -a "mdm_password=<ip>-<password>"
   ```

Results

The SDC is installed on the AIX-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the recommended post-deployment tasks that are described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Windows-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).

About this task

The following procedure explains how to manually install an external SDC on a Windows-based server. Alternatively, you can install the external SDC using the PowerFlex Installer. For more information about adding system components to an existing system, see the *Dell PowerFlex and PowerFlex Manager Install and Upgrade Guide*.

NOTE: This procedure requires a server reboot.

Steps

1. On the Windows-based server on which you are installing the SDC, run the following command in the command line:

   ```
   msiexec /i <SDC_PATH>.msi MDM_IP=<LIST_VIP_MDM_IPS>
   ```

   Where
   - *<SDC_PATH>* is the path where the SDC installation package is located.
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not add tiebreaker addresses here.
2. Get permission to reboot the Windows-based server, and perform a reboot to load the SDC driver on the server.

Results

The SDC is installed on the Windows-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks that are described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Performing SDC operations

Many SDC operations use `drv_cfg`. The `drv_cfg` command line is a local CLI utility that affects only the client on which the SDC is running.

Possible SDC operations include updating the SDC driver with IP changes, detecting new volumes, querying volumes, loading a configuration file, adding an MDM, modifying an MDM IP address, and enabling support of PDL state.

Performing SDC operations

Many SDC operations use `drv_cfg`. The `drv_cfg` command line is a local CLI utility that affects only the client on which the SDC is running.

Possible SDC operations include updating the SDC driver with IP changes, detecting new volumes, querying volumes, loading a configuration file, adding an MDM, modifying an MDM IP address, and enabling support of PDL state.

Update the SDC driver with IP address changes

Use the drv_cfg to update the IP address.

Steps

1. Edit `drv_cfg.txt` and change the IP address in the last line to the new IP address.

   Location of `drv_cfg.txt`:

- Linux: `/etc/emc/scaleio/`
- Windows: In the following registry key - `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\services\scini\Parameters\mdms`

NOTE:

On ESXi, GUID and MDM lists are stored as module parameters, and not in a `drv_cfg.txt` file. To modify these parameters, use `esxcli` commands.

2. Save and close the file.
3. Type the following command:

```
/etc/init.d/scini restart
```

NOTE: ESXi guidelines are described in the PowerFlex plug-in chapter.

Detecting new volumes

Volumes are always exposed to the operating system as devices with the prefix `scini` (such as `/dev/scinia`, `/dev/scinib` and so on). Unique names can be found under `/dev/disk/by-id/`.

PowerFlex periodically scans the system to detect new volumes. You can initiate a scan for the most up-to-date status on a particular SDC node. This command is unique because it is not a CLI command, but rather a command issued on the specific SDC.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: See the vSphere Client VMware guidelines on how to detect new storage. If troubleshooting is needed, contact Customer support.

Command

drv_cfg --rescan

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --rescan
```

Query volumes using drv_cfg

This utility retrieves information about all known active volume objects in KVM mode. You can use this utility to determine which volumes are mapped, and the ID of each volume in PowerFlex.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

Query volumes using drv_cfg

This utility retrieves information about all known active volume objects in KVM mode. You can use this utility to determine which volumes are mapped, and the ID of each volume in PowerFlex.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

Command

`drv_cfg --query_vols`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_vols
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_vols
```

Query tgt objects using drv_cfg

This utility retrieves information about all known active tgt objects (storage nodes) in kernel mode.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

Command

`drv_cfg --query_tgts`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_tgts
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_tgts
```

Query GUID using drv_cfg

This utility retrieves the unique ID of the kernel module. The utility can be used to verify that all SDC GUIDs in the system are unique.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

NOTE:

If the SDC was removed and reinstalled, the GUID of the SDC is different than its original GUID. In such a case, you may have to remove the SDC, if two SDCs now have the same GUID.

Command

drv_cfg --query_guid

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_guid
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_guid
```

Query MDMs using drv_cfg

This utility retrieves information about all known MDM objects in KVM mode. This utility is typically used to determine to which MDM an SDC is connected.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

Command

drv_cfg --query_mdms

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_mdms
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_mdms
```

Loading a configuration file using drv_cfg

This utility reads a configuration file containing MDM IP addresses, and calls the KVM to connect to them.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

The configuration file that is loaded when using the `drv_cfg --load_cfg_file` utility is not persistent; when you restart the SDC, the changes will be lost.

To make the changes persistent, perform either of the following:

- Install the SDC on every server that exposes volumes to the application running, by running the following command:

  ```
  MDM_IP=<IP of the MDM> rpm -i <full rpm file path>
  ```

- Use the following `drv_cfg` command:

  ```
  /opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
  --ip <EXISTING_MDM_IP_ADDRESS> --new_mdm_ip
  <NEW_MDM_IP_ADDRESSES>
  ```

Command

drv_cfg --load_cfg_file

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

This command can not be used on ESXi servers. Instead, follow the steps described in "Modify parameters on ESXi servers".

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg
--load_cfg_file <FILE_NAME>
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg
--load_cfg_file /etc/emc/scaleio/drv_cfg.txt
```

Adding an MDM using drv_cfg

This utility calls the kernel module to connect to an MDM. This command is typically used in cases where an SDC is connected to more than one PowerFlex system.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

NOTE: Extending your PowerFlex system with another MDM requires that you update all SDCs in your system with the new MDM IP address. It is recommended that you use virtual MDM IP address, which helps minimize the necessity to update all SDCs when MDM details change. Run the `drv_cfg` utility with the `--add_mdm` option, and to make the change persistent, use the `--file` parameter. In addition, any additional objects or systems which interface with the MDM must also be updated. For more information, see *Modifying an MDM IP address using drv_cfg*.

Command

`drv_cfg --add_mdm`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

This command cannot be used on ESXi servers.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
--ip <MDM_IP_ADDRESS_LIST>
```

Parameters

Parameters

`--ip <MDM_IP_ADDRESS_LIST>`
List of IP addresses (comma delimited) for this primary, secondary, or virtual MDM.

--file <CONFIG_FILE_NAME> (optional)
Name of the configuration file to which the MDM information should be written.

`--only_cfg (optional)`
Do not call the kernel to actually connect.

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
--ip 10.100.22.20,10.100.22.30
--file /etc/emc/scaleio/drv_cfg.txt
```

Modifying an MDM IP address using drv_cfg

This utility calls the kernel to modify the IP address list of an MDM. It is typically used in cases when an MDM IP address has changed, or when MDMs are added/removed from/to the system. The command must be run on every SDC in the system. A restart is required for the changes to be effective.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

NOTE: Extending your PowerFlex system with another MDM requires that you update all SDCs in your system with the new MDM IP address. It is recommended that you use virtual MDM IP address, which helps minimize the necessity to update all SDCs when MDM details change. Run the `drv_cfg` utility with the `--add_mdm` option, and to make the change persistent, use the `--file` parameter. In addition, any additional objects or systems which interface with the MDM must also be updated.

Command

drv_cfg --mod_mdm_ip

NOTE: This is not a CLI command, but rather an executable that is run on the SDC server. This command cannot be used on ESXi servers. Instead, follow the steps that are described in *Modify parameters in ESXi servers*.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip <EXISTING_MDM_IP_ADDRESS>
--new_mdm_ip <NEW_MDM_IP_ADDRESSES> [--file <CONFIG_FILE_NAME>]["-""-"only_cfg]
```

Parameters

Parameters

--ip <EXISTING_MDM_IP_ADDRESS>

One of the existing MDM IP addresses.

--new_mdm_ip <NEW_MDM_IP_ADDRESSES>

The new IP address list (comma delimited) for this MDM. If you want to retain the existing address(es), include them in this list.

--file <CONFIG_FILE_NAME> (optional)

The name of the configuration file to which the MDM information should be written.

--only_cfg (optional)

Do not call the KVM to actually connect.

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip 10.100.20.20
--new_mdm_ip 10.100.20.20,10.100.20.30,10.100.20.40
```

Permanent device loss state

When the MDM has disconnected from the SDC and a volume that is mapped to this SDC has experienced an I/O error, the ESXi host continuously sends I/Os to the device to determine if the device has become accessible. This can subsequently cause a high I/O-error load that can lead to the host freezing. In cases where the device is disconnected long-term, such as when the entire MDM cluster is down during a network upgrade, the SDC can change the volume state to permanent device loss (PDL) to prevent more I/O errors coming from the ESXi.

PDL is an ESXi state, which, once enabled, is supported in PowerFlex. Once the ESXi host loses connectivity with a device, if a timeout value is reached, the ESXi is notified that the device is in a PDL state. The timeout value can be manually set. Once a device is in a PDL state, the ESXi host no longer attempts to re-establish connectivity or issue commands to the device.

Recovering a device from PDL state is described in the VMware documentation for your operating system version.

Use the instructions that match your environment.

Enable support of PDL state on the VMware ESXi

Enable support of permanent device loss (PDL) state on the ESXi host.

Steps

1. On the ESXi host:

```
esxcli system module parameters set -m scini "<<PREVIOUS_MODULE_PARAMS>>
bBlkDevIsPdlActive=1 blkDevPdlTimeoutMillis=<TIMEOUT_VALUE>"
```

   where:
   - *<<PREVIOUS_MODULE_PARAMS>>* is any previous module parameters being used for this ESXi host.
   - *<TIMEOUT_VALUE>* is the timeout time in milliseconds. Its value can be 1000-3600000 (default is 60000) and including it in the command is optional.
2. Reboot the host.

Disable support of PDL state on the ESXi

Disable support of permanent device loss (PDL) state on the ESXi host.

Steps

1. On the ESXi host:

```
esxcli system module parameters set -m scini "<<PREVIOUS_MODULE_PARAMS>>
bBlkDevIsPdlActive=0"
```

   Where *<<PREVIOUS_MODULE_PARAMS>>* is any previous module parameters being used for this ESXi host.
2. Reboot the host.

Run the SDC driver on Red Hat Enterprise Linux or SUSE Enterprise Linux Enterprise Server with multipath enabled

Run the SDC driver on Red Hat Enterprise Linux or SUSE Enterprise Linux Enterprise Server with multipath enabled

Steps

1. On the machine where the SDC is running, open `/etc/udev/rules.d/20-scini.rules` for editing.
2. In the first line at the end add:

```
ENV{ID_WWN}="%c"
```

   The line should read like this:

```
KERNEL=="scini*[!0-9]", SUBSYSTEM=="block", PROGRAM="/bin/emc/scaleio/drv_cfg --
query_block_device_id --block_device $tempnode", SYMLINK+="disk/by-id/emc-vol-%c",
ENV{ID_BUS}="scsi", ENV{ID_SERIAL}="%c", ENV{ID_WWN}="%c"
```

3. Open `/etc/multipath.conf` and add the following additional settings to the configuration file:

```
defaults {
          retain_attached_hw_handler "no"
                find_multipaths no
}
```

```
blacklist_exceptions {
                    devnode "scini"
}
```

NOTE: From the command line, run `dracut -f.` This includes the changes in initrd.
`[retain_attached_hw_handler no]` is the default setting in Red Hat Enterprise Linux 7.x or 8.x.

4. Restart the PowerFlex service and multipathd service:

```
systemctl restart scini
```

```
systemctl restart multipathd.service
```

NOTE: If LUNs are in use and a restart is not possible, restart the node.

SDC authentication

This feature ensures secured connectivity between the SDC and the storage cluster over the data path by applying Challenge-Handshake Authentication Protocol (CHAP) based authentication of the SDC to the MDM for access to the system in general and to mapped volumes in particular. This prevents the SDC from accessing the cluster if it is not admitted as well as unauthorized volumes. Once enabled, the SDC internally performs mutual CHAP authentication with the storage nodes with no manual intervention.

Prerequisites

Enable SDC authentication according to the following rules:

- For each SDC, a CHAP authentication password is generated by the MDM.
- All SDCs must be configured with their generated passwords
- Run the `--check_sdc_authentication_status` command, to check the status of the SDCs and whether they are ready to authenticate

About this task

This procedure describes how to enable SDC authentication.

NOTE: Using CHAP authentication with SDC also means that an SDC can only perform I/O operations on volumes that are explicitly mapped to it. The storage node blocks SDC I/O operations on unmapped volumes. CHAP authentication is also used internally for I/O authentication to the storage node, however it is always enabled and not controlled by the user.

Steps

1. Get the shared generated password for SDC from the MDM using the command:

```
scli --generate_sdc_password --(sdc_id <ID> | sdc_name <NAME) | sdc_guid <GUID> |
sdc_ip <IP>) [--reason <REASON>]
```

Use the reason parameter to verify that the SDC password is being reset and not changed by accident. The reason is stored in the MDM events log.

NOTE: SDCs not configured with a password are disconnected after the feature is enabled in step 3.

NOTE: You must comply with organization password protection policy rules when performing this operation.

Copy the password that was generated in `<SDC_PASSWORD_STRING>`, used in the next step.

2. On the SDC, run the following command:
   - Linux:

```
/opt/emc/scaleio/sdc/bin/drv_cfg --set_mdm_password --ip <MDM_IP> --password
<SDC_PASSWORD_STRING> --file/etc/emc/scaleio/drv_cfg.txt
```

NOTE: The file option is required for password persistency, for cases such as service scini restart or SDC reboot. Open the file to verify the `<SDC_PASSWORD_STRING>` is logged at the end of the MDM line.

- ESXi:
  a. `cat /etc/vmware/esx.conf | grep scini | grep options`

     A string is returned representing all the ESXi configuration parameters set. Copy the string with the enclosing quotation marks and paste in a text editor for editing.
  b. At the end of the string, add the following text, within the quotation marks:

     ```
     IoctlMdmPasswordStr=<MDM_IP>-<MDM_PASSWORD>
     ```

     where:
     - *<MDM_IP>* is the MDM IP address.
     - *<MDM_PASSWORD>* is the MDM password.

     For example:

     ```
     "IoctlIniGuidStr=cd069ce3-bf2a-5dea-b50a-1a5ebc8ef3de
     IoctlMdmIPStr=192.169.217.165,172.17.217.165,192.169.217.166,172.17.217.166,192.1
     69.217.167,172.17.217.167 IoctlMdmPasswordStr=192.169.217.165-AQAAAAAAAADu/
     10fXW3BS1wPBDgnkR06tdneGoUK7VQ"
     ```

  c. Run the following command with the string that is appended to the end:

     ```
     esxcli system module parameters set -m scini -p <STRING>
     ```

     For example:

     ```
     esxcli system module parameters set -m scini -p "IoctlIniGuidStr=cd069ce3-
     bf2a-5dea-b50a-1a5ebc8ef3de
     IoctlMdmIPStr=192.169.217.165,172.17.217.165,192.169.217.166,172.17.217.166,192.1
     69.217.167,172.17.217.167 IoctlMdmPasswordStr=192.169.217.165-AQAAAAAAAADu/
     10fXW3BS1wPBDgnkR06tdneGoUK7VQ"
     ```

3. To check SDC readiness for all SDCs in the system, before enabling SDC authentication, run the following command:

   NOTE: It is important to complete the previous steps for all SDCs before running the command.

   ```
   scli --check_sdc_authentication_status [--run_test] [--file_name <FILENAME>]
   ```

   Where:

   - `--run_test` runs a connectivity test to check whether the SDCs can successfully authenticate using CHAP
   - `--filename <FILENAME>` is the full file name and path for the generated report.

   The command sends a report that includes the SDCs authentication password status.

   NOTE: The SDCs are disconnected for a short period from the MDM when running this command. This does not interrupt running I/Os or have any impact on MDM/SDC activity. It is recommended to run the command when the system is in a healthy state and not during rebalancing or rebuilding operations.

4. To enable SDC authentication, run the following command:

   ```
   scli --set_sdc_authentication --enable
   ```

5. To disable SDC authentication, run the following command:

   ```
   scli --set_sdc_authentication --disable
   ```

6. Reboot the ESXi for the configuration to take effect. For more information, see *Gracefully shut down or reboot a node* in the *Dell PowerFlex and PowerFlex Manager Administration Guide*.

Results

SDC authentication is enabled or disabled.

Manual SDC configuration and verification

If there are problems connecting to a PowerFlex system after a reboot, you can modify the `drv_cfg.txt` file or use the `drv_cfg` utility to ensure that the SDC reconnects to the MDM automatically upon node restart or reboot.

NOTE: Use virtual MDM IP address to reduce the necessity to update the `drv_cfg.txt` file whenever the IP addresses of the physical MDM components change.

To modify the `drv_cfg.txt` file, perform one of the following procedures:

- Edit `/etc/emc/scaleio/drv_cfg.txt` and change the #MDM line to include each IP address for any node(s) containing an MDM in your cluster.

  For example, change the line:

  From:

  ```
  #mdm 10.20.30.40
  ```

  To:

  ```
  mdm 10.108.158.48,10.108.158.49
  ```

  where `.48` is the primary node, and `.49` is the secondary node. The tiebreaker node should not be included.

- Use the drv_cfg binary or drv_cfg.exe on Windows to re-attach the MDM.
- Rescan all volumes using the drv_cfg utility.

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SDT commands

This section contains commands for managing storage data targets (SDT).

add_sdt

Add an SDT IP address.

Syntax

```
scli --add_sdt --sdt_ip <IP> [--sdt_ip_role <ROLE>] [--storage_port <PORT>] [--nvme_port
<PORT>] [--discovery_port <PORT>] [--sdt_name <NAME>] (--protection_domain_id <ID> | --
protection_domain_name <NAME>) [--profile <PROFILE>] [--force_clean [--i_am_sure]]
```

Parameters

--sdt_ip *<IP>*

A comma separated list of IP addresses assigned to the SDT

--sdt_ip_role *<ROLE>*

A comma separated list of roles assigned to each SDT IP address

Role options: storage_only, host_only, or storage_and_host

--storage_port *<PORT>*

Port assigned to the SDT (default: 12200)

--nvme_port *<PORT>*

Port used by the NVMe hosts (default: 4420)

--discovery_port *<PORT>*

Port used by the NVMe hosts for discovery (default: 8009). Set to 1 in order to indicate no use of discovery port.

--sdt_name *<NAME>*

A name assigned to the SDT

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--profile *<PROFILE>*

Set the performance profile from the following options: compact | high_performance

The default is high_performance

--force_clean

Clean a previous SDT configuration

--i_am_sure

Preemptive approval

Example

```
scli --add_sdt --sdt_name sdt1 --sdt_ip 10.0.0.1,10.0.1.1 --sdt_ip_role
storage_and_host, storage_and_host --protection_domain_name pd1
```

remove_sdt

Start the removal of an SDT.

Syntax

```
scli --remove_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--force
Force removing IP regardless the number of SDTs left on protection domains.

Example

```
scli --remove_sdt --sdt_name sdt1
```

rename_sdt

Name or rename an SDT.

Syntax

```
scli --rename_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --new_name <NAME>
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--new_name *<NAME>*

A new name to be assigned to the SDT

Example

```
scli --rename_sdt sdt1 --new_name someSdt
```

add_sdt_ip

Add an IP address to an SDT.

Syntax

```
scli --add_sdt_ip (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --new_sdt_ip <IP> [--sdt_ip_role <ROLE>]
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--new_sdt_ip *<IP>*

New IP address to be assigned to the SDT

--sdt_ip_role *<ROLE>*

The role assigned to the new SDT IP address

Role options: storage_only, host_only, or storage_and_host

Example

```
scli --add_sdt_ip  --sdt_name sdt1 --new_sdt_ip 10.0.3.1 --sdt_ip_role storage_and_host
```

remove_sdt_ip

Remove an IP address from an SDT.

Syntax

```
scli --remove_sdt_ip (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --sdt_ip_to_remove <IP> --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--sdt_ip_to_remove *<IP>*
IP address to remove from the SDT

--force
Force removing IP regardless the number of hosts connections

Example

```
scli --remove_sdt_ip --sdt_name sdt1 --sdt_ip_to_remove 10.0.3.1
```

modify_sdt_ip_role

Modify the role of an SDT IP address.

Syntax

```
scli --modify_sdt_ip_role (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--sdt_ip_to_modify <IP>] --new_sdt_ip_role <ROLE> --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--sdt_ip_to_modify *<IP>*
IP address for which to modify the role. Required if --sdt_ip is not specified

--new_sdt_ip_role <ROLE> *<ROLE>*

The new role assigned to the modified SDT IP address

Role options: storage_only, host_only, or storage_and_host

--force

Force modifying IP role regardless the number of hosts connections

Example

```
scli --modify_sdt_ip_role --sdt_name sdt1 --sdt_ip_to_modify 10.0.3.1 -sdt_ip_role
host_only
```

modify_sdt_port

Modify the port of an SDT.

Syntax

```
scli --modify_sdt_port (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--storage_port <PORT>] [--nvme_port <PORT>] [--discovery_port <PORT>]
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--storage_port *<PORT>*

Port assigned to the SDT

--nvme_port *<PORT>*

Port used by the NVMe hosts

--discovery_port *<PORT>*

Port used by the NVMe hosts for discovery.

Example

```
scli --modify_sdt_port (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--storage_port <PORT>] [--nvme_port <PORT>] [--discovery_port <PORT>]
```

query_sdt

Retrieve information about an SDT.

Syntax

```
scli --query_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>])
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

Example

```
scli --quert_sdt --sdt_name sdt1
```

query_all_sdt

Retrieve information about all SDTs.

Syntax

```
scli --query_all_sdt
```

enter_sdt_maintenance_mode

Move the SDT into maintenance mode.

Syntax

```
scli --enter_sdt_maintenance_mode (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--
sdt_port <PORT>]) --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--force

Force entering maintenance mode regardless the number of SDTs in a protection domain.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --enter_sdt_maintenance_mode --sdt_name sdt1
```

exit_sdt_maintenance_mode

Move the SDT out of maintenance mode.

Syntax

```
scli --exit_sdt_maintenance_mode (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--
sdt_port <PORT>])
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --exit_sdt_maintenance_mode --sdt_name sdt1
```

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Storage node commands

This section contains commands for managing storage nodes.

add_storage_node

Add a storage node to the specified protection domain.

Syntax

```
scli --add_storage_node (--protection_domain_id <ID> | --protection_domain_name <NAME>)
[--storage_node_name <NAME>] --storage_node_ip <IP> [--storage_node_ip_role <ROLE>] [--
pds_port <PORT>] [--dgwt_port <PORT>] [--core_count <COUNT>] [--socket_count <COUNT>]
```

Parameters

--protection_domain_id <ID>

Protection domain ID

--protection_domain_name <NAME>

Protection domain name

--storage_node_name <NAME>

A name to be assigned to the storage node

--storage_node_ip <IP>

A comma-separated list of IP addresses assigned to the storage node

--storage_node_ip_role <ROLE>

The role assigned to each storage node IP address. Each role can be:

- application_only
- storage_only
- all

When adding multiple IP addresses, use a comma-separated list for IP roles.

--pds_port <PORT>

Port assigned to the PDS on this storage node

--dgwt_port <PORT>

Port assigned to the device gateway target (DGWT) on this storage node.

--core_count <COUNT>

Number of cores in this storage node. Valid range: 32-256. Default: 64

--socket_count <COUNT>

Number of sockets in this storage node. Valid range: 1-8. Default: 2

Modify_storage_node

Modify a storage node.

Syntax

```
scli --modify_storage_node (--storage_node_id <ID> | --storage_node_name <NAME> | --
storage_node_ip <IP> [--storage_node_port <PORT>])
```

Parameters

--storage_node_name <NAME>

Storage node name

--storage_node_id <ID>

Storage node ID

--storage_node_ip <IP>

Storage node IP address

--storage_node_port <PORT>

Port assigned to the storage node

Options: Choose at least one of the following category to modify:

- Modify storage node port
  --new_pds_port <PORT>

  New port to be assigned to the PDS on this storage node

  --new_dgwt_port <PORT>

  New port to be assigned to the DGWT on this storage node
- Modify storage node ip role
  --storage_node_ip_to_modify <IP>

  IP address for which to modify the role. Optional if --storage node_ip not specified

  --new_storage_node_ip_role <ROLE>

  The new role assigned to the modified storage node IP address. Role can be application_only, storage_only, or all

  --force_role_modification

  Force the modification of the IP's role even if the IP could not be verified
- Modify storage node name
  --new_name <NAME>

  New name to be assigned to the storage node
- Modify storage node core count and socket count
  --new_core_count <COUNT>

  New core count for the storage node. Valid range: 32-256. Default: 64

  --new_socket_count <COUNT>

  New socket count for the storage node. Valid range: 1-8. Default: 2

remove_storage_node

Removes a storage node.

Syntax

```
scli --remove_storage_node (--storage_node_id <ID> | --storage_node_name <NAME> | --
storage_node_ip <IP> [--storage_node_port <PORT>])
```

Parameters

--storage_node_name <NAME>
Storage node name

--storage_node_id <ID>
Storage node ID

--storage_node_ip <IP>
Storage node IP address

--storage_node_port <PORT>
Port assigned to the storage node

validate_remove_storage_node

Validate removal of a storage node.

Syntax

```
 scli --validate_remove_storage_node (--storage_node_id <ID> | --storage_node_name
<NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])
```

Parameters

--storage_node_name <NAME>
Storage node name

--storage_node_id <ID>
Storage node ID

--storage_node_ip <IP>
Storage node IP address

--storage_node_port <PORT>
Port assigned to the storage node

add_storage_node_ip

Adds an IP address to a storage node.

Syntax

```
scli --add_storage_node_ip (--storage_node_id <ID> | --storage_node_name <NAME> |
--storage_node_ip <IP> [--storage_node_port <PORT>]) --new_storage_node_ip <IP> [--
storage_node_ip_role <ROLE>] [--force_add_ip]
```

Parameters

--storage_node_name <NAME>
: Storage node name

--storage_node_id <ID>
: Storage node ID

--storage_node_ip <IP>
: Storage node IP address

--storage_node_port <PORT>
: Port assigned to the storage node

--new_storage_node_ip <IP>
: New IP to be assigned to the storage node

--storage_node_ip_role <ROLE>
: The role assigned to the new storage node IP address. Role can be application_only, storage_only, or all

--force_add_ip
: Force the addition of the new IP even if it could not be verified

remove_storage_node_ip

Remove an IP address from a storage node.

Syntax

```
scli --remove_storage_node_ip (--storage_node_id <ID> | --storage_node_name <NAME> | --
storage_node_ip <IP> [--storage_node_port <PORT>]) --storage_node_ip_to_remove <IP>
```

Parameters

--storage_node_name <NAME>
: Storage node name

--storage_node_id <ID>
: Storage node ID

--storage_node_ip <IP>
: Storage node IP address

--storage_node_port <PORT>
: Port assigned to the storage node

--storage_node_ip_to_remove <IP>

IP address to remove from the storage node

query_storage_node

Retrieve information about a storage node.

Syntax

```
scli --query_storage_node
```

Parameters

--storage_node_name <NAME>
Storage node name.

--storage_node_id <ID>
Storage node ID.

--storage_node_ip <IP>
Storage node IP address.

--storage_node_port <PORT>
Port assigned to the storage node.

query_all_storage_nodes

Retrieves information about all storage nodes.

Syntax

```
scli --query_all_storage_nodes
```

enter_storage_node_maintenance_mode

Move one storage node to maintenance mode (includes PDS, DGWT).

Syntax

```
scli --enter_storage_node_maintenance_mode ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port
<PORT>])) [--force_others_are_in_maintenance_mode] [--force_degraded_or_failed_data][--
force_failed_storage_node]
```

Parameters

--storage_node_name <NAME>
Storage node name.

--storage_node_id <ID>
Storage node ID.

--storage_node_ip <IP>

Storage node IP address.

--storage_node_port <PORT>

Port assigned to the storage node.

--force_others_are_in_maintenance_mode

Allow entering maintenance mode in case another storage node is already in maintenance mode.

--force_degraded_or_failed_data

Allow entering maintenance mode when there is degraded or failed data.

--force_failed_storage_node

Allow entering maintenance mode when storage node down.

exit_storage_node_maintenance_mode

Moves a storage node out of maintenance mode (includes PDS, DGWT).

Syntax

```
scli --exit_storage_node_maintenance_mode ((--storage_node_id <ID> | --
storage_node_name <NAME> | --storage_node_ip <IP> [--storage_node_port <PORT>])) [--
force_failed_storage_node]
```

Parameters

--storage_node_name <NAME>

Storage node name

--storage_node_id <ID>

Storage node ID

--storage_node_ip <IP>

Storage node IP address

--storage_node_port <PORT>

Port assigned to the storage node

--force_failed_storage_node

Allow exit maintenance mode when storage node is down

update_device_original_paths

Change the original path configuration of all devices to the path assigned to the devices in the storage node.

Syntax

```
scli --update_device_original_paths (--storage_node_id <ID> | --storage_node_name <NAME>
| --storage_node_ip <IP> [--storage_node_port <PORT>]) [--force_failed_devices]
```

Parameters

--sds_id *<ID>*

Storage node ID

--sds_name *<NAME>*

Storage node name

--sds_ip *<IP>*

Storage node IP address

--sds_port *<PORT>*

Storage node port

--storage_node_name*<NAME>*

Storage node name (EC only)

--storage_node_id *<ID>*

Storage node ID (EC only)

--storage_node_ip *<IP>*

Storage node IP address (EC only)

--storage_node_port *<PORT>*

Port assigned to the storage node (EC only)

--force_failed_devices

Force path update even for devices in error

Example

```
scli --update_sds_original_paths --sds_name sds1 --force_failed_devices
```

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Storage pool commands

This section contains commands for managing storage pools.

disable_background_device_scanner

Disable the background device scanner on the specified storage pool.

Syntax

```
scli --disable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --disable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp180
```

enable_background_device_scanner

Enable the background device scanner on the devices in the specified storage pool to check for errors. .

Syntax

```
scli --enable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --
read_error_action <ACTION>  --compare_error_action <ACTION>
  [--scanner_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--read_error_action
Read error handling. Can be one of the following:
--report_and_fix
Perform read operations and automatically fix from peer on errors.

--scanner_bandwidth_limit *<LIMIT>*
Bandwidth limit in KB per second per device. The value should be in the range of 10 KB to 10 MB. Default: 3 MB.

NOTE: High bandwidth should be used very carefully for extreme cases only (such as an urgent need to check some devices), as it may create negative impact on the system performance. Setting the background device scanner bandwidth should take into account maximum bandwidth of the devices.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --enable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp18 --read_error_action --report_only --compare_error_action --
report_only --scanner_bandwidth_limit 2000
```

Errors

To check for errors detected by the background device scanner, query storage nodes using the `--query_sds` command. Error information is provided in event reports.

Bandwidth limits

High bandwidth may create a negative impact on system performance and should be used carefully and in extreme cases only—for example, when there is an urgent need to check certain devices. When setting the background device scanner bandwidth, you should consider the maximum bandwidth of the devices.

modify_compression_method

Modify the compression method of a storage pool or a V-Tree (VTree/volume options are supported only in mirroring protection domains).

Syntax

```
scli --modify_compression_method ( (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) | --
vtree_id <ID> | --volume_id <ID> | --volume_name <NAME> ) --compression_method <METHOD>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

--storage_pool_id *<ID>*

Storage pool ID

--vtree_id *<ID>*

V-Tree ID

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

Example

```
scli --modify_compression_method --protection_domain_name pd2 --storage_pool_name sp21 --
compression_method normal
```

modify_storage_pool

Modify a storage pool.

Syntax

```
scli --modify_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) [--size_gb <SIZE>
| --use_all_available_capacity] [--new_name <NAME>] [--over_provisioning_factor
<PERCENTAGE>] [--force_insufficient_spare_or_free_space] [--compression_method]
```

Parameters

--protection_domain_id <ID>

Protection domain ID

--protection_domain_name <NAME>

Protection domain name

--storage_pool_name <NAME>

Storage pool name

--storage_pool_id <ID>

Storage pool ID

--size_gb <SIZE>

Physical size in GBs for volumes in the storage pool

--use_all_available_capacity

Expand the storage pool to all capacity available in the device group

--new_name <NAME>

New name to be assigned to the storage pool

--force_insufficient_spare_or_free_space

To force reduce allocation below 120% of in-use space

--over_provisioning_factor <PERCENTAGE>

Over provisioning factor. Value treated as percentage. Zero is a special value that means no limit. Valid range: 100-10000

--compression_method <METHOD>

Indicates the compression method to be used. Can be one of the following:

- none - No compression applied
- normal - Normal compression applied

query_storage_pool

Retrieve aggregated information about all objects in the specified storage pool.

Syntax

```
scli --query_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --query_storage_pool --storage_pool_name sampleSP --protection_domain_name samplePD
```

Sample output

```
Storage Pool sampleSP (Id: 9520a65800000000) has 1 volumes, 0 clones, 0 snapshots.
        Logical Provisioning:
                1500% Overprovisioning Factor
```

```
                5.6 TB (5760 GB) total volume logical provisioning limit
                4.6 TB (4735 GB) remaining volume logical provisioning
                1.0 TB (1025 GB) volume logical provisioned (17.8% of total)
        Data layout: Erasure Coding
        Compression method: Normal
        Device Group id:805e6ea900000000
        WrC Device Group id:805e6eaa00000001
        Data protection scheme: 2+2
        SP health state: PROTECTED
        Capacity alert thresholds: High: 80, Critical: 90
Physical capacity:
        384.0 GB (393216 MB) total
        0 Bytes used
        381.5 GB (390656 MB) free
Logical capacity:
        1.0 TB (1025 GB) provisioned
        0 Bytes logical Used
        0 Bytes logical (owned)

Bandwidth:
        total WrC read bandwidth                    0 IOPS 0 Bytes per second
        total WrC write bandwidth                   0 IOPS 0 Bytes per second
        host read bandwidth                         0 IOPS 0 Bytes per second
        host write bandwidth                        0 IOPS 0 Bytes per second
        host trim bandwidth                         0 IOPS 0 Bytes per second
        storage fe read bandwidth                   0 IOPS 0 Bytes per second
        storage fe write bandwidth                  0 IOPS 0 Bytes per second
        storage fe trim bandwidth                   0 IOPS 0 Bytes per second
```

remove_storage_pool

Remove a storage pool from a protection domain.

NOTE: Each protection domain must contain at least one storage pool, which can be either the default or a user-defined storage pool. You can only remove all storage pools from a protection domain if they are empty and do not contain any devices.

Syntax

```
scli --remove_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--storage_pool_name *<NAME>*
: Storage pool name

--storage_pool_id *<ID>*
: Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --remove_storage_pool
--protection_domain_name rack_1.1
--storage_pool_name sp_1.1_rack_1.1
```

rename_storage_pool

Name, or rename, a storage pool.

Syntax

```
scli --rename_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--new_name *<NAME>*
New name to associate with the storage pool

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --rename_storage_pool --protection_domain_name rack_1.1 --
storage_pool_name sp_1.1_rack_1.1 --new_name sp_2.2_rack_1.1
```

Storage pool names

Assign the storage pool a meaningful name in order to facilitate future object identification. The name should conform to the following rules:

1. Contains fewer than 32 characters.
2. Contains only alphanumeric and punctuation characters.
3. Is unique within the object type.

set_capacity_alerts_threshold

Set the threshold for triggering capacity usage alerts.

Alerts thresholds are calculated from each storage pool capacity after deducting the defined amount of spare capacity. You can set thresholds for a single storage pool or for all storage pools.

Syntax

```
scli --set_capacity_alerts_threshold --capacity_high_threshold <PERCENT> --
capacity_critical_threshold <PERCENT> (--all_storage_pools | --system_default | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
|--storage_pool_id <ID>))
```

Parameters

--capacity_low_threshold *<PERCENT>*

Threshold in percent of capacity of the storage pool that triggers a low priority alert (EC only). Setting a zero threshold for a storage pool resets it to use the global threshold.

--capacity_high_threshold *<PERCENT>*

Threshold in percent of capacity of the storage pool that triggers a high-priority alert, expressed as a percentage. Setting a zero threshold for a storage pool resets it to use the global threshold.

--capacity_critical_threshold *<PERCENT>*

Threshold in percent of capacity of the storage pool that triggers a critical-priority alert, expressed as a percentage. Setting a zero threshold for a storage pool resets it to use the global threshold.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--all_storage_pools

Thresholds apply to all storage pools. (mirroring only)

--system_default

Thresholds apply to newly created storage pools

Example

```
scli --mdm_ip 192.168.1.200 --set_capacity_alerts_threshold --system_default --
capacity_high_threshold 80 --capacity_critical_threshold 90
```

31

System commands

This section contains commands for managing the system.

abort_upgrade

Stop an upgrade that is in process.

Syntax

```
scli --abort_upgrade
```

Example

```
scli --abort_upgrade
```

add_certificate

Add a certificate authority (CA) signed certificate to the trusted certificates list of the CLI. CLI automatically trusts all applied certificates that the CA has signed.

Syntax

```
scli ---add_certificate --certificate_file <FILE>
```

Parameters

--certificate_file *<FILE>*

File name of the root or trusted certificate

Example

```
scli --add_certificate --certificate_file CAcert-cacert.pem
```

add_standby_mdm

Add a standby MDM to the system. This MDM can later be added to the cluster.

Syntax

```
scli --add_standby_mdm --mdm_role {manager    | tb} --new_mdm_ip <IP>
  [--new_mdm_port <PORT>]
  [--new_mdm_management_ip <IP>]
  [--new_mdm_virtual_ip_interface <INTF>]
  [--new_mdm_name <NAME>]
  [--allow_asymmetric_ips]
  [--force_clean]
  [--i_am_sure]
```

Designate the standby MDM's role, either manager or tiebreaker (default). You can give the standby a unique name.

Parameters

--mdm_role {manager | tb}

Designates whether the MDM role is a manager (primary MDM of the cluster) or will only serve as a tiebreaker in the cluster primary MDM election process

--new_mdm_ip *<IP>*

Comma-separated list of up to four IP addresses assigned to the new MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_port *<PORT>*

Port of the new MDM. Default: 9011.

--new_mdm_management_ip *<IP>*

Comma-separated list of up to four IP addresses used to manage the MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of virtual IP addresses assigned to the cluster. Omit the space after each comma.

--new_mdm_name *<NAME>*

Name of the new MDM

--allow_asymmetric_ips

Allow the added node to have a different number of IP addresses from the primary node.

--force_clean

Clean the previous configuration of the MDM.

--i_am_sure

Skip the safety questions for command execution, and give preemptive approval.

Example

```
scli --add_standby_mdm --mdm_role manager --new_mdm_ip 192.168.1.153 --
new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth4
```

allow_commands_during_upgrade

Allows cluster commands to be sent during an upgrade for this session.

Syntax

```
scli --allow_commands_during_upgrade
```

Example

```
scli --allow_commands_during_upgrade
```

create_mdm_cluster

Create an MDM cluster from the selected primary MDM.

This command does not require the user to be logged in before running it. When the command is executed, the MDM cluster operates in single mode.

Syntax

```
scli --create_mdm_cluster --primary_mdm_ip <IP>
  [--primary_mdm_management_ip <IP>]
  [--cluster_virtual_ip <IP>]
  [--primary_mdm_virtual_ip_interface <INTF>]
  [--primary_mdm_name <NAME>]
  [--accept_license]
  [--disable_client_secure_communication]
  [--approve_certificate_once]
  [--approve_certificate]
```

Parameters

--primary_mdm_ip*<IP>*

Comma- separated list of IP addresses assigned to the primary MDM, to be used for MDM cluster internal control communications. Omit the space after each comma.

--primary_mdm_management_ip *<IP>*

Comma-separated list of IP addresses, used to manage the MDM. Omit the space after each comma.

--cluster_virtual_ip *<IP>*

Comma-separated list of virtual IP addresses to be used for the cluster. Omit the space after each comma.

--primary_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of the virtual IP addresses assigned to the cluster. Omit the space after each comma.

--primary_mdm_name*<NAME>*

MDM name. Each MDM name must be unique. Ensure that each server on which the MDM is installed has a unique hostname.

--accept_license

Accept the license agreement.

--disable_client_secure_communication

Create the cluster without management client secure communications.

--approve_certificate_once

One-time approval of the MDM's certificate (without adding the certificate to the truststore).

--approve_certificate

Automatic approval of the MDM's certificate for the next commands (adds the certificate to the truststore).

Example

```
scli --create_mdm_cluster --primary_mdm_ip 192.168.1.152 --cluster_virtual_ip
192.168.100.152 --primary_mdm_virtual_ip_interface eth4
```

finalize_upgrade

Finalize the upgrade process.

Syntax

```
scli --finalize_upgrade
```

Example

```
scli --finalize_upgrade
```

generate_certificate

Generate a new certificate and private key for the SDT, or a storage node. The system signs the certificate and key.

Syntax

```
scli --generate_certificate ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) | (--storage_node_id <ID> | --storage_node_name <NAME> | --storage_node_ip <IP>
[--storage_node_port <PORT>]))
```

Parameters

--sds_id *<ID>*

Storage node ID

--sds_name *<NAME>*

Storage node name

--sds_ip *<IP>*

Storage node IP address

--sds_port *<PORT>*

Port assigned to the storage node

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--storage_node_name *<NAME>*

Storage node name

--storage_node_id *<ID>*

Storage node ID

--storage_node_ip *<IP>*

Storage node IP address

--storage_node_port *<PORT>*

Port assigned to the storage node

generate_mdm_certificate

Create a new self-signed certificate and private key for the specfied MDM.

NOTE: Running this command on the primary MDM causes the MDM to restart. This may cause the MDM to switch ownership.

Syntax

```
scli --generate_mdm_certificate (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*

ID of the MDM for which the certificate and private key are required

--target_mdm_ip *<IP>*

IP address of the MDM for which the certificate and private key are required

--target_mdm_port *<PORT>*

Port of the MDM for which the certificate and private key are required

--target_mdm_name *<NAME>*

Name of the MDM for which the certificate and private key are required

Example

```
scli --generate_mdm_certificate --target_mdm_ip 198.168.1.2 --target_mdm_port 9011
```

generate_mdm_csr_file

Create a certificate signing request (CSR) file for the specified MDM.

The file is used to obtain security certificates from a certificate authority and is saved in the specified MDM directory:

- Linux: `/opt/emc/scaleio/mdm/cfg`
- Windows: `C:\Program Files\emc\scaleio\mdm\cfg`

Syntax

```
scli --generate_mdm_csr_file (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
  [OPTIONS]
```

Parameters

`--target_mdm_id <ID>`

ID of the MDM for which the certificate and private key are required

`--target_mdm_ip <IP>`

IP address of the MDM for which the certificate and private key are required

`--target_mdm_port <PORT>`

Port of the MDM for which the certificate and private key are required

`--target_mdm_name <NAME>`

Name of the MDM for which the certificate and private key are required

`--common_name <NAME>`

Common name to appear in the CSR. The default is the host name of the machine.

Options

NOTE: If you use any of these options, the default values will be disabled for all of them.

`--country <NAME>`

Country to appear in the CSR

`--state <NAME>`

State to appear in the CSR

`--location <NAME>`

Location to appear in the CSR

`--organization <NAME>`

Organization to appear in the CSR

`--organizational_unit <NAME>`

Organizational unit to appear in the CSR

`--email_address <EMAIL>`

Email address to appear in the CSR

Example

```
scli --generate_mdm_csr_file --target_mdm_ip 192.168.1.2 --target_mdm_port 9011 --
country australia --state victoria --location melbourne --organization example_pty_ltd
--organizational_unit sales --email_address sales@example.com.au
```

modify_cluster_virtual_ips

Set or modify the virtual IP address of an MDM cluster.

NOTE: After changing or adding a virtual IP address, you are required to update the SDC with the new virtual IP address.

Notes

For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM (when the MDM is installed on a Linux server), disable the IPv6 DAD setting on the server's relevant interfaces, using the command: `sysctl net.ipv6.conf.<interface_name>.dad_transmits=0`

Syntax

```
scli --modify_cluster_virtual_ips (--cluster_virtual_ip <IP> | --clear_all)
```

Parameters

--cluster_virtual_ip *<IP>*

Comma-separated list of IP addresses to be used for the cluster. Omit the space after each comma.

--clear_all

Clear all virtual IP addresses.

Example

```
scli --modify_cluster_virtual_ips --cluster_virtual_ip 192.168.100.152,192.168.100.153
```

modify_management_ip

Modify MDM node management IP addresses.

If you are using this command to add additional management IP addresses, you must enter all the existing IP addresses in the command, and add the new ones at the end. Up to four IP addresses are supported.

Syntax

```
scli --modify_management_ip (--new_mdm_management_ip <IP> | --clear_all) (--
target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --target_mdm_name
<NAME>)
  [--allow_duplicate_management_ips]
  [--i_am_sure]
```

Parameters

--new_mdm_management_ip *<IP>*

Comma-separated list of IP address to be used to manage the MDM. Omit the space after each comma.

--clear_all

Clear all management IP addresses.

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

--allow_duplicate_management_ips

Allow duplicate management IP addresses.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --modify_management_ip --target_mdm_id 0x34e8f1df6c84a410 --new_mdm_management_ip
fd00::192:168:1:17,10.103.110.17
```

Implementing the changes

After modifying MDM IP addresses, ensure that you implement the change in all components and interfaces that are mapped to the MDM:

1. PowerFlex Manager: In the **Login** window, log in using the new IP address, then clear the old IP address from the history if it is now obsolete.
2. REST Gateway and OpenStack: In the REST Gateway, modify the mapping to the new management IP address.
3. On each SDC, perform the following locally, in the command line with root access permissions:
   - If for any reason, the management IP address is not updated in the SDC, run:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
     ```
   - If the SDC is mapped to more than one PowerFlex system, run:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
     ```
4. vSphere: Using the vSphere web plug-in interface, unregister PowerFlex and reregister it with the new IP address.

modify_virtual_ip_interfaces

Set the virtual IP interfaces of a node in the MDM cluster.

Syntax

```
scli --modify_virtual_ip_interfaces (--new_mdm_virtual_ip_interface <INTF> | --
clear_all) (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --
target_mdm_name <NAME>)
```

Parameters

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP interfaces. Omit the space after each comma. The order of interfaces must match the order of virtual IPs assigned to the cluster.

--clear_all

Clear all virtual IP interfaces.

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

Example

```
scli --modify_virtual_ip_interfaces --target_mdm_ip 192.168.1.152 --
new_mdm_virtual_ip_interface eth4
```

query_all

Retrieve general information about the system.

Syntax

```
scli --query_all
```

Example

```
scli --mdm_ip 192.168.1.200 --query_all
```

Sample output

```
System Info:
        Product: DellEMC PowerFlex Version: R5_0.0.891
        ID: 2b8275a55ee1a80f
        Manager ID: 0000000000000000
        Name: sys_1
        NQN: nqn.1988-11.com.dell:powerflex:00:2b8275a55ee1a80f
License info:
        Installation ID: 019dc78b569ac58b
        SWID:
        Maximum capacity: Unlimited
        Usage time left: 90 days *** Non-Production License ***
        Enterprise features: Enabled
        The system was activated 0 days ago
System settings:
        Capacity alert thresholds: High: 80, Critical: 90
        MDM restricted SDC mode: NONE
        Management Clients secure communication: Enabled
        TLS version: TLSv1.2
        CLI preemptive login banner acceptance via shell: Enabled
        CLI login banner acceptance by password: Enabled
        User authentication method: M&O
```

```
        MDM connection authentication: Disabled
        SDC refresh with MDM IP addresses: disabled
        Restart connection to realign network path: Disabled
SDC-PDS connectivity status: All connected
Raw capacity:
        125.8 TB (128771 GB) total capacity
                115.1 TB (117900 GB) used capacity
                10.6 TB (10870 GB) system capacity
                738.0 MB (755712 KB) free capacity
Physical capacity:
        92.1 TB (94320 GB) total capacity
                79.4 TB (81304 GB) used capacity
                1.6 TB (1618 GB) system capacity
                11.1 TB (11397 GB) free capacity
Capacity Health:
                114.9 TB (117608 GB) protected capacity
                0 Bytes failed capacity
                0 Bytes degraded capacity
                0 Bytes degraded critical capacity
Logical capacity:
        90.8 TB (92995 GB) logical provisioned
        78.2 TB (80031 GB) logical used
        79.0 TB (80849 GB) unreducible data
        Ratio counters:
                1.1:1 efficiency ratio
                1.0:1 reducible ratio
                1.0:1 data reduction ratio
                0.0:1 data reduction reducible ratio
                1.0:1 compression ratio
                0.0:1 compression reducible ratio
                1.0:1 patterns saving ratio
                Patterns saving reducible ratio: Not applicable
                1.0:1 snapshot saving ratio
                1.2:1 thin provisioning ratio
Data Movement:
        0 Bytes raw rebuild
        0 Bytes raw rebalance planned
        0 Bytes raw rebuild planned
Bandwidth:
        total device Pmem read bandwidth         0 IOPS 0 Bytes per second
        total device Pmem write bandwidth        0 IOPS 0 Bytes per second
        total WrC read bandwidth                 0 IOPS 0 Bytes per second
        total WrC write bandwidth                0 IOPS 0 Bytes per second
        total device read bandwidth              11771 IOPS 587.2 MB (601244 KB)
per-second
        total device write bandwidth             50 IOPS 897.6 KB (919142 Bytes)
per-second
        host read bandwidth                      0 IOPS 0 Bytes per second
        host write bandwidth                     0 IOPS 0 Bytes per second
        host trim bandwidth                      0 IOPS 0 Bytes per second
        storage fe read bandwidth                0 IOPS 0 Bytes per second
        storage fe write bandwidth               0 IOPS 0 Bytes per second
        storage fe trim bandwidth                0 IOPS 0 Bytes per second
Query all returned 1 Protection Domain:
Protection Domain pd1 , (Id: eeab748300000000), Type: EC, has 1 Device Groups, 1 storage
pools, 12 Storage Nodes, 12 PDSs, 12 DGWTs, and 773 volumes
        Raw free capacity: 738.0 MB (755712 KB), remaining volume logical provisioning:
4.0 EB (4095 PB), total volume logical provisioning limit: 4.0 EB (4095 PB)
        Operational state is Active
Maximum IO Limits:
        Overall concurrent IO limit                          : 4 per device.
        Bandwidth limit overall IOs                          : 400 MB per device.
        Bandwidth limit background device scanner            : 10 MB per device.
        Bandwidth limit garbage collector                    : No limit - aligning
with the general secondary IO limit.
        Bandwidth limit degraded                             : 400 MB per device.
        Bandwidth limit critically degraded                  : 400 MB per device.
        Bandwidth limit rebalance                            : 40 MB per device.
        Bandwidth limit of all other secondary IOs           : 10 MB per device.
        Bandwidth limit of Node Network                      : 25 Gb per Node.
Secondary I/O policy: favor_application
Rebuild Enabled: TRUE
Rebalance Enabled: TRUE
```

```
Storage Pool pd1_sp1 (Id: 8562fa1b00000000) has 773 volumes, 0 clones, 0 snapshots.
        Logical Provisioning:
                >10000% Overprovisioning Factor
                4.0 EB (4095 PB) total volume logical provisioning limit
                4.0 EB (4095 PB) remaining volume logical provisioning
                90.8 TB (92995 GB) volume logical provisioned (0.00% of total)
        Data layout: Erasure Coding
        Compression method: Normal
        Device Group id:4b7e114e00000000
        WrC Device Group id:4b7e114e00000000
        Data protection scheme: 8+2
        SP health state: PROTECTED
        Capacity alert thresholds: High: 80, Critical: 90
        Physical capacity:
                92.1 TB (94320 GB) total capacity
                        79.4 TB (81304 GB) used capacity
                        11.1 TB (11397 GB) free capacity
                        1.6 TB (1618 GB) system capacity
Device Group 4b7e114e00000000 Name: dg_YQtxBf
        Media Type: SSD
        Protection Domain id:eeab748300000000
        Raw Capacity:
                Total                                   : 125.8 TB (128771 GB)
                Used                                    : 115.1 TB (117900 GB)
                Free                                    : 738.0 MB (755712 KB)
                Rebuild used                            : 0 Bytes
                Spare                                   : 10.6 TB (10835 GB)
                Spare Node Count:                       : 1
                Spare Device Count:                     : 0
        Capacity Health State: Protected
        Node Quantity: 12
        Device Quantity: 72
        Device HW Checks Enabled: TRUE
        Physical Capacity:
                Total                                   : 92.1 TB (94320 GB)
                Used                                    : 79.4 TB (81304 GB)
                Free                                    : 11.1 TB (11397 GB)
                System                                  : 1.6 TB (1618 GB)
        Rebuild and Rebalance:
                Rebuild Required:                       : FALSE
                Rebuild Is Running:                     : FALSE
                Rebalance Is Running:                   : FALSE
        Raw Capacity Health:
                Raw health failed                       : 0 Bytes
                Raw health degraded                     : 0 Bytes
                Raw health degraded critical            : 0 Bytes
        Bandwidth:
                Backward-rebuild-reads                  : 0
                Backward-rebuild-writes                 : 0
                Forward-rebuild-reads                   : 0
                Forward-rebuild-writes                  : 0
                Rebalance-reads                         : 0
                Rebalance-writes                        : 0
Volumes summary:
        773 volumes. Total logical used size: 78.2 TB (80031 GB)
        90.8 TB (92995 GB) all volumes size
        0 snapshot policiesSDT Summary:
        Total 0 SDT NodesHost Summary:
        Total 12 SDC Hosts
        Total 0 NVMe Hosts
```

Output description

The following tables explain the output for `query_all`.

Table 13. System information and description

System information	Description
Product	The PowerFlex version of the system.

Table 13. System information and description (continued)

System information	Description
ID	The system ID number.
Manager ID	The ID of the management application that controls the MDM.

Table 14. System settings

System settings	Description
Volumes are/are not obfuscated	The obfuscation setting for all volumes in the system. Default: not obfuscated.
Capacity alert thresholds	The alert thresholds, as percentages, currently configured for capacity usage.
Thick volume reservation	The amount of capacity, as a percentage, reserved for thick volumes.
MDM restricted SDC mode	Indicates whether users on remote clients are restricted to read-only access: ● enabled: read-only access ● disabled: full read-write access
TLS version	N/A
User authentication method	N/A
Storage nodes connection authentication	The status of the ..... : enabled/disabled

Table 15. Protection domain details

Protection domain	Description
Protection domain summary	The following items are displayed separately for each protection domain: ● Name ● ID ● Number of storage pools in the protection domain ● Number of storage nodes in the protection domain ● Number of volumes associated with the protection domain ● Amount of capacity in the protection domain that is available for volume allocation
Operational state	Indicates the protection domain's current state, such as active, inactive, and so on.

Table 16. Storage pool details

Storage pool	Description
Storage pool summary	Lists the following information for each storage pool: ● Name ● ID ● Number of volumes associated with the storage pool ● Amount of capacity in the storage pool that is available for volume allocation
The number of parallel rebuild/rebalance jobs	The number of parallel rebuild and rebalance jobs currently existing in the system
Rebuild is enabled/disabled	Indicates whether the rebuild feature is enabled or disabled. (In normal production conditions, rebuild should be enabled.) When enabled, the rebuild policy configured in the system is displayed, along with the relevant parameters.

Table 16. Storage pool details (continued)

Storage pool	Description
Rebalance is enabled/disabled	Indicates whether the rebalance feature is enabled or disabled. (In normal production conditions, rebalance should be enabled.) When enabled, the rebalance policy configured in the system is displayed, along with the relevant parameters.
Background device scanner	Displays the mode and the bandwidth limit per device.
Zero padding is enabled/disabled	Indicates whether zero padding is enabled or disabled.
Spare policy	The percentage of total capacity reserved as spare.
Read RAM cache	Indicates whether the read RAM cache feature is used in the storage pool. If the read RAM cache is used, it displays the write handling mode currently in use (passthrough or cached).
Capacity alert thresholds	The alert thresholds, as percentages, configured for capacity usage. Thresholds may be configured for the entire system, or per storage pool.

Table 17. Storage node details

Storage nodes summary	Description
Total	The total number of storage nodes.
n storage nodes have membership state 'Joined'	The number of storage nodes that are connected to the cluster, can receive I/O, and hold primary and secondary copies of data. It takes some time from actual disconnection until the storage node is disconnected from the cluster, or the reverse. There are also up-pending and down-pending states.
n storage nodes have connection state 'Connected'	The number of storage nodes currently connected to the MDM.
x TB (*y* GB) total capacity	The total amount of available raw storage. This does not represent the total capacity available for volume allocation.
x TB (*y* GB) unused capacity	The quantity of raw capacity in the system that can be earmarked for specific purposes, such as Spare, or used for new volume creation.
x Bytes snapshots capacity	The quantity of capacity used for storing snapshots.
x TB (*y* GB) in-use capacity	The total quantity of healthy, degraded, and failed capacity.
x Bytes thin capacity	The quantity of capacity currently needed for storage purposes.
x TB (*y* GB) protected capacity	The quantity of capacity that is fully protected (primary and secondary copies of the data exist).
x Bytes failed capacity	The quantity of capacity that is not available at all (neither primary, nor secondary copies).
x Bytes degraded-failed capacity	The quantity of degraded-failed capacity. When a storage node fails, all of its capacity is defined as degraded-failed, and the secondary copies (which are spread across the protection domain) are defined as degraded-healthy.
x Bytes degraded-healthy capacity	The quantity of degraded-healthy capacity. When a storage node fails, all of its capacity is defined as degraded-failed, and the secondary copies (which are spread across the protection domain) are defined as degraded-healthy.

Table 17. Storage node details (continued)

Storage nodes summary	Description
x Bytes unreachable-unused capacity	The quantity of capacity not configured for any use type in the system that is currently unavailable.
x Bytes active rebalance capacity	The quantity of capacity that is currently being migrated to a different location for load balancing purposes.
x Bytes pending rebalance capacity	The quantity of capacity that is waiting in the job queue for migration to a different location for load balancing purposes.
x Bytes active forward-rebuild capacity	The quantity of capacity for which one copy of data exists and a second copy is currently being created.
x Bytes pending forward-rebuild capacity	The quantity of capacity for which one copy of data exists, and a job for the creation of a second copy is waiting in the job queue.
x Bytes active backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are currently being synchronized in that copy.
x Bytes pending backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are waiting in the job queue to be synchronized in that copy.
x Bytes rebalance capacity	The total quantity of capacity that is either currently rebalancing or is pending rebalance.
x Bytes forward-rebuild capacity	The total quantity of degraded capacity that is either currently in forward rebuild state or is pending forward rebuild.
x Bytes backward-rebuild capacity	The total quantity of degraded capacity that is either currently in backward rebuild state or is pending backward rebuild.
x Bytes active moving capacity	The quantity of capacity that is currently being migrated from one location to another.
x Bytes pending moving capacity	The quantity of capacity that is waiting in the job queue for migration from one location to another.
x Bytes total moving capacity	The total quantity of active and pending moving (migrating) capacity.
x TB (*y* GB) spare capacity	The quantity of capacity that is reserved for system use when recovery from failure is required. This capacity cannot be used for storage purposes.
x TB (*y* GB) at-rest capacity	The quantity of capacity that is fully protected and not in a rebuild or rebalance state.
x Bytes decreased capacity	The quantity of decreased capacity that was deducted from devices (using the **Set Device Capacity Limit** GUI command or `modify_sds_device_capacity` CLI command) and cannot be used for any purpose
Primary-reads *x* IOPS *y* bytes per second	Number of primary data copy read IOPS, and read bandwidth.
Primary-writes *x* IOPS *y* bytes per second	Number of primary data copy write IOPS, and write bandwidth.
Secondary-reads *x* IOPS *y* bytes per second	Number of secondary data copy (protection) read IOPS, and read bandwidth.
Secondary-writes *x* IOPS *y* bytes per second	Number of secondary copy (protection) write IOPS, and write bandwidth.
Backward-rebuild-reads *x* IOPS *y* bytes per second	Number of backward rebuild read IOPS, and read bandwidth.
Backward-rebuild-writes *x* IOPS *y* bytes per second	Number of backward rebuild write IOPS, and write bandwidth.

Table 17. Storage node details (continued)

Storage nodes summary	Description
Forward-rebuild-reads *x* IOPS *y* bytes per second	Number of forward rebuild read IOPS, and read bandwidth.
Forward-rebuild-writes *x* IOPS *y* bytes per second	Number of forward rebuild write IOPS, and write bandwidth.
Rebalance-reads *x* IOPS *y* bytes per second	Number of rebalance read IOPS, and read bandwidth.
Rebalance-writes *x* IOPS *y* bytes per second	Number of rebalance write IOPS, and write bandwidth.

Table 18. Volumes summary

Volumes summary	Description
n thick-provisioned volumes	The number of thick-provisioned volumes, and the total size in TB and GB.
n volumes mapped to all SDC nodes	The number of volumes mapped to SDC nodes.

query_cluster

Retrieve MDM cluster information.

Syntax

```
scli --query_cluster [--show_certificate_info]
```

Parameters

--show_certificate_info
Display certificates for all the MDMs

Example

```
scli --mdm_ip 192.168.1.200 --query_cluster --show_certificate_info
```

Sample output

The following is sample output for a five-node cluster:

```
scli --query_cluster
Cluster:
    Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
    Virtual IP Addresses: N/A
Primary MDM:
    Name: MDM_116, ID: 0x155c8db3025d5f01
        IP Addresses: 10.55.137.116, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Issuer:     /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: 2E:68:A4:48:A3:BC:DF:77:D9:8B:1C:77:01:99:04:5B:1D:F9:77:B6
Secondary MDMs:
    Name: MDM_114, ID: 0x75c68cb264723f00
```

```
        IP Addresses: 10.55.137.114, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Issuer:     /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: F7:B6:37:3C:45:EF:98:5B:64:57:5E:D3:7D:A1:B7:AD:E4:7B:5A:EA
    Name: MDM_115, ID: 0x7386c66b551fdf02
        IP Addresses: 10.55.137.115, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Issuer:     /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Valid From: Jan 30 09:34:18 2021 GMT
            Valid To:   Jan 29 10:34:18 2031 GMT
            Thumbprint: A8:FB:1E:A2:6C:92:32:6C:42:FB:A0:A3:50:1D:6F:90:06:E2:1B:A0
```

Possible values for items in the sample output

Table 19. Possible values for items in sample output

Item	Possible values
Mode	Cluster mode: • 1_node • 3_node • 5_node
State	State of the cluster: • Normal • Degraded (one or more cluster members are down or inactive)
Active	The number of cluster members that are active. If all members are in normal state, this number equals the number of cluster members.
Replicas	The number of MDM repository holders (the primary MDM, plus the number of secondary MDMs in normal state)
Secondary MDM status	• Normal • Error • Disconnected • Not synchronized
Tiebreaker status	• Normal • Error • Disconnected
Standby MDM	Role (manager or tiebreaker)

query_performance_parameters

Retrieve performance parameters. The output displays the configurations of performance-related parameters for the performance profile that is active. When no parameters are added to the command, the name of the active profile is shown.

Syntax

```
scli --query_performance_parameters [--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>] | --all_sds] [--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID>
| --sdc_ip <IP> | --all_sdc] [--peer_system_id <ID> | --peer_system_name <NAME> |
```

```
--peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID> | --
all_peer_system] [ --all_storage_node] [--print_all]
```

Parameters

--sds_id *<ID>*
Storage node ID.

--sds_name *<NAME>*
Storage node name.

--sds_ip *<IP>*
Storage node IP address.

--sds_port *<PORT>*
Port associated with the storage node.

--all_sds
Query storage node performance parameters from all storage nodes.

--sdc_id *<ID>*
SDC ID.

--sdc_name *<NAME>*
SDC name.

--sdc_guid *<GUID>*
SDC global unique identifier.

--sdc_ip *<IP>*
SDC IP address.

--all_sdc
Query SDC performance parameters from all SDCs.

--storage_node_name <NAME>
Storage node name.

--storage_node_id <ID>
Storage node ID.

--storage_node_ip <IP>
Storage node IP address.

--storage_node_port <PORT>
Port assigned to the storage node.

--all_storage_node
Query storage node performance parameters from all storage nodes.

--print_all
Print all performance parameters, including those with active profile default values.

Example

```
scli --query_performance_parameters --print_all
```

query_properties

Retrieve any set of properties on any set of objects of the specified type.

The properties that can be queried depend on the type of the object that is being queried. For example, a storage node has different attributes than a volume.

Several preset combinations are available for use with this command. A preset is a predefined set of properties, usually with some logical relationship, which specifies a group of commonly queried properties. The available presets are also dependent on the object type. All object types have at least one preset called `all`, which queries all the properties for that type.

Syntax

```
scli --query_properties --object_type <TYPE> (--object_id <ID> | --all_objects) (--
properties <PROPS> | --preset <PRESET>)
  [--group_by_property]
  [--display_property_options]
  [--flat_list]
```

Parameters

--object_type *<TYPE>*

Object type to query. One of:

SYSTEM

PROTECTION_DOMAIN

STORAGE_POOL

STORAGE NODE

HOST

VOLUME

DEVICE

VTREE

MDM

SNAPSHOT_POLICY

PEER_MDM

SDT

REMOTE_SYSTEM

SNAPSHOT_COPY_TASK

PDS

DEVICE_GROUP

NODE

DGWT

NOTE: To view detailed presets and properties options for a type, enter `scli --query_properties --object_type <TYPE>`, where <*TYPE*> is one of the above options.

--object_id *<ID>*

Comma-separated list of object ID used for querying one or more specific objects. Omit the space after each comma.

--all_objects

Query all objects of the specified type.

--properties *<PROPS>*

Query a comma-separated list of properties. Omit the space after each comma. See the scli help output for a list of properties that can be queried per object type

--preset *<PRESET>*

Query a predefined set of properties. Omit the space after each comma. See the scli help output for the list of presets that can be used per object type.

--group_by_property

Group results by property type, rather than by object (the default).

--display_property_options

Display all available properties grouped by object type.

--flat_list

Print results as a flat list rather than by group (better for searching).

NOTE: Use `scli --query_properties --object_type <TYPE>` with one of the valid types above, to get detailed presets and properties options.

Example

```
scli2 --query_properties --object_type NODE --all_objects --properties
ID,NAME,PROTECTION_DOMAIN_ID,DEVICE_ID_LIST,ACCUMULATED_TOTAL_READ_BWC,ACCUMULATED_TOTAL_
WRITE_BWC
```

Sample output

```
NODE 2db4f15000000000:
        ID                                                  2db4f15000000000
        NAME                                                node_0
        PROTECTION_DOMAIN_ID                                90dbe59a00000001
        DEVICE_ID_LIST
fff7975400000000,fff7975500000001,fff7975600000002,fff7975700000003
        ACCUMULATED_TOTAL_READ_BWC                          0 IOs, 0 Bytes
        ACCUMULATED_TOTAL_WRITE_BWC                         256 IOs, 16.7 MB (17088 KB)

NODE 2db4f15100000001:
        ID                                                  2db4f15100000001
        NAME                                                node_1
        PROTECTION_DOMAIN_ID                                90dbe59a00000001
        DEVICE_ID_LIST
fff7975800010000,fff7975900010001,fff7975a00010002,fff7975b00010003
        ACCUMULATED_TOTAL_READ_BWC                          0 IOs, 0 Bytes
        ACCUMULATED_TOTAL_WRITE_BWC                         256 IOs, 16.7 MB (17088 KB)

NODE 2db4f15200000002:
        ID                                                  2db4f15200000002
        NAME                                                node_2
        PROTECTION_DOMAIN_ID                                90dbe59a00000001
        DEVICE_ID_LIST
fff7975c00020000,fff7975d00020001,fff7975e00020002,fff7975f00020003
        ACCUMULATED_TOTAL_READ_BWC                          0 IOs, 0 Bytes
        ACCUMULATED_TOTAL_WRITE_BWC                         256 IOs, 16.7 MB (17088 KB)
```

Object presets

The following table provides a list of presets that can be used per object type.

Table 20. Object type and presets

Object type	ALL	CAPACITY	RPL_JOURNAL_CAPACITY	RPL_BW_COUNTERS	I/O	RAM_CACHE	PERFORMANCE_PARAMETERS
SYSTEM	Yes	Yes	No	No	Yes	Yes	Yes
PROTECTION_DOMAIN	Yes	Yes	No	No	Yes	Yes	No
STORAGE_POOL	Yes	Yes	Yes	No	Yes	Yes	Yes

Table 20. Object type and presets (continued)

Object type	ALL	CAPACITY	RPL_JOURNAL_CAPACITY	RPL_BW_COUNTERS	I/O	RAM_CACHE	PERFORMANCE_PARAMETERS
STORAGE NODE	Yes	Yes	No	No	Yes	Yes	Yes
SDC	Yes	No	No	No	Yes	No	Yes
VOLUME	Yes	No	No	No	Yes	Yes	No
DEVICE	Yes	Yes	No	No	Yes	No	No
VTREE	Yes	Yes	No	No	No	No	No
MDM	Yes	No	No	No	No	No	No
SP_STORAGE NODE	Yes	Yes	No	No	Yes	No	No
SNAPSHOT_POLICY	Yes	No	No	No	No	No	No
PEER_MDM	Yes	No	No	No	No	No	Yes

Object properties

TheNofollowingNotableNoprovidesNoaNolistNoofNopropertiesNothatNocanNobeNoqueriedNoperNoobjectNotypeNoNoNo

NOTE: Sys=System, PD=Protection domain, SP=Storage pool, Vol=Volume,NoDev=DeviceNo

Table 21. Object types and properties

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
CAPACITY_LIMIT_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
MAX_CAPACITY_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
CAPACITY_IN_USE_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
THICK_CAPACITY_IN_USE_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
THIN_CAPACITY_IN_USE_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
SNAP_CAPACITY_IN_USE_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
UNREACHABLE_UNUSED_CAPACITY_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
UNUSED_CAPACITY_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
SNAP_CAPACITY_IN_USE_OCCUPIED_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
THIN_CAPACITY_ALLOCATED_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
SPARE_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
AVAILABLE_FOR_THICK_ALLOCATION_IN_KB	Yes	Yes	Yes	No	No	No	No	No
PROTECTED_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
DEGRADED_HEALTHY_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No

Table 21. Object types and properties (continued)

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
DEGRADED_FAILED_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
FAILED_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
PROTECTED_VAC_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
DEGRADED_HEALTHY_VAC_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
DEGRADED_FAILED_VAC_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
FAILED_VAC_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
MOVING_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
ACTIVE_MOVING_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
PENDING_MOVING_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
FWD_REBUILD_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
ACTIVE_FWD_REBUILD_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
PENDING_FWD_REBUILD_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
BCK_REBUILD_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
ACTIVE_BCK_REBUILD_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
PENDING_BCK_REBUILD_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
REBALANCE_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
ACTIVE_REBALANCE_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
PENDING_REBALANCE_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
AT_REST_CAPACITY_IN_KB	Yes	Yes	Yes	No	No	No	No	No
ACTIVE_MOVING_IN_FWD_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
ACTIVE_MOVING_IN_BCK_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
ACTIVE_MOVING_IN_REBALANCE_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
ACTIVE_MOVING_OUT_FWD_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
ACTIVE_MOVING_OUT_BCK_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
ACTIVE_MOVING_OUT_REBALANCE_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
PENDING_MOVING_IN_FWD_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
PENDING_MOVING_IN_BCK_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
PENDING_MOVING_IN_REBALANCE_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No

Table 21. Object types and properties (continued)

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
PENDING_MOVING_OUT_FWD_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
PENDING_MOVING_OUT_BCK_REBUILD_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
PENDING_MOVING_OUT_REBALANCE_JOBS	Yes	Yes	Yes	Yes	No	No	Yes	No
IN_USE_VAC_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
PRIMARY_VAC_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
SECONDARY_VAC_IN_KB	Yes	Yes	Yes	Yes	No	No	Yes	No
REBUILD_WAIT_SEND_Q_LENGTH	Yes	Yes	No	Yes	No	No	No	No
REBALANCE_WAIT_SEND_Q_LENGTH	Yes	Yes	No	Yes	No	No	No	No
REBUILD_PER_RECEIVE_JOB_NET_THROTTLING_IN_KBPS	Yes	Yes	No	Yes	No	No	No	No
REBALANCE_PER_RECEIVE_JOB_NET_THROTTLING_IN_KBPS	Yes	Yes	No	Yes	No	No	No	No
FIXED_READ_ERROR_COUNT	Yes	Yes	Yes	Yes	No	No	Yes	No
PRIMARY_READ_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
PRIMARY_READ_FROM_DEV_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
PRIMARY_WRITE_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
SECONDARY_READ_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
SECONDARY_READ_FROM_DEV_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
SECONDARY_WRITE_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
FWD_REBUILD_READ_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
FWD_REBUILD_WRITE_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
BCK_REBUILD_READ_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
BCK_REBUILD_WRITE_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
REBALANCE_READ_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
REBALANCE_WRITE_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
TOTAL_READ_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
TOTAL_WRITE_BWC	Yes	Yes	Yes	Yes	No	No	Yes	No
USER_DATA_READ_BWC	Yes	Yes	Yes	No	Yes	Yes	No	No
USER_DATA_WRITE_BWC	Yes	Yes	Yes	No	Yes	Yes	No	No
RMCACHE_SIZE_IN_KB	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_SIZE_IN_USE_IN_KB	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_ENTRY_EVICTION_SIZE_COUNT_IN_KB	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_BIG_BLOCK_EVICTION_SIZE_COUNT_IN_KB	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_NUM_OF_4KB_ENTRIES	Yes	Yes	No	Yes	No	No	No	No

Table 21. Object types and properties (continued)

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
• RMCACHE_NUM_OF_8KB_ENTRIES-No • RMCACHE_NUM_OF_16KB_ENTRIES-No • RMCACHE_NUM_OF_32KB_ENTRIES-No • RMCACHE_NUM_OF_64KB_ENTRIES-No • RMCACHE_NUM_OF_128KB_ENTRIES-No								
RMCACHE_4KB_ENTRY_COUNT • RMCACHE_8KB_ENTRY_COUNT-No • RMCACHE_16KB_ENTRY_COUNT-No • RMCACHE_32KB_ENTRY_COUNT-No • RMCACHE_64KB_ENTRY_COUNT-No • RMCACHE_128KB_ENTRY_COUNT-No	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_ENTRY_EVICTION_COUNT	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_BIG_BLOCK_EVICTION_COUNT	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_NO_EVICTION_COUNT	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_SKIP_COUNT_LARGE_IO	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_SKIP_COUNT_UNALIGNED_4KB_IO	Yes	Yes	No	Yes	No	No	No	No
RMCACHE_SKIP_COUNT_CACHE_ALL_BUSY	Yes	Yes	No	Yes	No	No	No	No
NUM_OF_UNMAPPED_VOLUMES	Yes	Yes	Yes	No	No	No	No	No
NUM_OF_MAPPED_TO_ALL_VOLUMESNo	Yes	Yes	Yes	No	No	No	No	No
NUM_OF_THICK_BASE_VOLUMES	Yes	Yes	Yes	No	No	No	No	No
NUM_OF_THIN_BASE_VOLUMES	Yes	Yes	Yes	No	No	No	No	No
NUM_OF_SNAPSHOTSNo	Yes	Yes	Yes	No	No	No	No	No
NUM_OF_VOLUMES_IN_DELETION	Yes	Yes	Yes	No	No	No	No	No
NUM_OF_DEVICES	Yes	No	Yes	Yes	No	No	No	No
NUM_OF_SDS	Yes	Yes	No	No	No	No	No	No
NUM_OF_STORAGE_POOLS	Yes	Yes	No	No	No	No	No	No
NUM_OF_VOLUMES	Yes	No	Yes	No	No	No	No	Yes
NUM_OF_VTREES	Yes	No	Yes	No	No	No	No	No
PROTECTION_DOMAIN_ID_LIST	Yes	No	No	No	No	No	No	No
NUM_OF_PROTECTION_DOMAINS	Yes	No	No	No	No	No	No	No
SDC_ID_LIST	Yes	No	No	No	No	No	No	No

Table 21. Object types and properties (continued)

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
NUM_OF_SDC	Yes	No	No	No	No	No	No	No
ID	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
NAME	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
VERSION_NAME	Yes	No	No	No	No	No	No	No
DEFAULT_VOL_OBFUSCATION	Yes	No	No	No	No	No	No	No
CAPACITY_ALERT_HIGH_THRESHOLD	Yes	No	No	No	No	No	No	No
CAPACITY_ALERT_CRITICAL_THRESHOLD	Yes	No	No	No	No	No	No	No
INSTALL_ID	Yes	No	No	No	No	No	No	No
SW_ID	Yes	No	No	No	No	No	No	No
DAYS_INSTALLED	Yes	No	No	No	No	No	No	No
MAX_LICENSED_CAPACITY	Yes	No	No	No	No	No	No	No
CAPACITY_DAYS_LEFT	Yes	No	No	No	No	No	No	No
OBFUSCATION_DAYS_LEFT	Yes	No	No	No	No	No	No	No
SNAPSHOTS_DAYS_LEFT	Yes	No	No	No	No	No	No	No
QOS_DAYS_LEFT	Yes	No	No	No	No	No	No	No
INITIAL_LICENSE	Yes	No	No	No	No	No	No	No
THICK_VOLUME_PERCENT	Yes	No	No	No	No	No	No	No
MDM_MODE	Yes	No	No	No	No	No	No	No
MDM_CLUSTER_STATE	Yes	No	No	No	No	No	No	No
PRIMARY_MDM_ACTOR_IPS	Yes	No	No	No	No	No	No	No
PRIMARY_MDM_ACTOR_PORT	Yes	No	No	No	No	No	No	No
SECONDARY_MDM_ACTOR_IPS	Yes	No	No	No	No	No	No	No
SECONDARY_MDM_ACTOR_PORT	Yes	No	No	No	No	No	No	No
TIEBREAKER_MDM_ACTOR_IPS	Yes	No	No	No	No	No	No	No
TIEBREAKER_MDM_ACTOR_PORT	Yes	No	No	No	No	No	No	No
MDM_MGMT_IPS	Yes	No	No	No	No	No	No	No
MDM_MGMT_PORT	Yes	No	No	No	No	No	No	No
RESTRICTED_SDC_MODE_ENABLED	Yes	No	No	No	No	No	No	No
SDS_ID_LIST	No	Yes	No	No	No	No	No	No
STORAGE_POOL_ID_LIST	No	Yes	No	No	No	No	No	No
STATE	No	Yes	No	Yes	No	No	Yes	No
REBUILD_NETWORK_THROTTLING_ENABLED	No	Yes	No	No	No	No	No	No
REBALANCE_NETWORK_THROTTLING_ENABLED	No	Yes	No	No	No	No	No	No
OVERALL_IO_NETWORK_THROTTLING_ENABLED	No	Yes	No	No	No	No	No	No

Table 21. Object types and properties (continued)

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
REBUILD_NETWORK_THROTTLING	No	Yes	No	No	No	No	No	No
REBALANCE_NETWORK_THROTTLING	No	Yes	No	No	No	No	No	No
OVERALL_IO_NETWORK_THROTTLING	No	Yes	No	No	No	No	No	No
DEVICE_ID_LIST	No	No	Yes	Yes	No	No	No	No
VOLUME_ID_LIST	No	No	Yes	No	Yes	No	No	No
VTREE_ID_LIST	No	No	Yes	No	No	No	No	No
SPARE_PERCENT	No	No	Yes	No	No	No	No	No
PROTECTION_DOMAIN_ID	No	No	Yes	Yes	No	No	No	No
ZERO_PAD_ENABLED	No	No	Yes	No	No	No	No	No
USE_RMCACHE	No	No	Yes	No	No	Yes	No	No
RMCACHE_WRITE_HANDLING_MODE	No	No	Yes	No	No	No	No	No
REBUILD_ENABLED	No	No	Yes	No	No	No	No	No
REBUILD_IO_PRIORITY_POLICY	No	No	Yes	No	No	No	No	No
NUM_REBUILD_IOPS_PER_DEVICE	No	No	Yes	No	No	No	No	No
REBUILD_BW_LIMIT_PER_DEVICE	No	No	Yes	No	No	No	No	No
REBUILD_APP_IOPS_PER_DEVICE_THRESHOLD	No	No	Yes	No	No	No	No	No
REBUILD_APP_BW_PER_DEVICE_THRESHOLD	No	No	Yes	No	No	No	No	No
REBUILD_QUIET_PERIOD	No	No	Yes	No	No	No	No	No
REBALANCE_ENABLED	No	No	Yes	No	No	No	No	No
REBALANCE_IO_PRIORITY_POLICY	No	No	Yes	No	No	No	No	No
NUM_REBALANCE_IOPS_PER_DEVICE	No	No	Yes	No	No	No	No	No
REBALANCE_BW_LIMIT_PER_DEVICE	No	No	Yes	No	No	No	No	No
REBALANCE_APP_IOPS_PER_DEVICE_THRESHOLD	No	No	Yes	No	No	No	No	No
REBALANCE_APP_BW_PER_DEVICE_THRESHOLD	No	No	Yes	No	No	No	No	No
REBALANCE_QUIET_PERIOD	No	No	Yes	No	No	No	No	No
NUM_PARALLEL_JOBS_PER_DEVICE	No	No	Yes	No	No	No	No	No
IPS	No	No	No	Yes	No	No	No	No
PORT	No	No	No	Yes	No	No	No	No
ON_VMWARE	No	No	No	Yes	No	No	No	No
PROTECTION_DOMAIN_ID	No	No	No	Yes	No	No	No	No
MEMBERSHIP_STATE	No	No	No	Yes	No	No	No	No
MDM_CONNECTION_STATE	No	No	No	Yes	Yes	No	No	No
DRL_MODE	No	No	No	Yes	No	No	No	No
RMCACHE_ENABLED	No	No	No	Yes	No	No	No	No

Table 21. Object types and properties (continued)

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
RMCACHE_SIZE	No	No	No	Yes	No	No	No	No
RMCACHE_FROZEN	No	No	No	Yes	No	No	No	No
RMCACHE_MEMORY_ALLOCATION_STATE	No	No	No	Yes	No	No	No	No
NUMBER_OF_IO_BUFFERS	No	No	No	Yes	No	No	No	No
NUM_OF_MAPPED_VOLUMES	No	No	No	No	Yes	No	No	No
GUID	No	No	No	No	Yes	No	No	No
IP	No	No	No	No	Yes	No	No	No
APPROVED	No	No	No	No	Yes	No	No	No
CHILD_VOLUME_ID_LIST	No	No	No	No	No	Yes	No	No
NUM_OF_CHILD_VOLUMES	No	No	No	No	No	Yes	No	No
DESCENDANT_VOLUME_ID_LIST	No	No	No	No	No	Yes	No	No
NUM_OF_DESCENDANT_VOLUMES	No	No	No	No	No	Yes	No	No
NUM_OF_MAPPED_SDCS	No	No	No	No	No	Yes	No	No
SIZE	No	No	No	No	No	Yes	No	No
OBFUSCATED	No	No	No	No	No	Yes	No	No
CREATION_TIME	No	No	No	No	No	Yes	No	No
TYPE	No	No	No	No	No	Yes	No	No
CONSISTENCY_GROUP_ID	No	No	No	No	No	Yes	No	No
STORAGE_POOL_ID	No	No	No	No	No	Yes	Yes	Yes
VTREE_ID	No	No	No	No	No	Yes	No	No
ANCESTOR_ID	No	No	No	No	No	Yes	No	No
SOURCE_DELETED	No	No	No	No	No	Yes	No	No
MAPPING_TO_ALL_SDCS_ENABLED	No	No	No	No	No	Yes	No	No
AVG_READ_SIZE_IN_BYTES	No	No	No	No	No	No	Yes	No
AVG_WRITE_SIZE_IN_BYTES	No	No	No	No	No	No	Yes	No
AVG_READ_LATENCY_IN_MICROSEC	No	No	No	No	No	No	Yes	No
AVG_WRITE_LATENCY_IN_MICROSEC	No	No	No	No	No	No	Yes	No
CURRENT_PATH	No	No	No	No	No	No	Yes	No
ORIGINAL_PATH	No	No	No	No	No	No	Yes	No
ERR_STATE	No	No	No	No	No	No	Yes	No
CAPACITY_LIMIT	No	No	No	No	No	No	Yes	No
MAX_CAPACITY	No	No	No	No	No	No	Yes	No
SDS_ID	No	No	No	No	No	No	Yes	No
NET_CAPACITY_IN_USE_IN_KB	No	No	No	No	No	No	No	Yes
BASE_NET_CAPACITY_IN_USE_IN_KB	No	No	No	No	No	No	No	No
SNAP_NET_CAPACITY_IN_USE_IN_KB	No	No	No	No	No	No	No	Yes

Table 21. Object types and properties (continued)

Property	Sys	PD	SP	STORAGE NODE	SDC	Vol	Dev	V-Tree
TRIMMED_CAPACITY_IN_KB	No	No	No	No	No	No	No	Yes
BASE_VOLUME_ID	No	No	No	No	No	No	No	Yes
IQN	No	No	No	No	No	No	No	No

query_remote_read_only_limit_state

Query the remote access restriction state of the MDM. If the state is enabled, remote users may only issue read-only commands to the MDM. If the state is disabled, all command types may be issued to the MDM by remote clients.

Syntax

```
scli --query_remote_read_only_limit_state
```

Example

```
scli --query_remote_read_only_limit_state
```

query_system_limits

Retrieve information about the system limits.

Syntax

```
scli --query_system_limits
```

query_upgrade

Retrieve information about the upgrade process.

Syntax

```
scli --query_upgrade
   [--show_sdc_info]
```

Parameters

--show_sdc_info

Include SDC information

Example

```
scli --query_upgrade --show_sdc_info
```

refresh_mdm_cluster_capabilities

Refresh the exposed cluster node capabilities.

This command is usually used to instruct the cluster nodes to check whether LDAP and SSL libraries are installed.

Syntax

```
scli --refresh_mdm_cluster_capabilities
```

Example

```
scli --refresh_mdm_cluster_capabilities --mdm_port 6611
```

remove_standby_mdm

Remove a standby MDM or standby tiebreaker, that is not part of the cluster, from the system.

Syntax

```
scli --remove_standby_mdm (--remove_mdm_id <ID> | --remove_mdm_ip <IP> [--
remove_mdm_port <PORT>] | --remove_mdm_name <NAME>)
```

Parameters

--remove_mdm_id *<ID>*
: ID of the MDM to remove

--remove_mdm_ip *<IP>*
: IP address of the MDM to remove

--remove_mdm_port *<PORT>*
: Port associated with the MDM to remove

--remove_mdm_name *<NAME>*
: Name of the MDM to remove

Example

```
scli --remove_standby_mdm --remove_mdm_name mdm153
```

remove_standby_mdm

Remove a standby MDM or standby tiebreaker, that is not part of the cluster, from the system.

Syntax

```
scli --remove_standby_mdm (--remove_mdm_id <ID> | --remove_mdm_ip <IP> [--
remove_mdm_port <PORT>] | --remove_mdm_name <NAME>)
```

Parameters

--remove_mdm_id *<ID>*
ID of the MDM to remove

--remove_mdm_ip *<IP>*
IP address of the MDM to remove

--remove_mdm_port *<PORT>*
Port associated with the MDM to remove

--remove_mdm_name *<NAME>*
Name of the MDM to remove

Example

```
scli --remove_standby_mdm --remove_mdm_name mdm153
```

rename_mdm

Rename an MDM cluster node.

Syntax

```
scli --rename_mdm (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port
<PORT>] | --target_mdm_name <NAME>) --new_mdm_name <NAME>
```

Parameters

--target_mdm_id *<ID>*
ID of the MDM to be renamed

--target_mdm_ip *<IP>*
IP address of the MDM to be renamed

--target_mdm_port *<PORT>*
Port of the MDM to be renamed

--target_mdm_name *<NAME>*
Current name of the MDM to be renamed

--new_mdm_name *<NAME>*
New MDM name. The MDM name must be unique. If a new name is not provided, the old name is deleted.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --rename_mdm --new_mdm_name mdm3 --target_mdm_name mdm2
```

rename_system

Name or rename the system.

To determine the current system name, use the `query_cluster` or `query_all` command.

Syntax

```
scli --rename_system --new_name <NAME>
```

Parameters

--new_name *<NAME>*

New name to associate with the system.

Example

```
scli --rename_system --new_name sio-prod-sys2
```

query_all_system_networks

Retrieve information about all system networks.

query_system_network

Retrieve information about a system network.

Syntax

```
scli --query_system_network (--network_id <ID> | --network_name <NAME>)
```

Parameters

--network_id *<ID>*

System network ID

--network_name *<NAME>*

System network name

add_system_network

Add a system network to the system.

Syntax

```
scli --add_system_network --network_ip <IP> (--network_mask <MASK> | --network_prefix
<PREFIX>) [--network_name <NAME>] [(--host_group_id <ID> | --host_group_name <NAME>)]
```

Parameters

--network_ip *<ID>*

The new network IP

--network_mask *<MASK>*

The new network mask

--network_prefix *<PREFIX>*

The new network prefix

--network_name *<NAME>*

A name to be assigned to the system network

--network_prefix *<PREFIX>*

The new network prefix

--host_group_id *<ID>*

A comma separated list of host group IDs that have access to the system network

--host_group_name *<NAME>*

A comma separated list of host group names that have access to the system network

rename_system_network

Name or rename a system network.

Syntax

```
scli --rename_system_network (--network_id <ID> | --network_name <NAME>) --new_name
<NAME>
```

Parameters

--network_id *<ID>*

System network ID

--network_name *<NAME>*

System network name

--new_name *<NAME>*

A new name to be assigned to the system network

remove_system_network

Remove a system network from the system.

Syntax

```
scli --remove_system_network (--network_id <ID> | --network_name <NAME>)
```

Parameters

--network_id *<ID>*
System network ID

--network_name *<NAME>*
System network name

replace_cluster_mdm

Replace members of the MDM cluster. Up to half (rounded down) of the cluster members can be replaced at one time.

Syntax

```
scli --replace_cluster_mdm ((--add_secondary_mdm_id <ID> |
        --add_secondary_mdm_ip <IP> [--add_secondary_mdm_port <PORT>] |
        --add_secondary_mdm_name <NAME>)| (--remove_secondary_mdm_id <ID> |
        --remove_secondary_mdm_ip <IP> [--remove_secondary_mdm_port <PORT>] |
        --remove_secondary_mdm_name <NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP>
        [--add_tb_port <PORT>] | --add_tb_name <NAME>) | (--remove_tb_id <ID> | --
remove_tb_ip
        <IP> [--remove_tb_port <PORT>] | --remove_tb_name <NAME>)) [--allow_leave_failed]
        [--i_am_sure]
```

Parameters

--add_secondary_mdm_id *<ID>*
Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_secondary_mdm_ip *<IP>*
Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_secondary_mdm_port *<PORT>*
Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_secondary_mdm_name *<NAME>*
Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_secondary_mdm_id *<ID>*
Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --replace_cluster_mdm --add_secondary_mdm_name mdm33 --remove_secondary_mdm_name
mdm31 --add_tb_name tb44 --remove_tb_name tb41
```

replace_mdm_security_files

Replace the security certificate and key of the MDM. The replacement certificate can be either a self-generated and self-signed certificate, or a certificate generated and signed by an external certificate authority. Security permissions are required in order to run this command.

NOTE: Running the command on the primary MDM causes the MDM to restart. It may also cause a brief, single point of failure period due to an MDM switch over.

Syntax

```
scli --replace_mdm_security_files (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

Example

```
scli --replace_mdm_security_files --target_mdm_name mdm153
```

set_management_client_communication

Enable or disable secure communications between the MDM and the management client.

Syntax

```
scli --set_management_client_communication (--enable_client_secure_communication | --
disable_client_secure_communication)
```

Parameters

--enable_client_secure_communication
Enable the use of security when communicating with management clients.

--disable_client_secure_communication
Disable the use of security when communicating with management clients.

Example

```
scli --set_management_client_communication --enable_client_secure_communication
```

set_performance_parameters

Apply performance profiles to system components. You can apply separate profiles for storage nodes, SDCs,and the MDM cluster.

When a container is provided in the command (protection domain, all storage nodes, all SDCs), the command configures all objects in the container. Objects that are added later are not configured.

NOTE: If you change the performance profile of a storage node on an SVM, you must afterwards perform manual memory allocation on the SVM.

Syntax

```
scli --set_performance_parameters ([--storage_node_id <ID> | --storage_node_name <NAME>
| --storage_node_ip <IP> [--storage_node_port <PORT>] | --protection_domain_id <ID> | --
protection_domain_name <NAME> | ((--protection_domain_id <ID> | --protection_domain_name
<NAME>) | --all_storage_node] [--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> |
--sdc_ip <IP> | --all_sdc] [(((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)] [--apply_to_mdm]) (--
profile <PROFILE>)
```

Parameters

--sds_id *<ID>*
Storage node ID

--sds_name *<NAME>*
Storage node name

--sds_ip *<IP>*
Storage node IP address

--sds_port *<PORT>*
Port associated with the storage node

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC global unique identifier

--sdc_ip *<IP>*
SDC IP address

--all_sdc
Set storage node-related parameters on all SDCs.

--storage_node_name <NAME>
Storage node name

--storage_node_id <ID>
Storage node ID

--storage_node_ip <IP>
Storage node IP address

--storage_node_port <PORT>
Port assigned to the storage node

--all_storage_node
Set storage node related parameters to all storage nodes

--apply_to_mdm
Apply the performance profile to the MDM. This flag is only applicable when setting the profile.

--profile {compact | high_performance}

Set the performance profile as compact or high_performance. The high performance profile configures a predefined set of parameters for very high performance use cases.

NOTE: For a complete list of parameters controlled by the profiles, see the *PowerFlex Performance Fine-Tuning Technical Notes*.

Example

```
scli --set_performance_parameters --all_sds --all_sdc --profile high_performance
```

start_upgrade

Start the upgrade process.

Syntax

```
scli --start_upgrade
```

Example

```
scli --start_upgrade
```

switch_cluster_mode

Switch the MDM cluster from one operating mode or configuration to another.

Syntax

```
scli --switch_cluster_mode --cluster_mode <MODE> ((--add_secondary_mdm_id <ID> |
        --add_secondary_mdm_ip <IP> [--add_secondary_mdm_port <PORT>] |
        --add_secondary_mdm_name <NAME>)| (--remove_secondary_mdm_id <ID> |
        --remove_secondary_mdm_ip <IP> [--remove_secondary_mdm_port <PORT>] |
        --remove_secondary_mdm_name <NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP>
        [--add_tb_port <PORT>] | --add_tb_name <NAME>) | (--remove_tb_id <ID> | --
remove_tb_ip
        <IP> [--remove_tb_port <PORT>] | --remove_tb_name <NAME>)) [--allow_leave_failed]
        [--i_am_sure]
```

Parameters

--cluster_mode *<MODE>*

Cluster operation mode. Can be one of the following:

1_node

The MDM cluster will operate in single-node mode.

NOTE: Single-node mode should only be used temporarily during maintenance activities. Single-node mode does not provide failover protection for the MDM.

3_node

The MDM cluster will operate in 3-node mode.

5_node

The MDM cluster will operate in 5-node mode.

--add_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --switch_cluster_mode --cluster_mode 5_node --add_secondary_mdm_ip 192.168.1.164 --
add_tb_ip 192.168.1.134
```

switch_mdm_ownership

Switch MDM cluster ownership from the current primary MDM to a different MDM.

Syntax

```
 scli --switch_mdm_ownership (--new_primary_mdm_id <ID> | --new_primary_mdm_ip
       <IP> [--new_primary_mdm_port <PORT>]| --new_primary_mdm_name <NAME>)
```

Parameters

--new_primary_mdm_id *<ID>*

MDM ID. Must be an MDM with manager role.

--new_primary_mdm_ip *<IP>*

MDM IP address. Must be an MDM with manager role.

--new_primary_mdm_port *<PORT>*

MDM port. Must be an MDM with manager role.

--new_primary_mdm_name *<NAME>*

MDM name. Must be an MDM with manager role.

Example

```
scli --switch_mdm_ownership --new_primary_mdm_ip 192.168.1.164
```

generate_mdm_mno_csr

Generate a certificate signing request (CSR) file for the given address. The file is saved at <generated CSR path> and also locally in the configuration directory of the MDM.

Syntax

```
scli --generate_mdm_mno_csr --csr_file_path <generated CSR path>
```

Parameters

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

replace_mdm_mno_certificate

Replace the old MDM certificate with a new certificate at a given address.

Syntax

```
scli --replace_mdm_mno_certificate (--target_mdm_id <ID> | --target_mdm_ip
<IP> [--target_mdm_port <PORT>] | --target_mdm_name <NAME>) --certificate_file
<MDM_CERT_FILENAME>
```

Parameters

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

query_all_network_sets

Retrieve information about all network sets.

query_network_set

Retrieve information about a network set.

Syntax

```
scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*

Network set ID

--network_set_name *<NAME>*

Network set name

modify_network_set

Modify the list of system networks of an existing network set.

Syntax

```
scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [--
network_id <ID> | --network_name <NAME>]
```

Parameters

--network_set_id *<ID>*
: Network set ID

--network_set_name *<NAME>*
: Network set name

--network_id *<ID>*
: A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*
: A comma separated list of system network names to be replaced with the old one

rename_network_set

Name or rename a network set.

Syntax

```
scli --rename_network_set (--network_set_id <ID> | --network_set_name <NAME>) --new_name
<NAME>
```

Parameters

--network_set_id *<ID>*
: Network set ID

--network_set_name *<NAME>*
: Network set name

--new_name *<NAME>*
: A new name to be assigned to the network set

remove_network_set

Remove a network set from the system.

Syntax

```
scli --remove_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

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Snapshot policy commands

This section contains commands for managing snapshot policy commands.

add_snapshot_policy

Create a snapshot policy to enable you to define the number of snapshots taken at a given time for one or more volumes.

Volumes snapshots are taken according to the rule defined in the snapshot policy. In a multilevel structure, define the time interval in between two rounds of snapshots and the number of snapshots to retain. For example, take snapshots every x minutes/hours/days/weeks. There are one to six levels, with the first level having the most frequent snapshots.

Syntax strings

```
scli --add_snapshot_policy [--snapshot_policy_name <NAME>] --snapshot_creation_cadence
<CADENCE> --number_of_snapshots_per_retention_level  <LEVELS> [--paused] [--
read_only_snapshots] [--secure_snapshots [--i_am_sure]]
```

Parameters

--snapshot_policy_name *<NAME>*

Snapshot policy name

--snapshot_creation_cadence *<CADENCE>*

The interval in minutes between two snapshots in the policy (minimum is 5).

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are one to six levels, and the first level has the most frequent snapshots. There can be up to 1021 snapshots in all levels (two consecutive levels overlap by one).

--paused

Indicates that the snapshot policy should pause on creation.

--secure_snapshots

Define all auto snapshots created as secure. It will not be possible to edit or remove them before the retention period is complete.

--i_am_sure

Preemptive approval for the creating the snapshots as secure.

assign_snapshot_policy

Add a source volume to a snapshot policy.

Syntax

```
scli --assign_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>)
```

Parameters

--snapshot_policy_id <ID>
: Snapshot policy ID

--snapshot_policy_name <NAME>
: Snapshot policy name

--source_vol_id <ID>
: The source volume ID

--source_vol_name <NAME>
: The source volume name

align_snapshot_policy_to_current_time

Align the specified snapshot policy's cadence cycles to current time.

Syntax

```
scli --align_snapshot_policy_to_current_time (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*
: Snapshot policy ID

--snapshot_policy_name *<NAME>*
: Snapshot policy name

Example

```
scli --mdm_ip 192.168.1.200 --align_snapshot_policy_to_current_time --
snapshot_policy_name SNP1
```

modify_snapshot_policy

Modify the number of snapshots that are taken per minute or the number of snapshots per retention level for a specific policy.

Syntax

```
scli --modify_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>)((--snapshot_creation_cadence <CADENCE> --number_of_snapshots_per_retention_level
<LEVELS>) |
([--set_secure_snapshots] [--apply_to_existing_snapshots])) [--i_am_sure]
```

Parameters

--snapshot_policy_id *<ID>*
: Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

--snapshot_creation_cadence *<CADENCE>*

The interval in minutes between two snapshots in the policy

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are six levels, and the first level has the most frequent snapshots. There can be up to 1021 snapshots total in all levels (note that two consecutive levels overlap by one).

--set_secure_snapshots

The auto snapshots will be created as secure. They cannot be edited or removed prior to their policy expiration time

--apply_to_existing_snapshots

Referring to --set_secure_snapshots:

apply the change on existing auto snapshots of all policies

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --modify_snapshot_policy --snapshot_policy_name SP_pol1 --snapshot_creation_cadence
2 --numer of snapshots_per_retention_level 5 --set_read_only_snapshots --i_am_sure
```

pause_snapshot_policy

Pause a snapshot policy so that no auto snapshots will be created or removed.

Syntax

```
scli --pause_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

Example

```
scli --pause_snapshot_policy --snapshot_policy_name snap_pol_1
```

query_all_snapshot_policies

Retrieve information about all snapshot policies.

Syntax

```
scli --query_all_snapshot_policies
```

Example

```
scli --query_all_snapshot_policies
```

Sample output

```
Query all snapshot policies returned 3 policies
Snapshot Policy policy1 (Id: 36a39c6200000000)
        Cadence: 30 minutes
        Number of snapshots per retention level: 1 (read-only)
        State: Active
        Time of last auto snapshot: N/A
        Next run time: 2020-11-12 09:56:17
        Number of source volumes: 0
        Number of auto snapshots: 0

Snapshot Policy policy2_secure (Id: 36a3c37200000001)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1 (secure, read-only)
        State: Active
        Time of last auto snapshot: 2020-11-12 09:47:18
        Next run time: 2020-11-12 09:52:17
        Number of source volumes: 1
        Number of auto snapshots: 1

Snapshot Policy policy3 (Id: 36a3ea8200000002)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1 (read-only)
        State: Active
        Time of last auto snapshot: 2020-11-12 09:44:18
        Next run time: 2020-11-12 09:49:17
        Number of source volumes: 1
        Number of auto snapshots: 1
```

query_snapshot_policy

Retrieve information about a snapshot policy.

Syntax

```
scli --query_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
[--show_auto_snapshots]
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

--show_auto_snapshots

Indicate whether to display the full list of auto snapshots

Example

```
scli --query_snapshot_policy --snapshot_policy_name policy1 --show_auto_snapshots
```

Sample output

```
Snapshot Policy policy1 (Id: aaa4278700000000)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1
        State: Active
        Time of last auto snapshot: 2021-02-10 13:39:13
        Next run time: 2021-02-10 13:44:13
        Number of auto snapshots: 2

Source volume information (total 2 volumes):
         1: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB)
         2: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB)
```

remove_snapshot_policy

Remove the snapshot policy specified by name or ID.

Syntax

```
scli --remove_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>) (--detach_auto_snapshots | (--remove_auto_snapshots ))
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

--detach_auto_snapshots

Detach all associated auto snapshots

--remove_auto_snapshots

Remove all associated auto snapshots

Example

```
scli --remove_snapshot_policy --snapshot_policy_name SP_pol_1
```

rename_snapshot_policy

Name or rename the snapshot policy specified by name or ID.

Syntax

```
scli --rename_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>) --new_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

--new_name *<NAME>*
New name of the snapshot policy

Example

```
scli --rename_snapshot_policy --snapshot_policy_name SP_pol_1 --new_name snap_pol_1
```

resume_snapshot_policy

Resume a paused snapshot policy so that auto snapshots will once again be created and removed.

Syntax

```
scli --resume_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

Example

```
scli --resume_snapshot_policy --snapshot_policy_name snap_pol_1
```

unassign_snapshot_policy

Remove a source volume from a snapshot policy.

When removing the source volume from the policy, you must choose how to handle the snapshots created by the policy (auto snapshots). If the source volume has auto snapshots, you can choose to detach all snapshots with the `--detach_auto_snapshots` flag. They become regular snapshots as if they were created manually. If you use the `--remove_auto_snapshots`, the snapshots are deleted.

Syntax

```
scli --unassign_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>) (--detach_auto_snapshots | (--
remove_auto_snapshots ))
```

Parameters

--snapshot_policy_id <ID>
: Snapshot policy ID

--snapshot_policy_name <NAME>
: Name of the snapshot policy

--source_vol_id <ID>
: The source volume ID

--source_vol_name <NAME>
: The source volume name

--detach_auto_snapshots
: Detach all associated auto snapshots

--remove_auto_snapshots
: Remove all associated auto snapshots

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Volume commands

This section contains commands for managing volumes.

add_volume

Add a volume to the PowerFlex system.

You can create a volume when the requested capacity is available and system capacity is balanced.

The created volume cannot be used until it is mapped to at least one SDC.

Syntax

```
scli --add_volume (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
storage_pool_name <NAME>) | --storage_pool_id <ID>) --size_gb <SIZE>
  [--volume_name <NAME>]
```

Parameters

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--storage_pool_name *<NAME>*
: Storage pool name

--storage_pool_id *<ID>*
: Storage pool ID

--size_gb *<SIZE>*
: Volume size, in GB. Basic allocation granularity is 1 GB.

--volume_name *<NAME>*
: Name to be associated with the added volume

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --add_volume --size_gb 8 --volume_name vol_1 --
protection_domain_name pd1 --storage_pool_name sp1
```

Volume names

Assign each volume a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the ID of the volume.

Each volume name should conform to the following rules:

1. Contains fewer than 32 characters.
2. Contains only alphanumeric and punctuation characters.
3. Is unique within the object type.

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

clear_scsi2_reservation

Clear SCSI2 reservation from an SDC.

Syntax

```
scli --clear_scsi2_reservation (--volume_ID <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id *<ID>*
ID of the volume where SCSI2 reservation from the SDC will be reset

--volume_name *<NAME>*
Name of the volume where SCSI2 reservation from the SDC will be reset

--wait
Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure
Skip the safety questions for command execution.

Example

```
scli --clear_scsi2_reservation --volume_name public --sdc_ip 10.76.60.233 --wait
```

clear_scsi3_reservation

Clear SCSI3 reservations from all SDCs.

Syntax

```
scli --clear_scsi3_reservation (--volume_id <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id *<ID>*
ID of the volume for which SCSI3 reservation will be reset

--volume_name *<NAME>*
Name of the volume for which SCSI3 reservation will be reset

--wait

Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --clear_scsi3_reservation --volume_name temp1 --i_am_sure
```

map_volume_to_host

Map a volume to SDC or to NVMe host.

Syntax

```
scli --map_volume_to_host (--volume_id <ID> | --volume_name <NAME> (--host_id <ID> |
--host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip <IP>) [--
access_mode <ACCESS_MODE>] [--allow_multi_map]
```

Add the attribute, `--volume class` to map volumes to host:

```
scli --map_volume_to_host --volume_id <ID> --sdc_guid <GUID> --volume_class nascluster
```

For example:

```
scli --map_volume_to_host --volume_id 2f362c3800000100 --sdc_guid 01D16FC2-342B-4CC2-
A050-DCD97B4E0GH7 --volume_class nascluster
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--access_mode *<ACCESS_MODE>*

Mapping access mode. Can be one of the following: read_write (default), read_only, no_access.

--allow_multi_map

Allow this volume to be mapped to more than one host.

unmap_volume_from_host

Unmap a volume from one host or all hosts.

Syntax

```
scli --unmap_volume_from_host (--volume_id <ID> | --volume_name <NAME>) (--host_id <ID>
| --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip <IP> | --
all_host) [--i_am_sure]
```

Add the attributes `--volume class` and `--sdc_guid` to unmap volumes from the host:

```
scli --unmap_volume_from_host --volume_id <ID> --sdc_guid <GUID> --volume_class
nascluster
```

For example:

```
scli --unmap_volume_from_host --volume_id 2f362c3800000100 --sdc_guid 01D16FC2-342B-4CC2-
A050-DCD97B4E0GH7 --volume_class nascluster
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe Host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--all_host

Unmap the volume from all hosts.

--i_am_sure

Preemptive approval

modify_volume_capacity

Increase the capacity of a volume. You can increase (but not decrease) a volume capacity at any time, as long as there is enough capacity for the volume size to grow.

Syntax

```
scli --modify_volume_capacity (--volume_id <ID> | --volume_name <NAME>) --size_gb <SIZE>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--size_gb *<SIZE>*

New volume size, in GB. Basic allocation granularity is 8 GB.

Example

```
scli --mdm_ip 192.168.1.200 --modify_volume_capacity
--volume_name vol_1 --size_gb 150000
```

query_all_volumes

Retrieve information about all volumes in the system. You can optionally filter the query results by protection domain or storage pool.

Syntax

```
scli --query_all_volumes [(((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)] [--show_snapshots]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--show_snapshots

Display EC read-only snapshots (not displayed by default)

Example

```
scli --mdm_ip 192.168.1.200 --query_all_volumes --show_snapshots
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Storage Pool 3df1cd9600000000 Name: sp1
 Volume ID: f6351e3700000003 Name: vol1_114 Size: 16.0 GB (16384 MB) Mapped to 1 Host
 Volume ID: f6351e3800000004 Name: vol2_114 Size: 8.0 GB (8192 MB) Not mapped
 Volume ID: f6351e3900000005 Name: vol3_pol1 Size: 8.0 GB (8192 MB) Not mapped
```

```
 Volume ID: f6683c2700000002 Name: N/A Size: 8.0 GB (8192 MB) Not mapped read-only

Protection Domain fcc9af5600000001 Name: pd2
Storage Pool 3df1cd9700000001 Name: sp2
 <No volumes defined>

Protection Domain fcc9d66500000002 Name: pd3
 <No volumes defined>
Query-all-volumes returned 4 volumes
```

query_volume

Retrieve information about the specified volume.

Syntax

```
scli --query_volume (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_volume --volume_id 102bbf0a0000000c
```

query_volume_copies

Retrieves the direct copies of a volume or a clone.

Syntax

```
scli --query_volume_copies (--volume_id <ID> | --volume_name <NAME>) [--snapshots_only]
[--clones_only] [--entire_volume_family]
```

Parameters

--volume_id <ID>
Volume ID

--volume_name <NAME>
Volume name

--snapshots_only
Display snapshots only

--clones_only
Display clones only

--entire_volume_family

Display entire volume family

Example

```
scli --query_volume_copies --volume_name vol_0
```

Sample output

```
Volume ID: 102bbf0500000007 Name: clone_0
      Size: 8.0 GB (8192 MB)
      Used: 0 Bytes
      Creation time: 2025-07-10 11:40:13

Volume ID: 102bbf0400000006 Name: clone_1
      Size: 8.0 GB (8192 MB)
      Used: 0 Bytes
      Creation time: 2025-07-10 11:40:13

Query volume copies returned 2 volumes
```

query_volume_sdc_meters

Query the IOPS, bandwidth, and latency counters for volume mappings. You can either specify a volume, an SDC, or both. For volume only, display metrics for all mappings of the specified volume. For SDC only, display metrics for all mappings of volumes to the specified SDC. For both volume and SDC, display metrics for a specific mapping.

Syntax

```
scli --query_volume_sdc_meters [--volume_id <ID> | --volume_name <NAME>] [--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --query_volume_sdc_meters --volume_id f6351e3700000003
```

Sample output

```
Information on 1 mapping:

Volume: f6351e3700000003 SDC: c27ab52600000001
Reads:  426 IOPS 212.8 MB (217907 KB) per-second
        213 IOPS 22.033 ms average latency
Writes: 450 IOPS 224.6 MB (230000 KB) per-second
        225 IOPS 261.137 ms average latency
```

query_volume_sdt_meters

Query the IOPS, bandwidth, and latency counters for volume mappings. You can either specify a volume, an SDT, or both. For volume only, display metrics for all mappings of the specified volume. For SDT only, display metrics for all mappings of volumes to the specifed SDT. For both volume and SDT, display metrics for a specific mapping.

Syntax

```
scli --query_volume_sdt_meters [(--volume_id <ID> | --volume_name <NAME>)] [(--sdt_id
<ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>])]
```

Parameters

--volume_id *<ID>*
: Volume ID

--volume_name *<NAME>*
: Volume name

--sdt_id *<ID>*
: SDC ID

--sdt_name *<NAME>*
: SDC name

--sdt_ip *<IP>*
: SDC IP address

--sdt_port *<PORT>*
: Port assigned to the SDT

remove_volume

Remove a PowerFlex volume.

NOTE: Removing a volume erases all the data on the corresponding volume.

Before removing a volume, you must ensure that it is not mapped to any SDCs. If the volume is mapped to SDCs, unmap the volume using the `unmap_volume_from_sdc` command or the PowerFlex Manager, or the vSphere plug-in (where applicable) before removing it.

By default EC volumes are removed along with their direct read-only snapshots.

Syntax

```
scli --remove_volume (--volume_id <ID> | --volume_name <NAME>)
{--remove_entire_snapshot_tree | --remove_with_descendant_snapshots | --
remove_descendant_snapshots_only}
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*

ID of the volume to remove

--volume_name *<NAME>*

Name of the volume to remove

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

[Options]

Additional optional parameters:

--remove_entire_tree

Removes the entire tree assigned to the specified volume including parent and sibling snapshots and clones

--remove_with_descendant_snapshots

Removes the specified volume and all clones and snapshots that were created from it or from one of its descendants

--remove_descendants_only

Removes all clones and snapshots created from the specified volume but does not remove the specified volume itself

NOTE: Before removing a V-Tree, you must unmap every volume in the V-Tree.

Example

```
scli --mdm_ip 192.168.1.200 --remove_volume --volume_name vol_1
```

rename_volume

Changes the name of a volume. You can use this command at any time.

Syntax

```
scli --rename_volume --volume_id <ID>| --volume_name <NAME> --new_name <NAME>
```

Parameters

--volume_id ***<ID>***

Volume ID

--volume_name ***<NAME>***

Volume name

--new_name *<NAME>*

New name to assign to the volume

Example

```
scli --mdm_ip 192.168.1.200 --rename_volume
--volume_name vol1 --new_name vol_new_1
```

set_volume_mapping_access_mode

Set the mapping access mode for the specified volume.

Syntax

```
scli --set_volume_mapping_access_mode (--volume_id <ID> | --volume_name <NAME>) (--
host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip
<IP> | --all_host) --access_mode <ACCESS_MODE>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--host_id *<ID>*

Host ID

--host_name *<NAME>*

host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC GUID

--sdc_ip *<IP>*

SDC IP address

--all_host

Use this flag to apply the volume access mode to all hosts

--access_mode {read_write | read_only | no_access}

The access mode

set_volume_access_mode_limit

Set the highest mapping access mode allowed for a volume.

Syntax

```
scli --set_volume_access_mode_limit (--volume_id <ID> | --volume_name <NAME>) --
access_mode_limit <ACCESS_MODE>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--access_mode_limit {read_write | read_only}

Access mode limit.

Example

```
scli --mdm_ip 192.168.1.200 set_volume_access_mode_limit --volume_id a4b5862000000000 --
access_mode_limit read_only
```

create_thin_clone

Create clones of one or several volumes or snapshots. The results are writable objects.

Syntax

```
scli --create_thin_clone (--from_volume_id <ID> | --from_volume_name <NAME> | --
from_snapshot_id <ID> | --from_snapshot_name <NAME>) [--new_clone_name <NAME>]
```

Parameters

--from_volume_id <ID>

A comma-separated list of source volume IDs

--from_volume_name <NAME>

A comma-separated list of source volume names

--from_snapshot_id <ID>

Source snapshot ID

--from_snapshot_name <NAME>

Source snapshot name

--new_clone_name <NAME>

A comma-separated list of names to be assigned to the created clones accordingly

query_vtree

Retrieve information about the entire V-Tree (a volume and all its snapshots) of the specified volume.

Syntax

```
scli --query_vtree (--vtree_id <ID> | --volume_id <ID> | --volume_name
```

Parameters

--vtree_id *<ID>*
V-Tree ID

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_vtree --volume_name TestVol0
```

create_snapshot

Create snapshots of one or several volumes. The results are read-only objects.

Syntax

```
scli --create_snapshot (--from_volume_id <ID> | --from_volume_name <NAME>) [--
new_snapshot_name <NAME>] [--retention_period <PERIOD>] [--i_am_sure]
```

Parameters

--from_volume_id <ID>
A comma-separated list of source volume IDs

--from_volume_name <NAME>
A comma-separated list of source volume names

--new_snapshot_name <NAME>
A comma-separated list of snapshot names to be assigned to the created snapshots accordingly

--retention_period <PERIOD>
The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m` | `h` | `D` | `W` | `M` | `Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days).

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

clear_snapshot_security

Clear the snapshot security for the specified volume to enable editing or removing the volume. This command can only be run with a signature file obtained from customer support. If you need to clear the snapshot security, contact Dell Technologies Support to obtain the signature file.

Syntax

```
scli --clear_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
signature_file <FILE> [--i_am_sure]
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--signature_file *<FILE>*
Signature file provided by customer support.

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --clear_snapshot_security --volume_name vol1 --
signature_file /opt/vol-dd6e34001.sig --i_am_sure
```

set_snapshot_security

Set the specified snapshot as secure. The snapshot is not editable or removable for the specified retention period. If the specified snapshot is already secure, the security expiration date may be postponed, but not configured to an earlier point in time.

Syntax

```
scli --set_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
retention_period <PERIOD> [--i_am_sure]
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--retention_period *<PERIOD>*

The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m` | `h` | `D` | `W` | `M` | `Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days).

`--i_am_sure`

Preemptive approval

Example

```
scli --mdm_ip 192.168.1.200 --set_snapshot_security --volume_id 022beb2500000006 --
retention_period 2Y --i_am_sure
```

refresh_data

Overwrite the data of one volume with the data of a selected volume or snapshot from the same volume family.

Syntax

```
scli --refresh_data (--from_volume_id <ID> | --from_volume_name <NAME> | --
from_snapshot_id <ID> | --from_snapshot_name <NAME>) (--to_volume_id <ID> | --
to_volume_name <NAME>) [--i_am_sure]
```

Parameters

--from_volume_id <ID>

Source volume ID

--from_volume_name <NAME>

Source volume name

--from_snapshot_id <ID>

Source Snapshot ID

--from_snapshot_name <NAME>

Source Snapshot name

--to_volume_id <ID>

Target volume ID

--to_volume_name <NAME>

Target volume name

--i_am_sure

Preemptive approval

restore_data

Restores data from a snapshot to its parent volume.

Syntax

```
scli --restore_data (--from_snapshot_id <ID> | --from_snapshot_name <NAME>) (--
to_volume_id <ID> | --to_volume_name <NAME>) [--i_am_sure]
```

Parameters

--from_snapshot_id <ID>
Source Snapshot ID

--from_snapshot_name <NAME>
Source Snapshot name

--to_volume_id <ID>
Target volume ID

--to_volume_name <NAME>
Target volume name

--i_am_sure
Preemptive approval

remove_snapshot

Removes a snapshot.

Syntax

```
scli --remove_snapshot (--snapshot_id <ID> | --snapshot_name <NAME>)  [--i_am_sure]
```

Parameters

--i_am_sure
Preemptive approval

--snapshot_id <ID>
Snapshot ID

--snapshot_name <NAME>
Snapshot name